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ATX

MS-7D09

Ver: 10

RLK Platform

CPU:

Rocket lake S

Onboard Chip:

SIO:NTC6687

LAN-Intel I225

Flash ROM: SPI 256 MB X1

HD Audio Codec:ALC897

Main Memory:

*DDRIV (5000MHz) * 4 (Dual Channel)*

ACPI:

LDO

Expansion Slots:

*PCI Express (X16) Slot * 1*

*PCI Express (X4) Slot * 1*

*PCI Express (X1) Slot * 2*

*M2 (X4) Slot * 3*

System Chipset:

Z590 PCH_H

PWM:

IMVP8 -RT3609

Other:

*SATA3.0 *6*

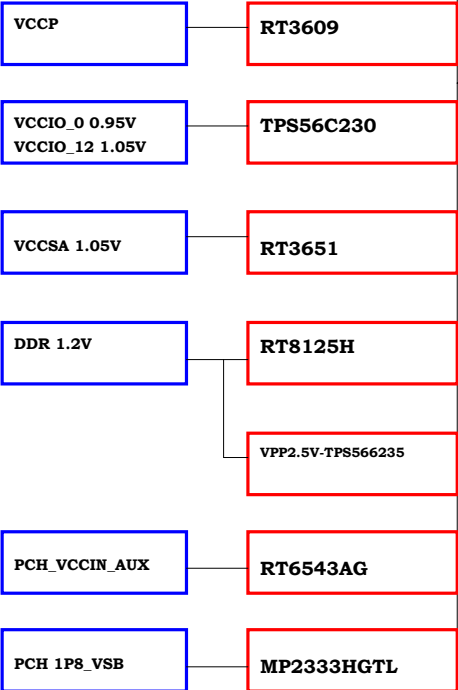
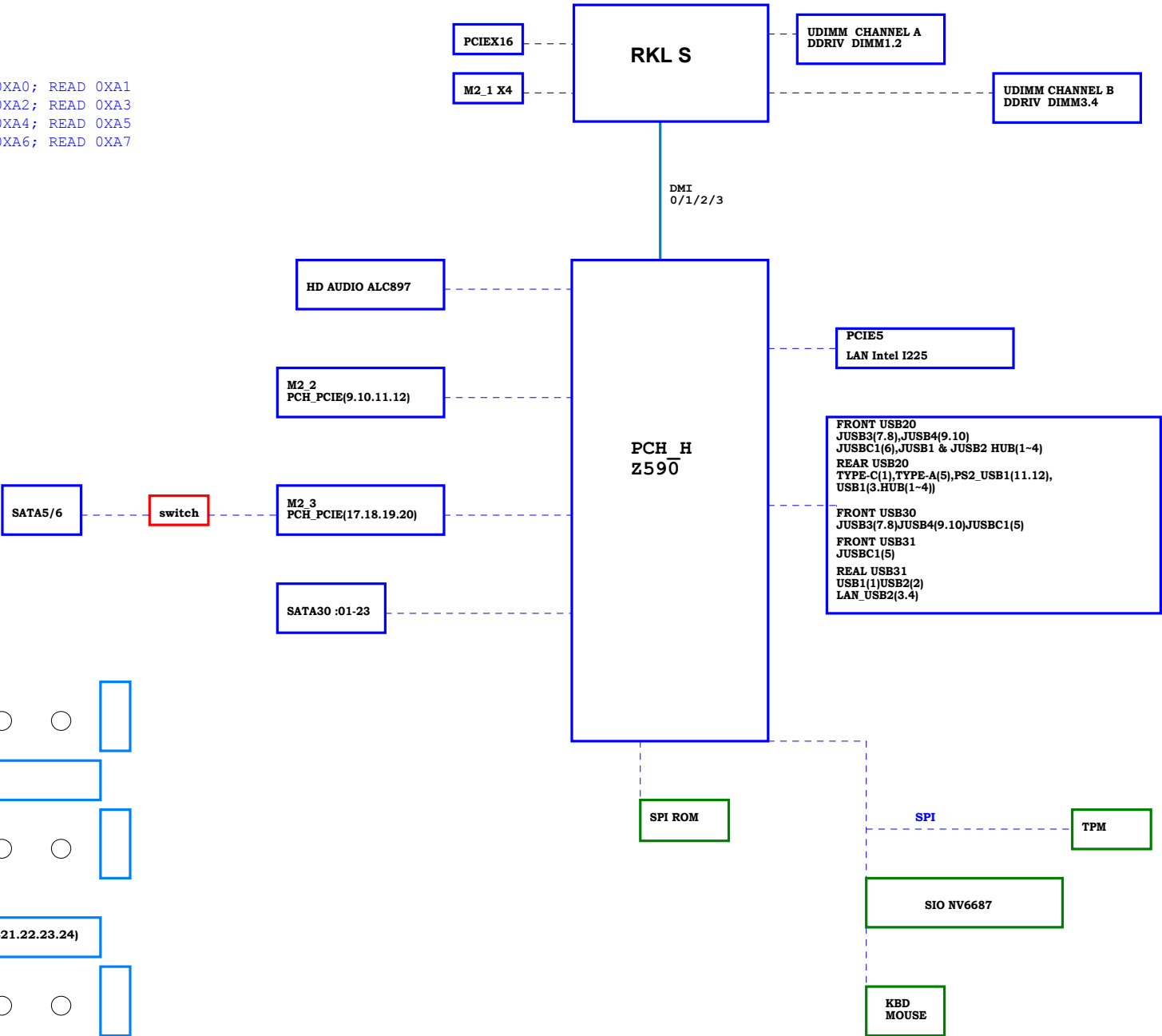
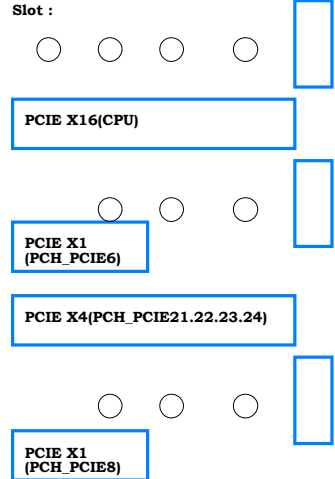
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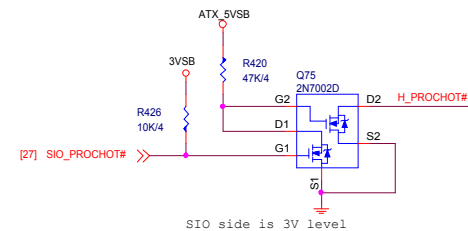
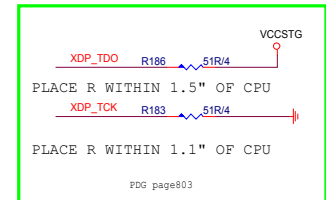
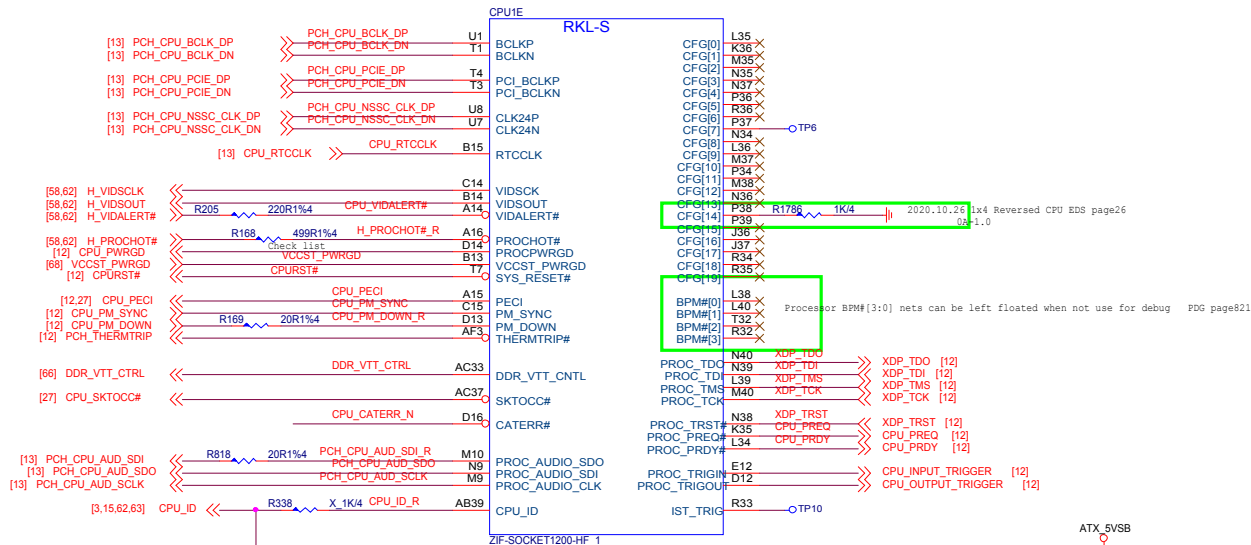
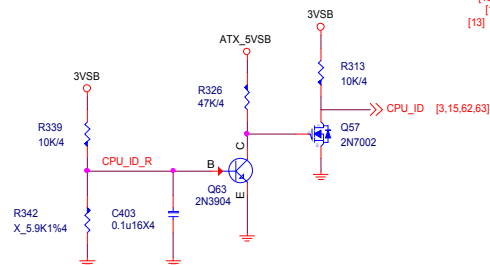
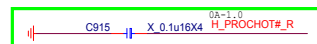
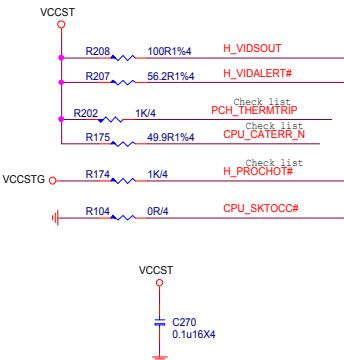
*REAL USB3.0 *2+TYPEC*1*

*FRONT USB3.0 *4+TYPEC*1*

Block Diagram

SMBUS ADDRESS:
VCORE_PWM:0X60
MCU:0X52
3933:0X20;0X26
DIMM_A0:WRITE 0XA0; READ 0XA1
DIMM_A1:WRITE 0XA2; READ 0XA3
DIMM_B0:WRITE 0XA4; READ 0XA5
DIMM_B1:WRITE 0XA6; READ 0XA7



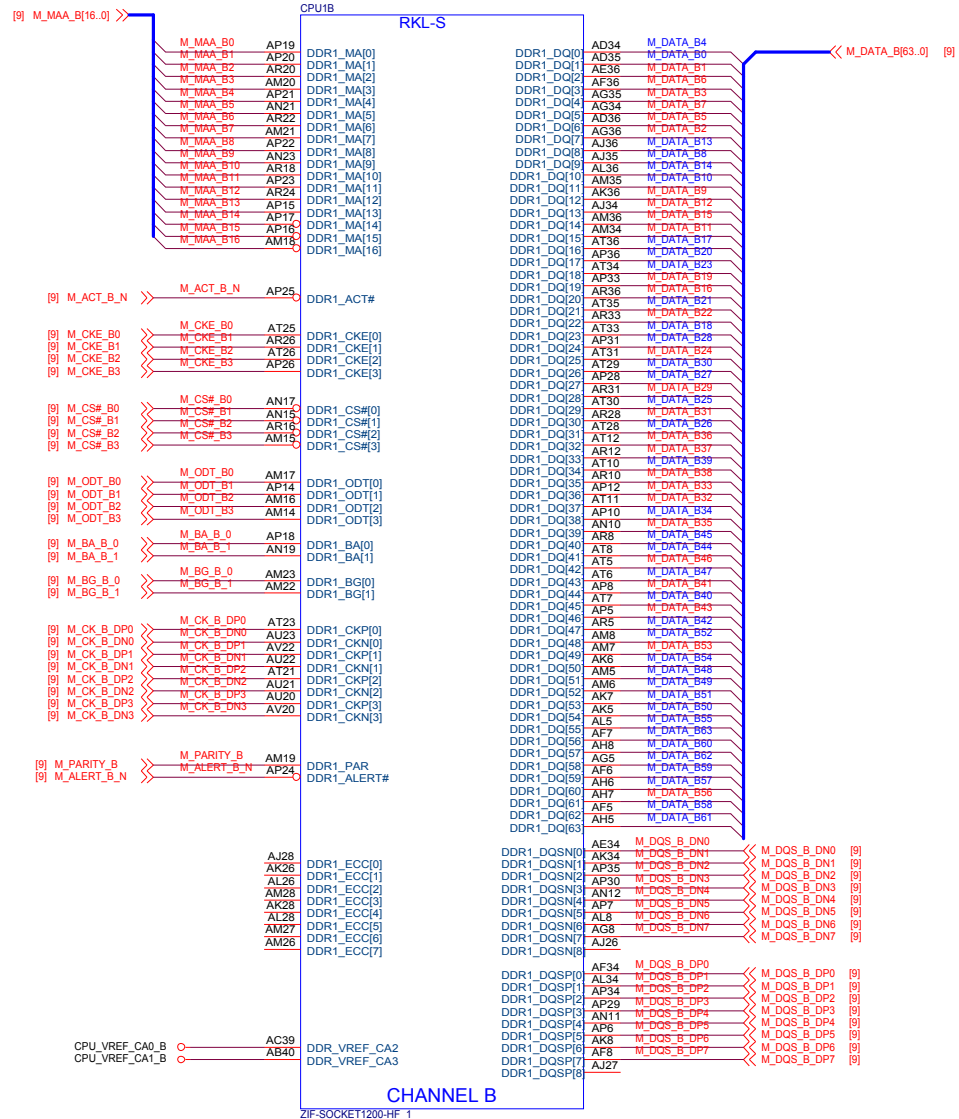
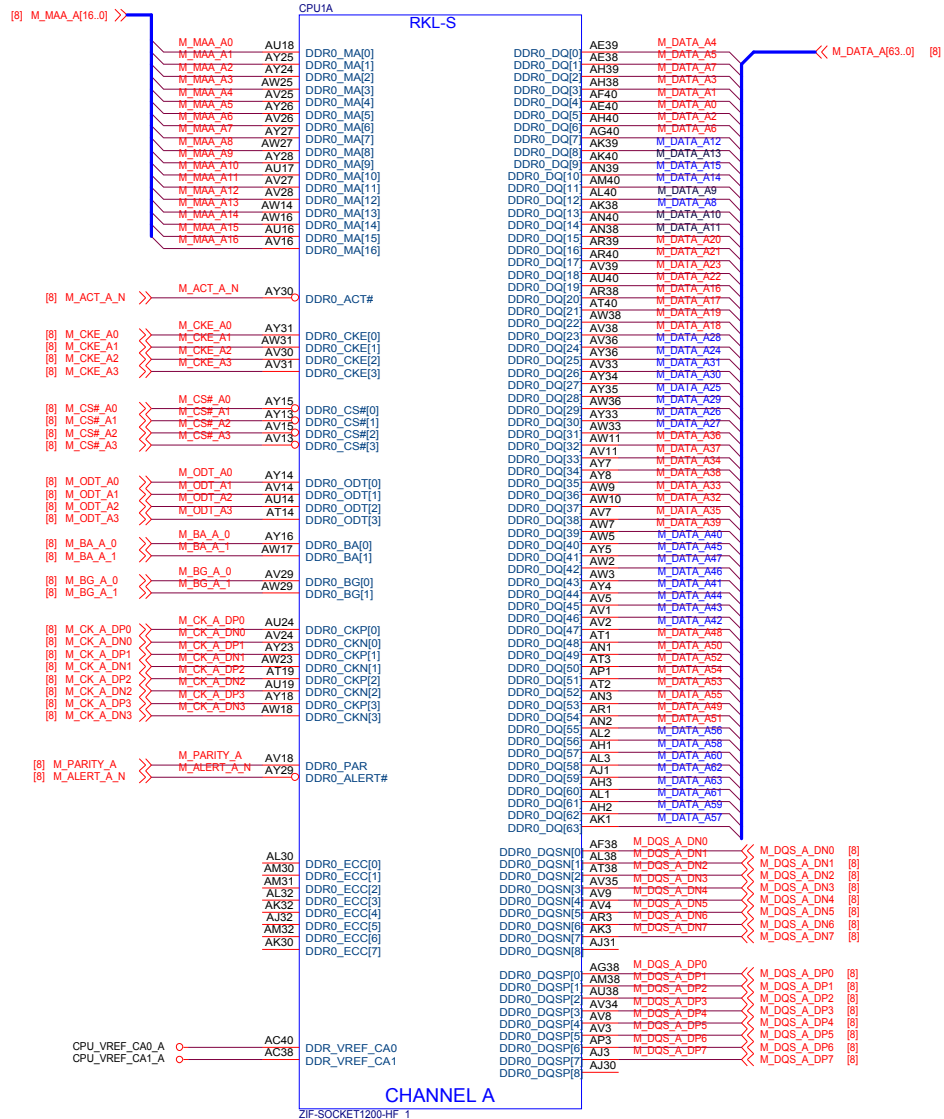


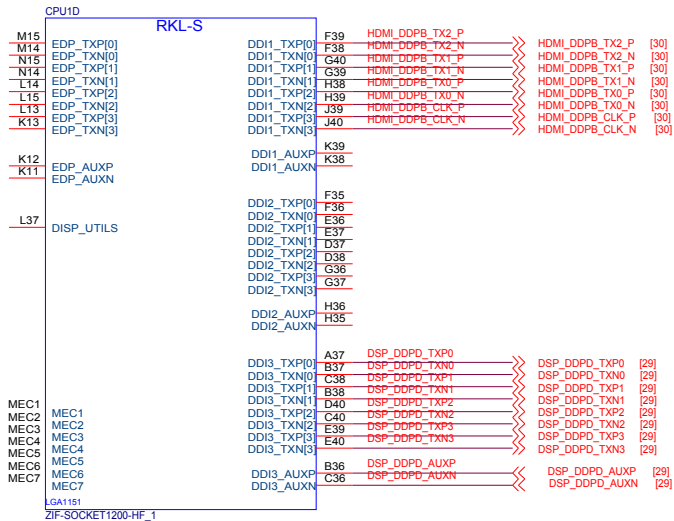
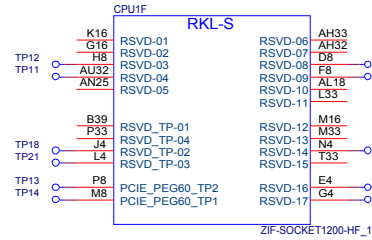
x8 and X4 Ctrl

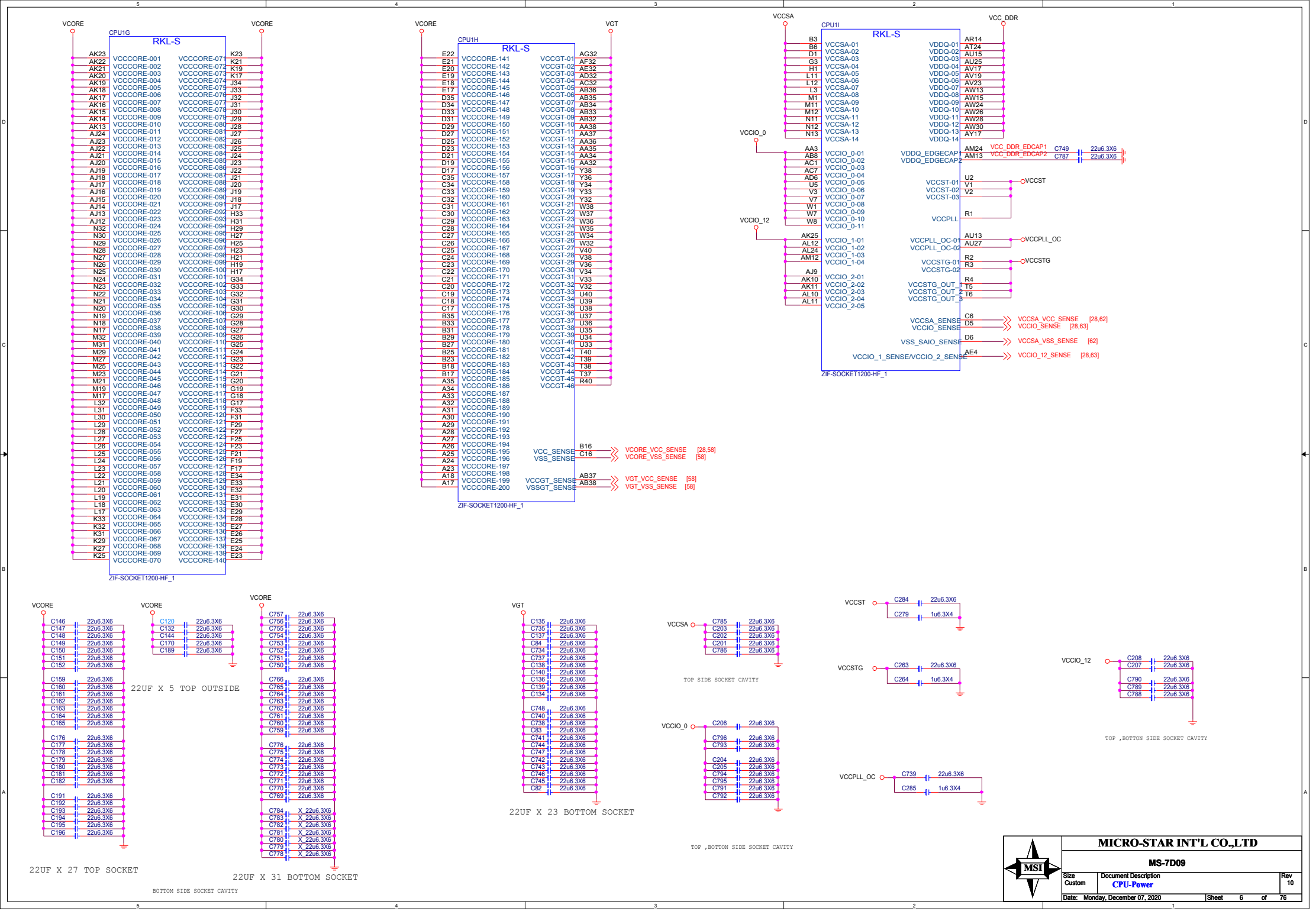
ENABLE#	SLOT1	SLOT4	SLOT6
X8	X4	X8	X4
0	0	X8	X4
0	1	X8	X0
1	0	None	None
1	1	X16	X0

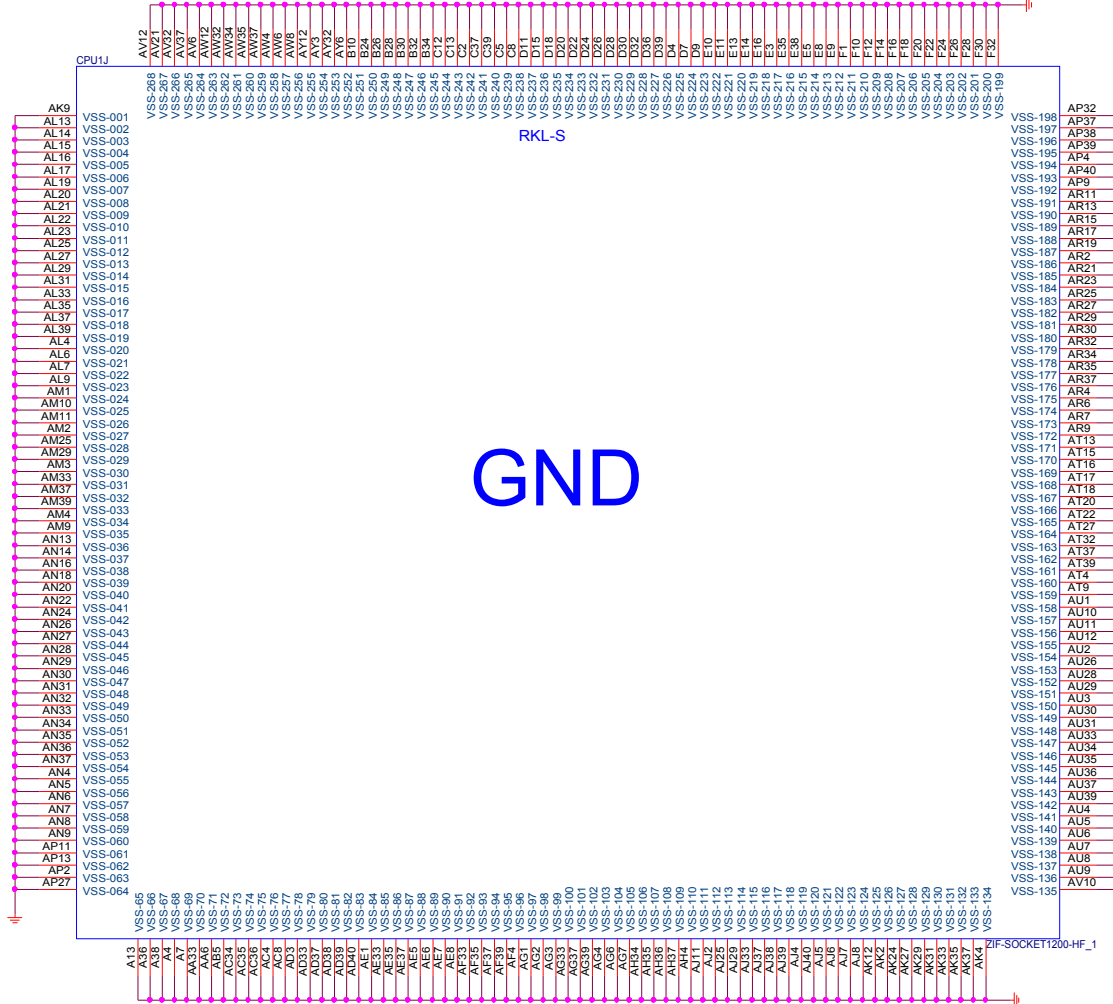
CFG Strap

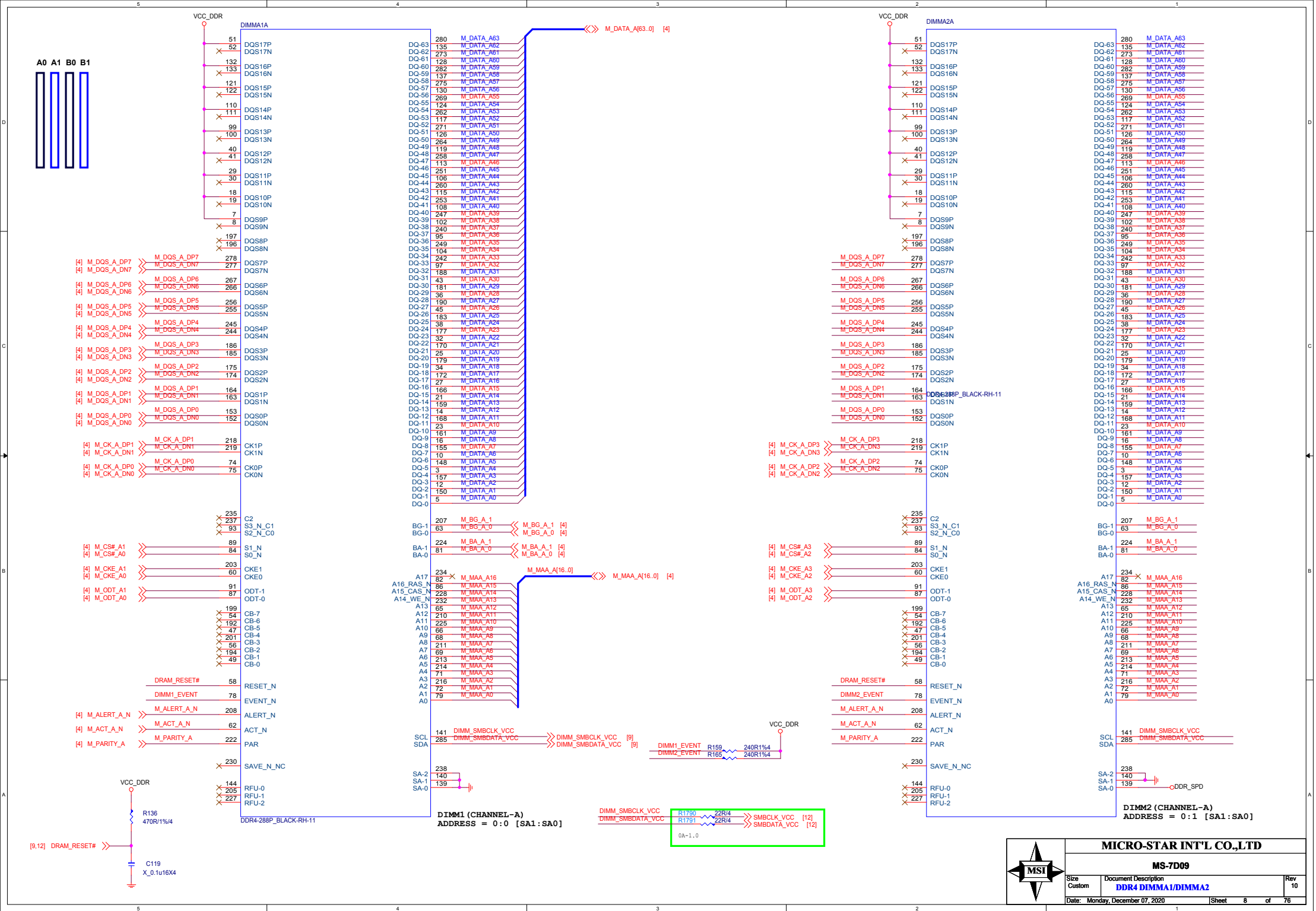
CFG Table			
HIGH	LOW	DESCRIPTION	
0	No Lock	Lock	PCU PLL lock
1			RSVD
2	NORM	REVERSE	PEG LANE REVERSAL
3			RSVD
4	DISABLE	ENABLE	eDP
5	DISABLE	ENABLE	PEG0CFGSEL[0]
6	DISABLE	ENABLE	PEG0CFGSEL[1]
7	RESET#	BIOS REQ	PEG DEFER TRAINING
8			RSVD
9	PRESENT	NO PRESENT	SVID PRESENT
10			RSVD
11			RSVD
12			RSVD
13			RSVD
14	1x4	1x4 Reversed	1x4 Pu down
15			RSVD
16			RSVD
17			RSVD
18			RSVD
19			RSVD



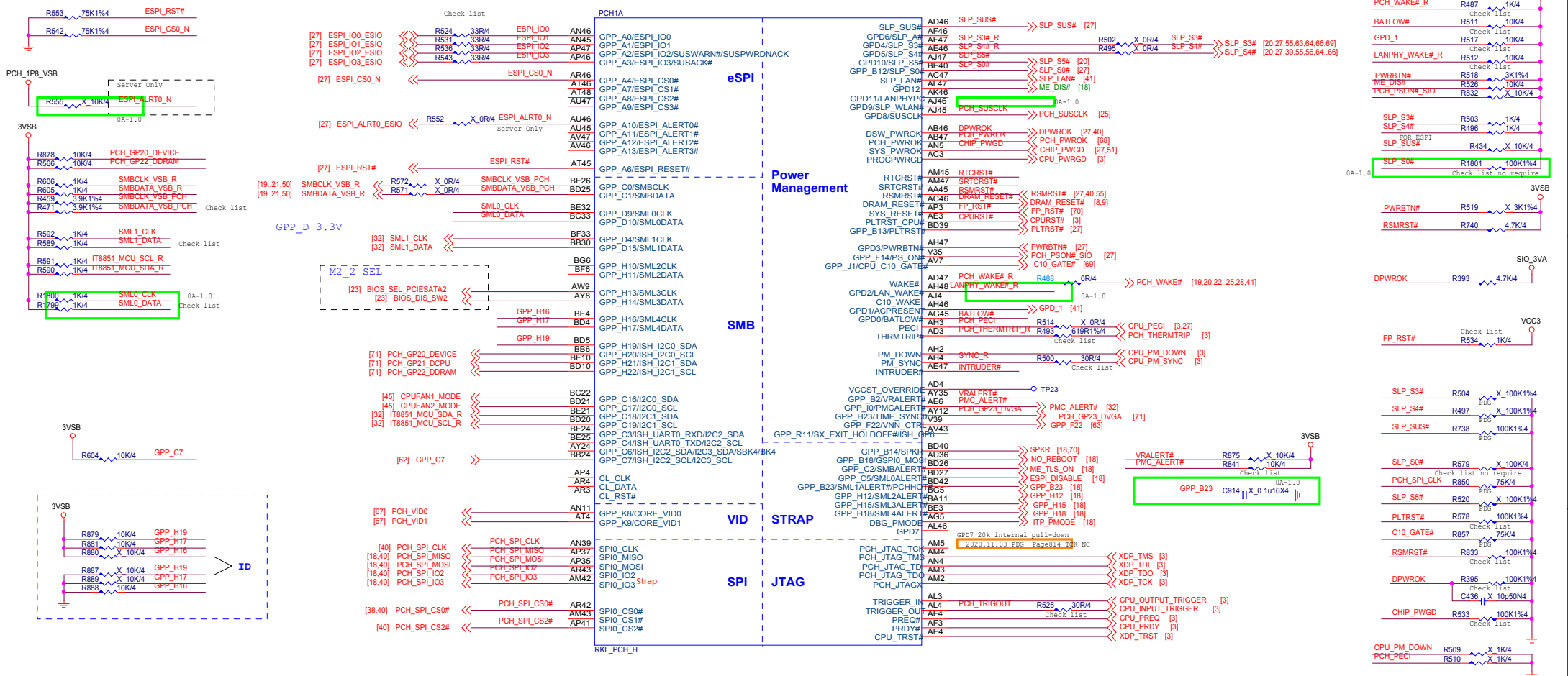




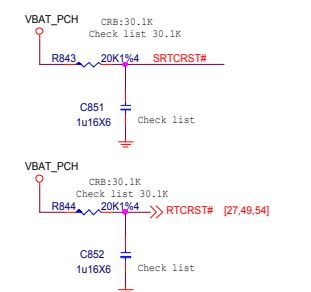




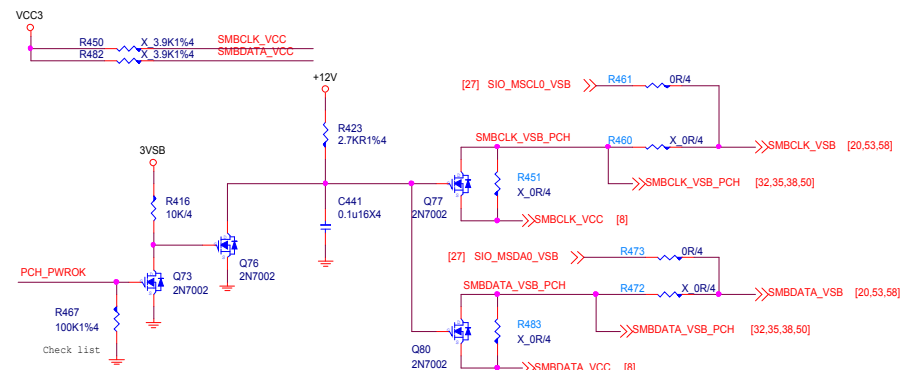
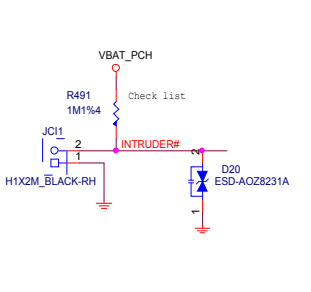




RTC



Chassis Intrusion



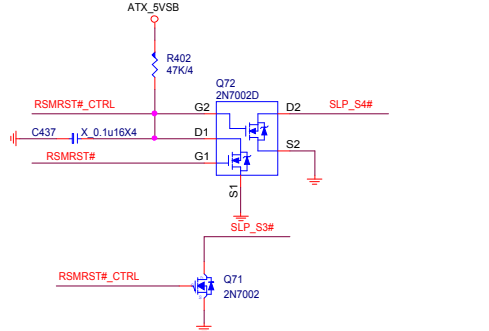


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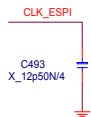
MS-7D09

Size Custom Document Description
PCH-LPC/SPI/SMBUS/MISC

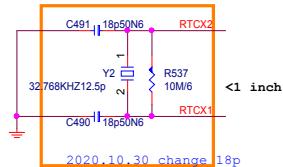
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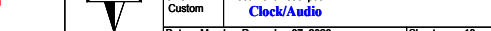
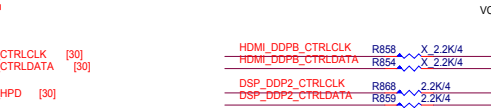
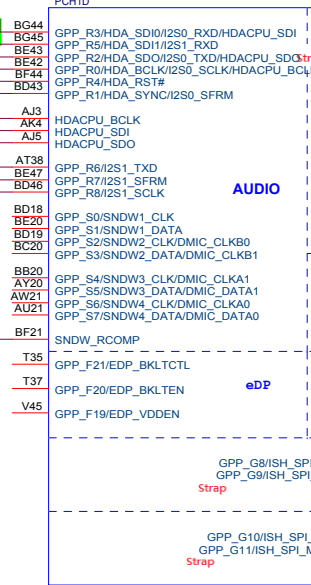
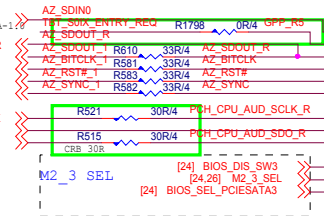
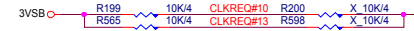
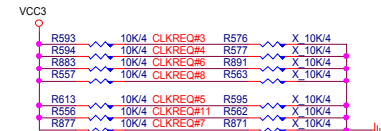
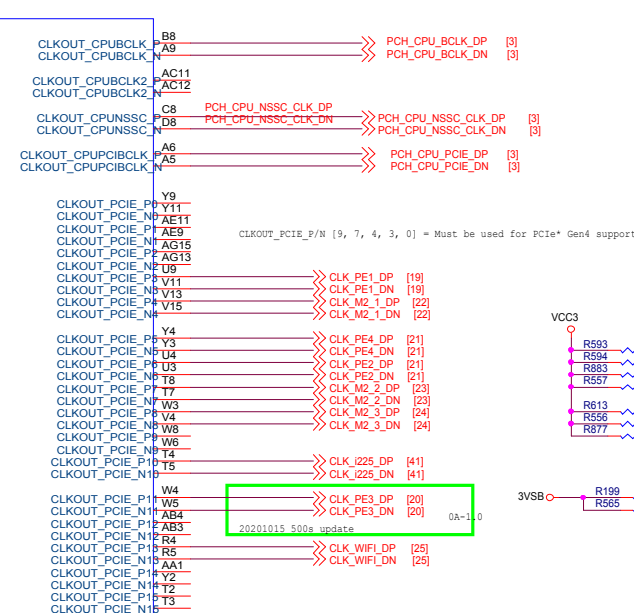
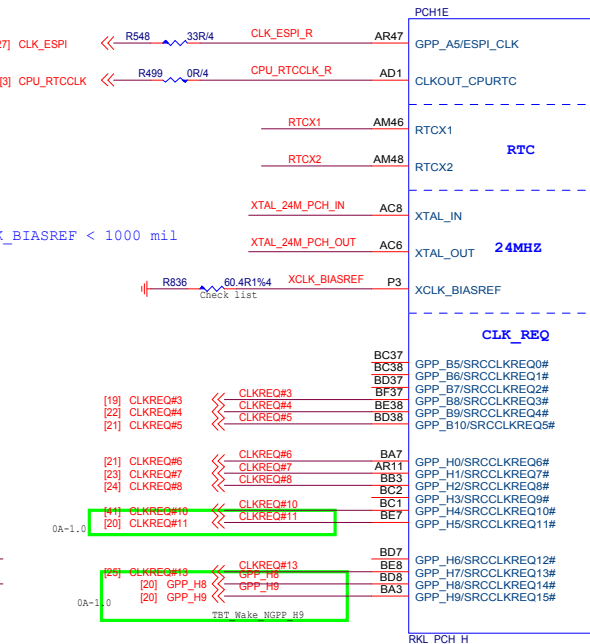
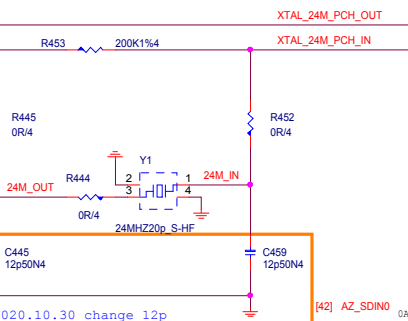
PCH_CLK



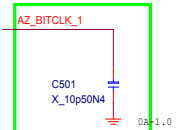
Close to PCH



XCLK_BIASREF < 1000 mil



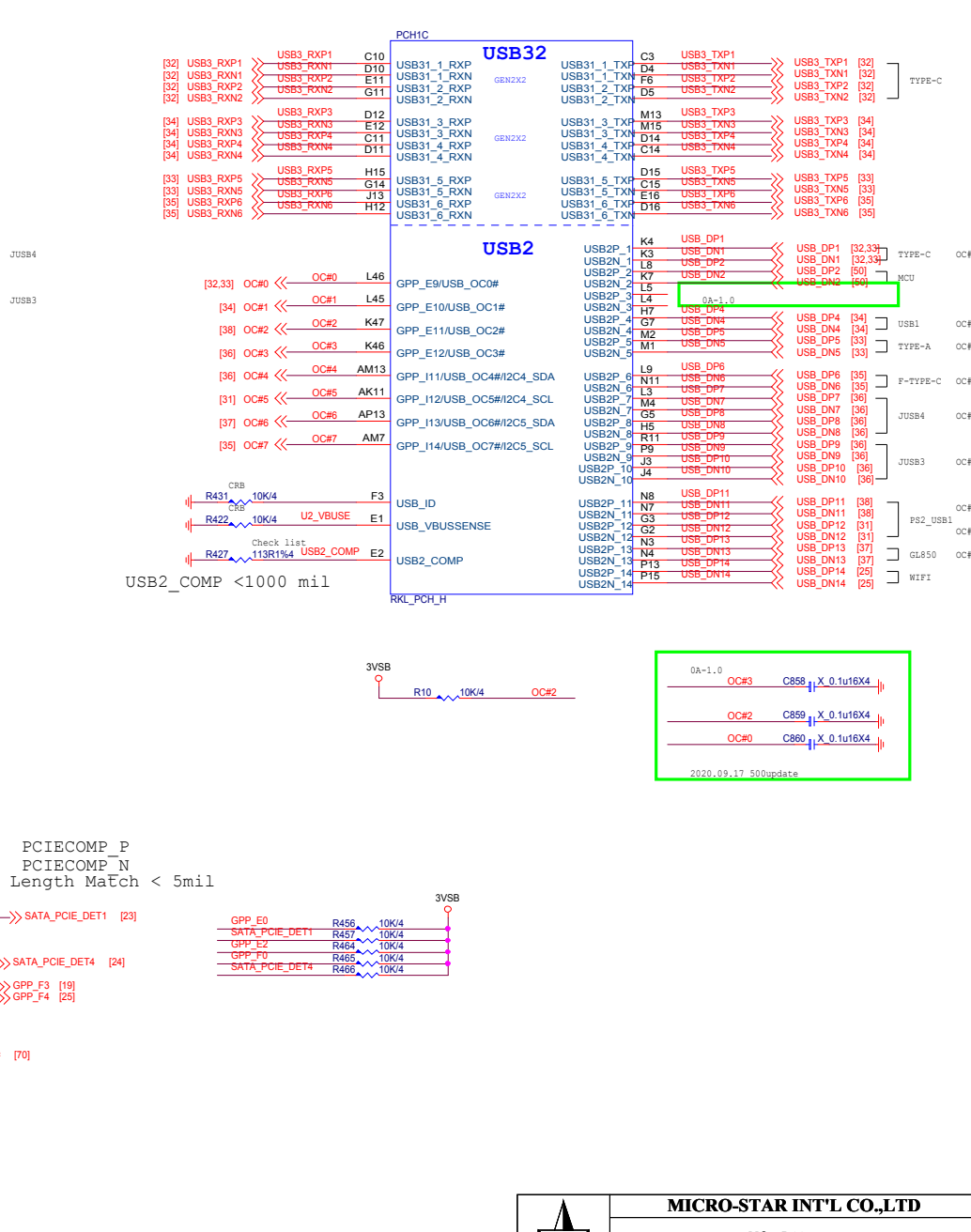
PCH_AUDIO

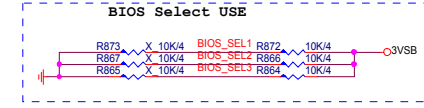
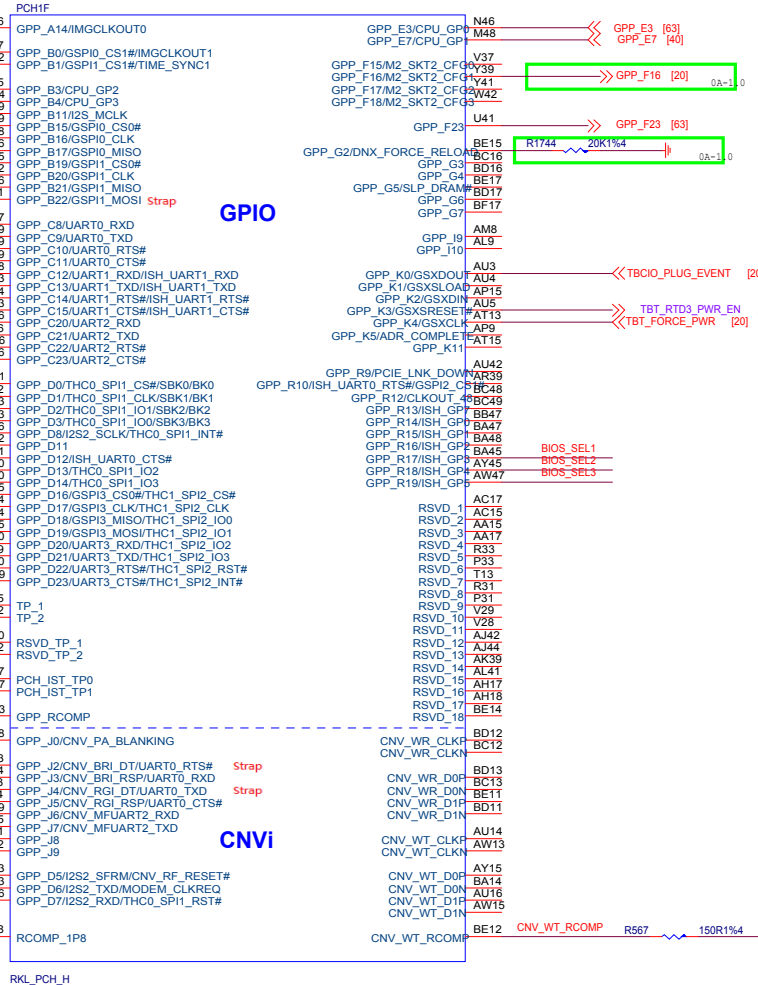
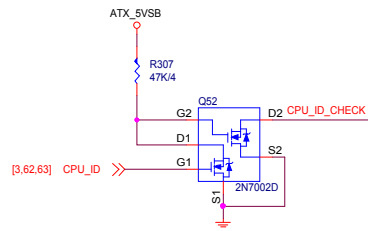
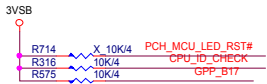


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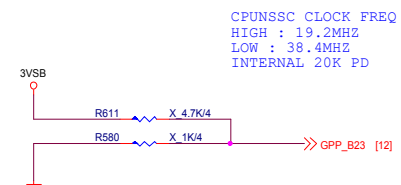
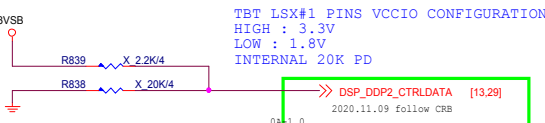
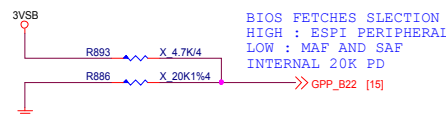
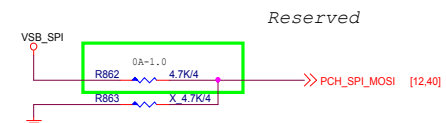
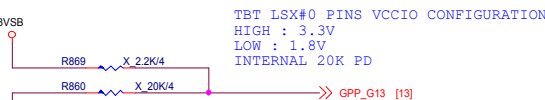
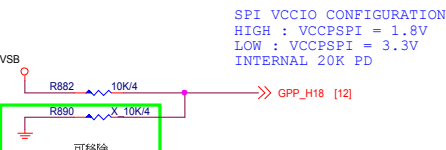
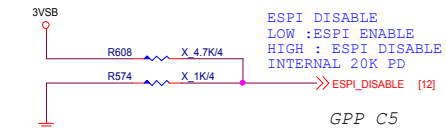
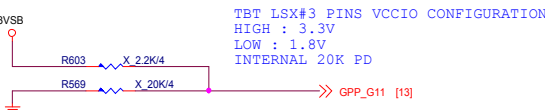
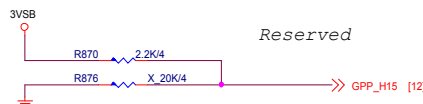
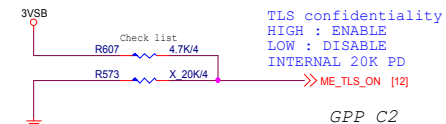
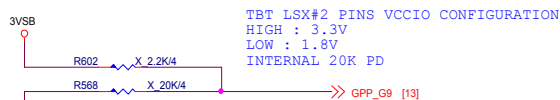
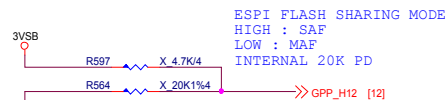
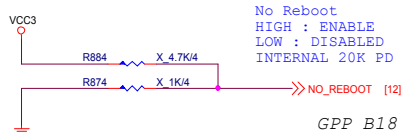
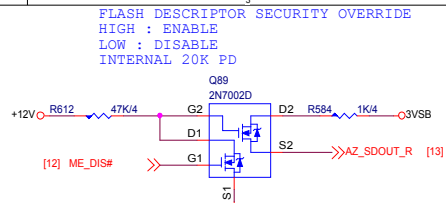
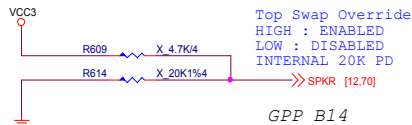
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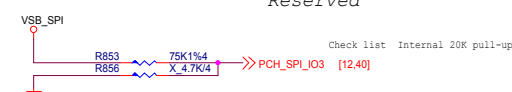
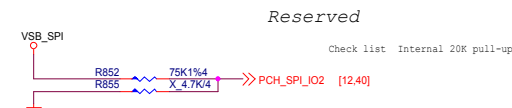
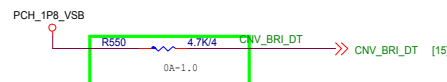
	BIOS_SEL1	BIOS_SEL2	BIOS_SEL3
WIFI	1	1	1
A PRO	0	1	1

VSS



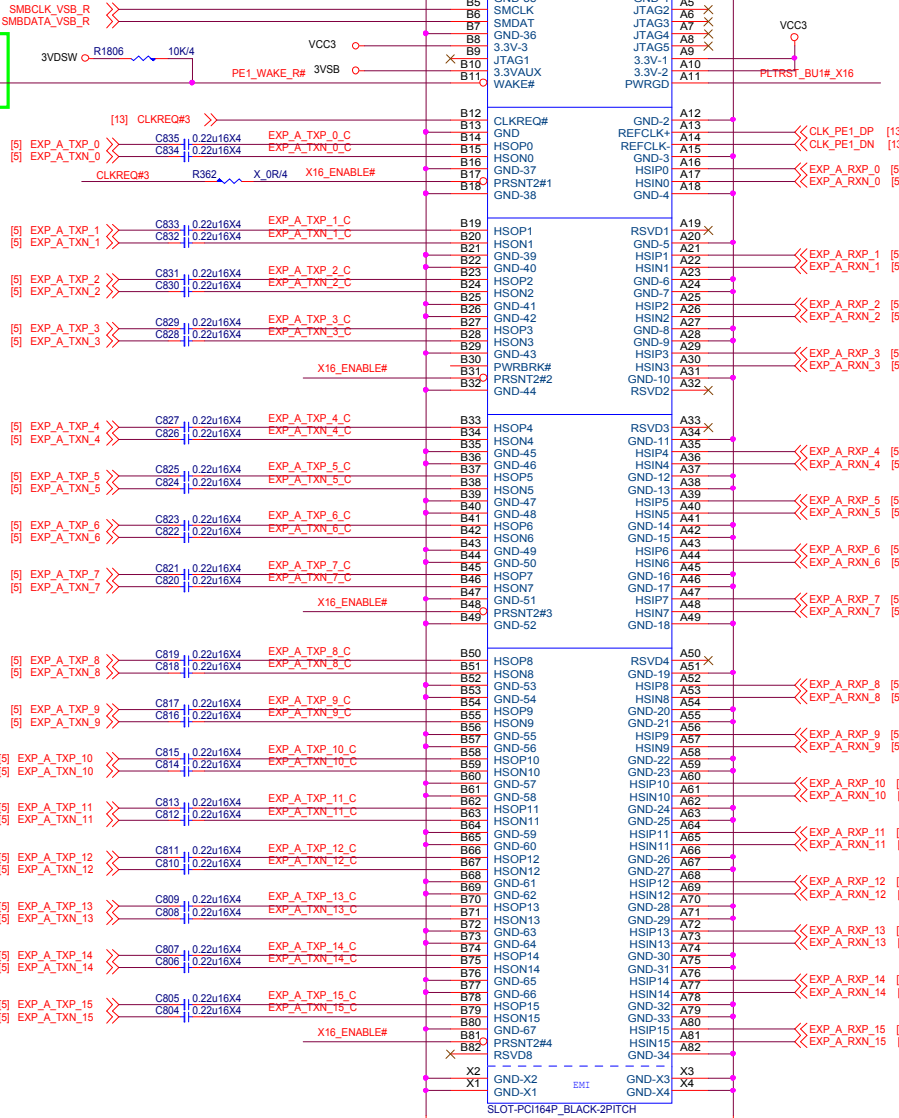
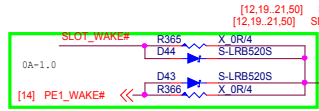
GPP J2 /
CNV_BRI_DT /
UART0_RTS#

XTAL SEL1
1 = 24MHZ
0 = 38.4/19.2MHZ
INTERNAL 20K PD

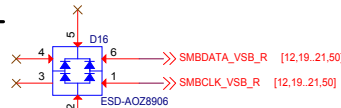


Note: When a RF companion chip is connected to the PCH CNVi interface, the device internal pull-down resistor will pull the strap low to enable CNVi interface.

12V - 5.5A
VCC3 - 3A
3VSB- 375mA

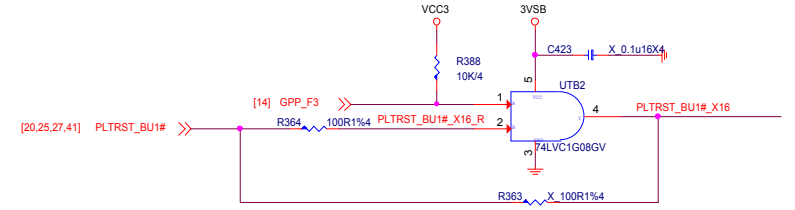


SMBUS ESD

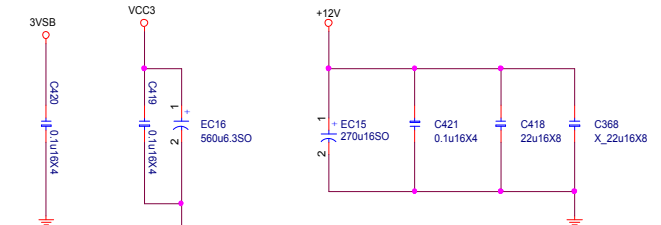
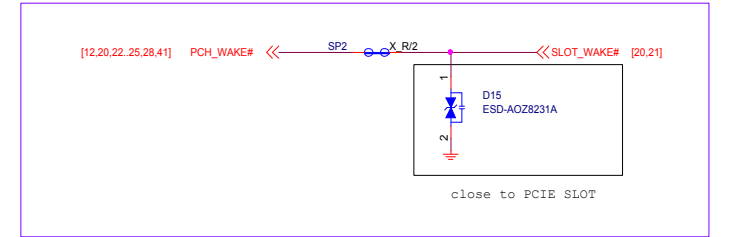


By Placement

Main:D0G-05A0529-A68
AVL:D0G-45B0510-I14

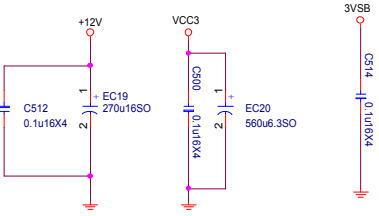
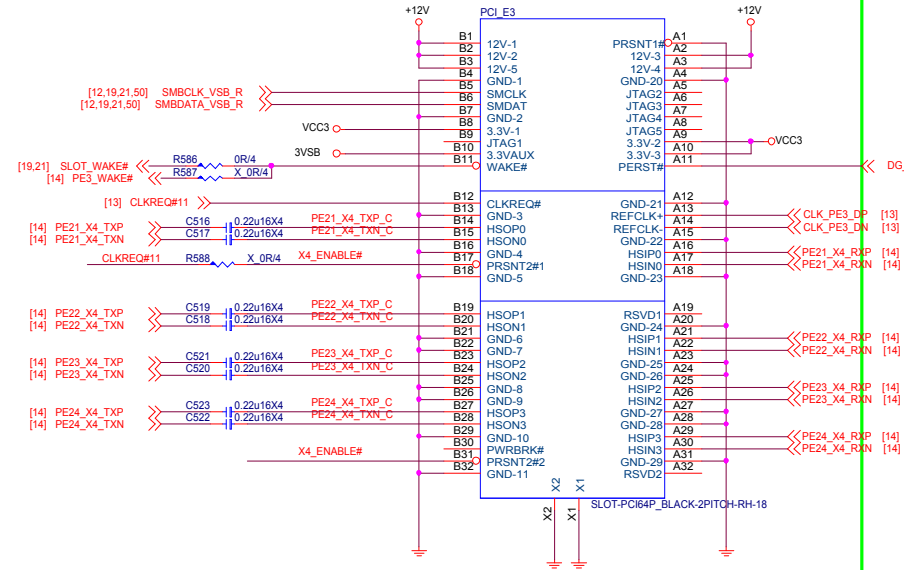


190909:Add TVS for PCIE SLOT

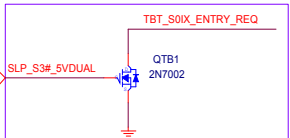
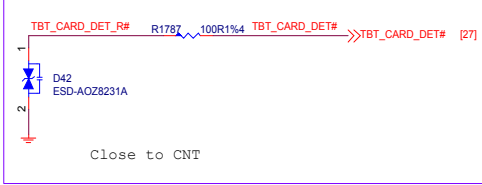
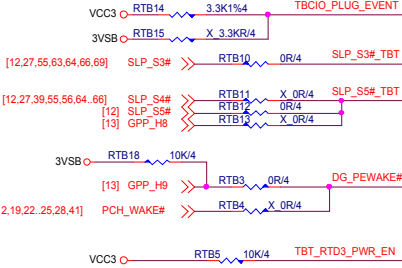
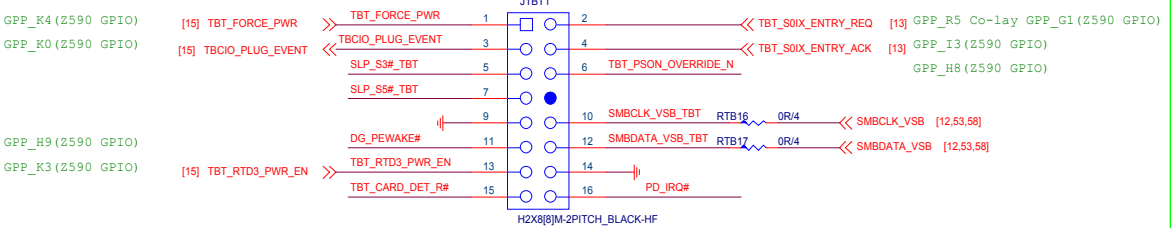


PCI Express X4 Slot

12V - 2.1A
VCC3 - 3A
3VSBV - 375mA



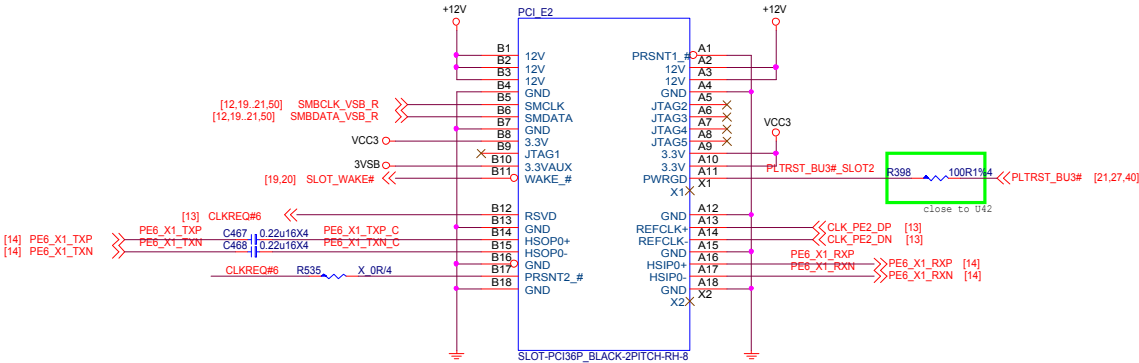
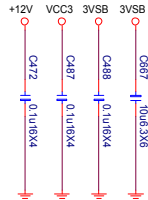
JTBT



0A-1.0

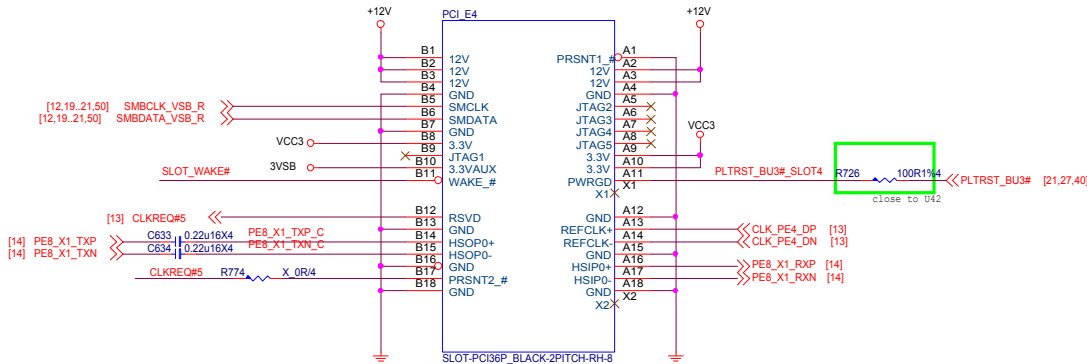
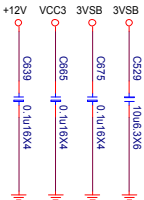
PCH PCIE X1 Slot

12V - 0.5A
VCC3 - 3A
3VSVB - 20mA



PCH PCIE X1 Slot

12V - 0.5A
VCC3 - 3A
3VSVB - 20mA

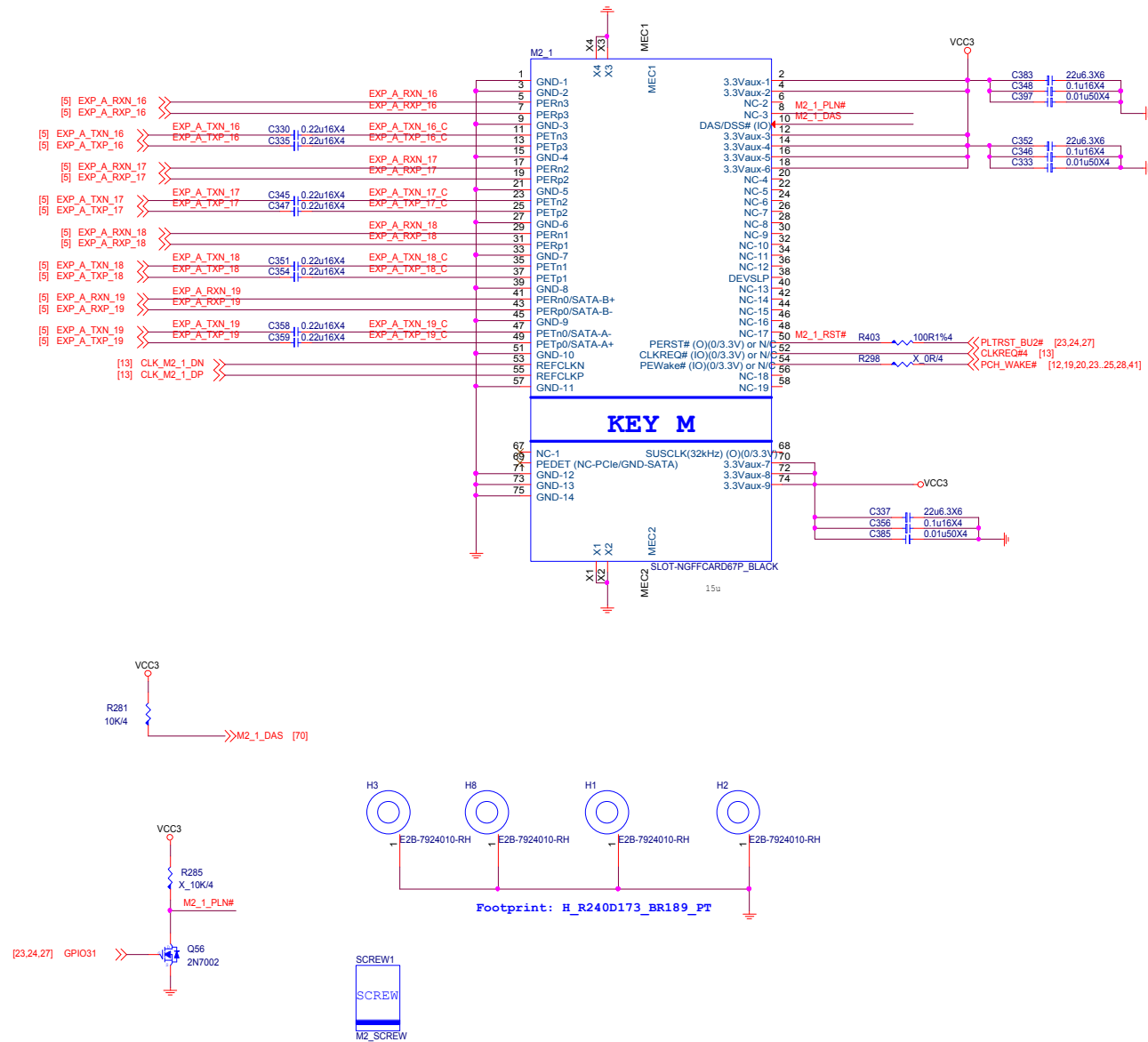


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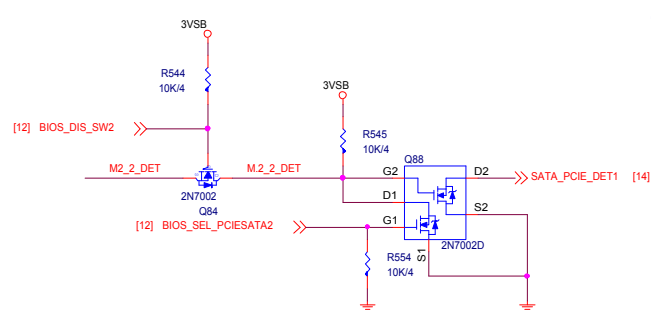
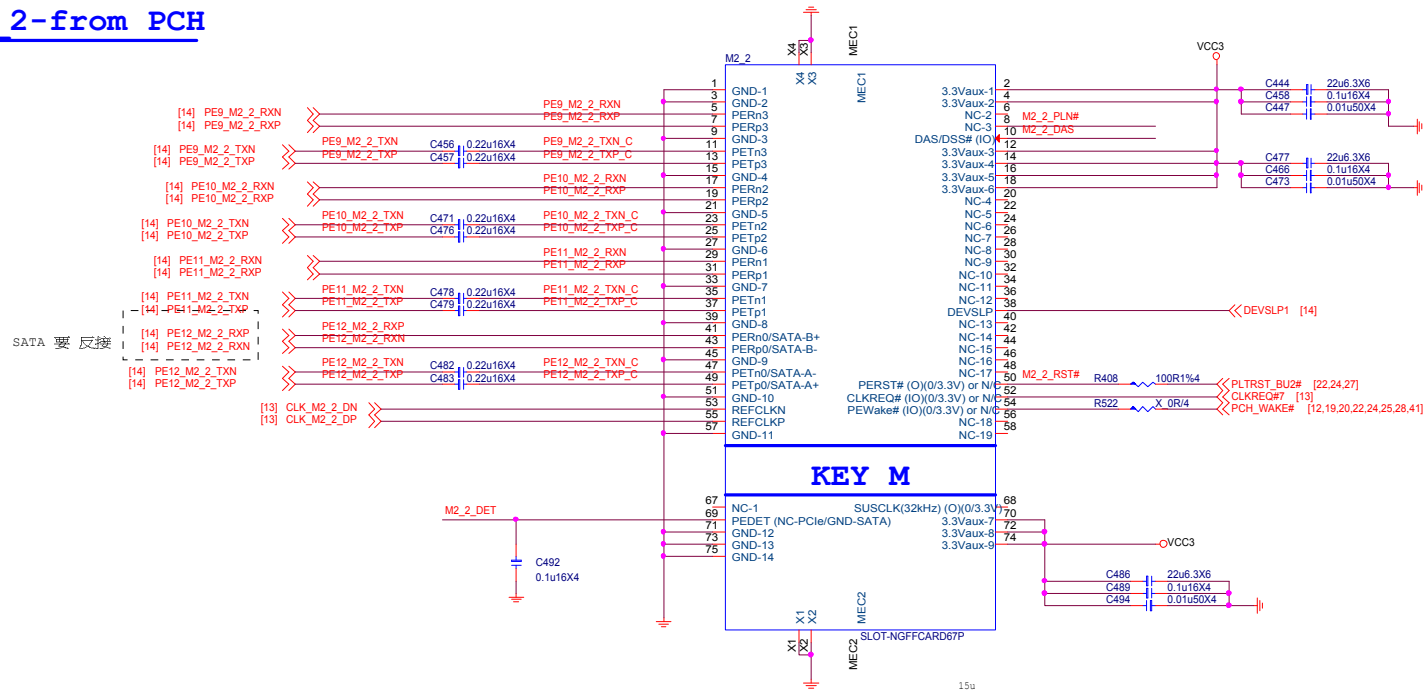
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Size	Document Description	Rev
Custom	PCIE SLOT-PCH (X1)	10
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M2 1-from CPU

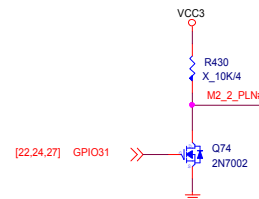
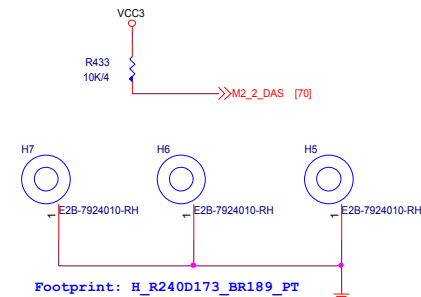


M2 2-from PCH

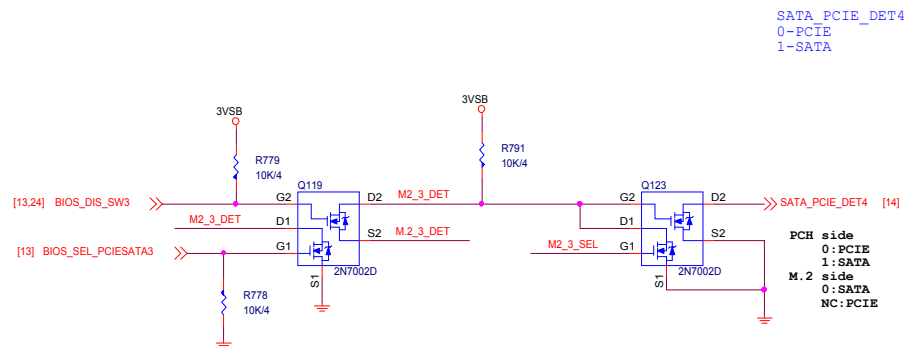
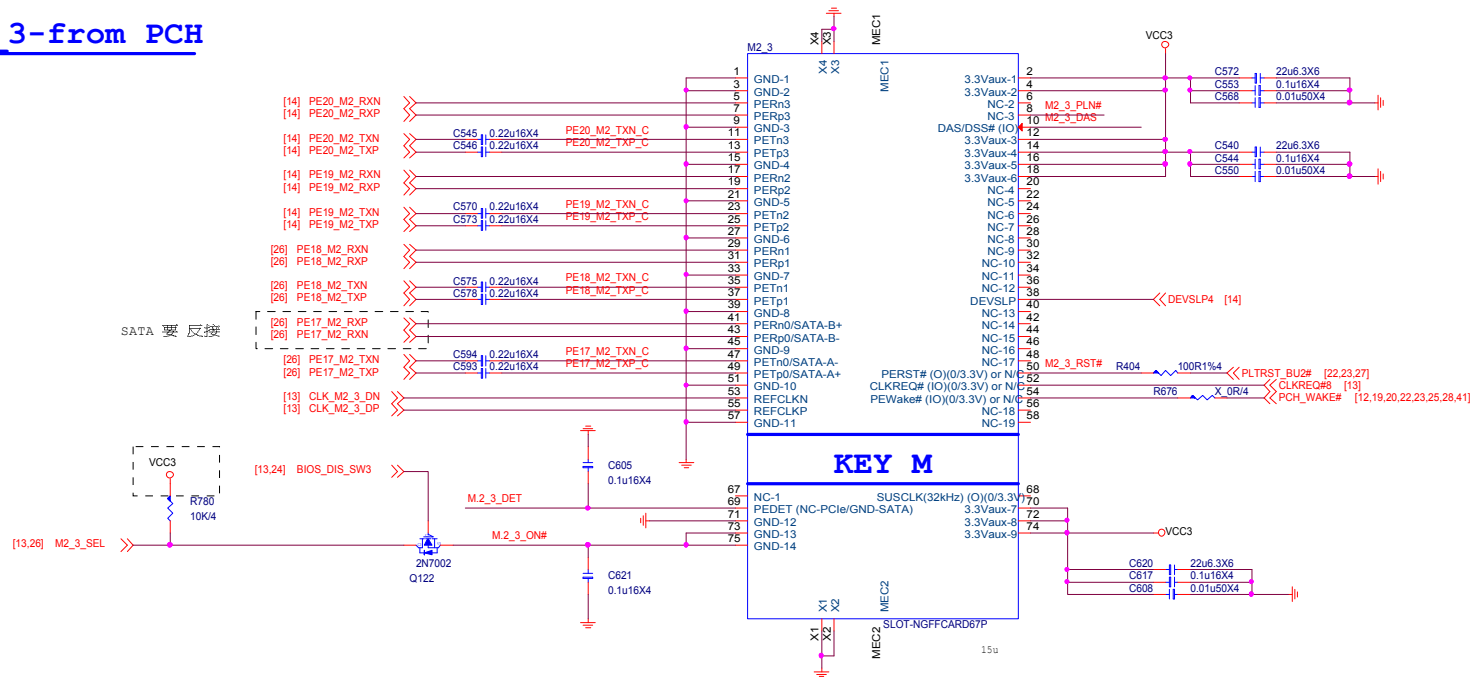


BIOS MODE

BIOS DIS SW2 GPP_H14	BIOS_SEL PCIESATA2 GPP_H13	Mode
GPO (0)	GPO (1)	M2-SATA
GPO (0)	GPO (0)	M2-PCIE
GPI	GPI	AUTO

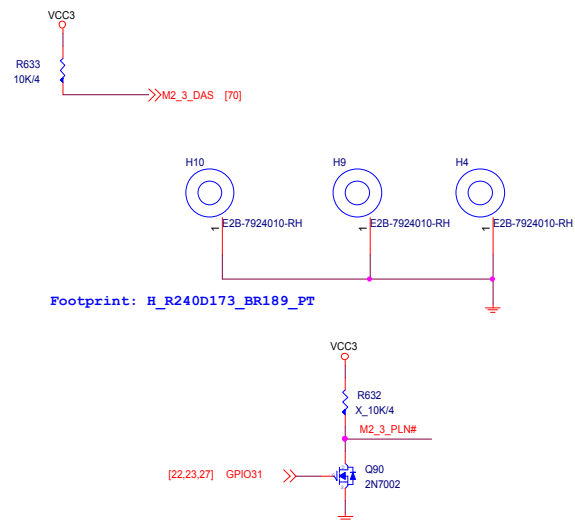


M2 3-from PCH



BIOS_MODE

GPP_R6	GPP_R7	GPP_R8	
BIOS_DIS_SW	M2_3_SEL	BIOS_SEL_PCIESATA3	Mode
GPI(1)	GPI(1)	GPI(0)	AUTO
0	1	0	SATA5
0	0	1	M2-SATA
0	0	0	M2-PCIE



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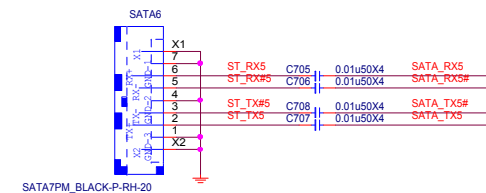
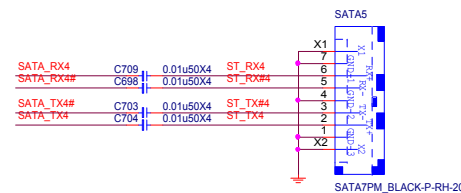
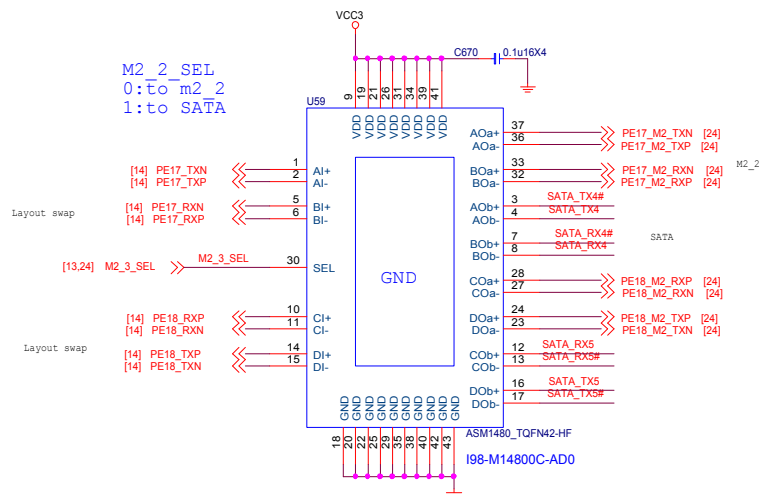
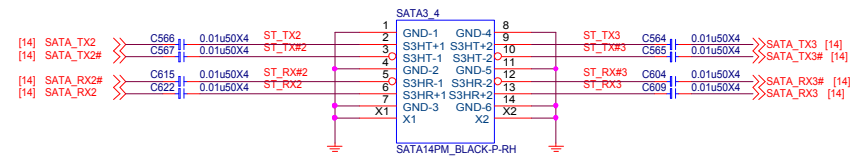
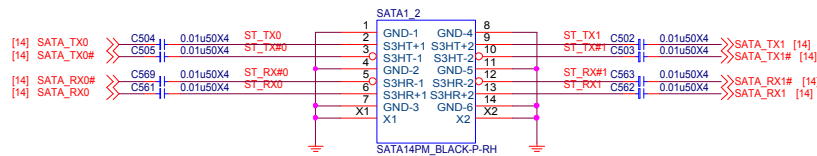
Size	Document Description
Custom	M.2_3-SLOT3

Date: Monday, December 07, 2020 Sheet 24 of 76

Rev
10

SATA 6G PORT 0.1

SATA 6G PORT 2.3



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Size
Custom

Document Description

SATA Connector

Rev
10

Date: Monday, December 07, 2020

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SIO HM Voltage Over 2V will Not Detect

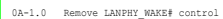
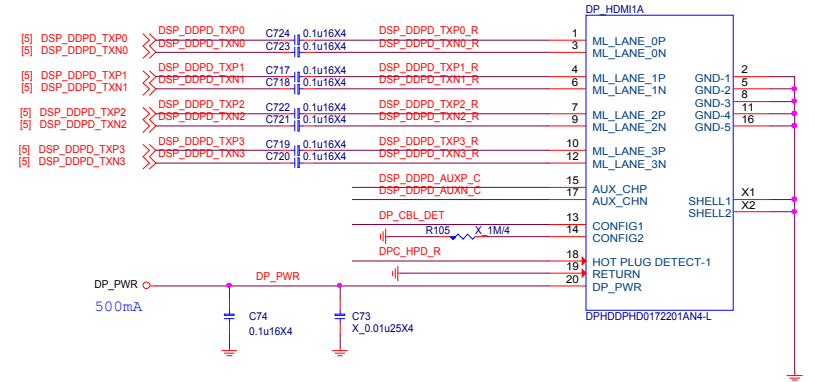
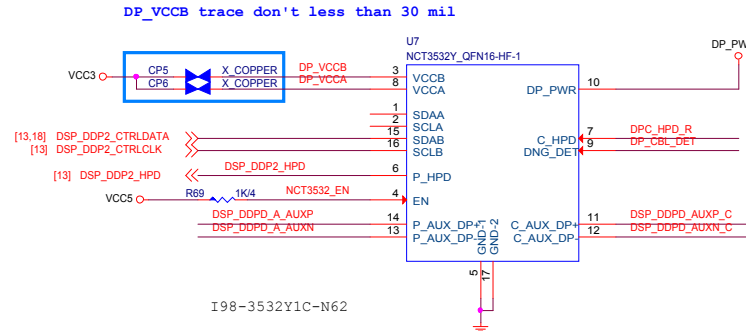
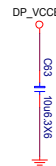
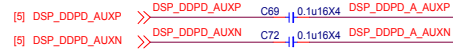
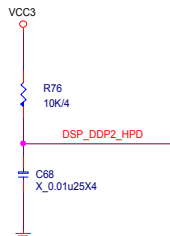
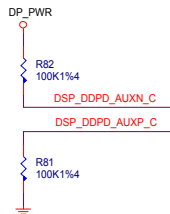
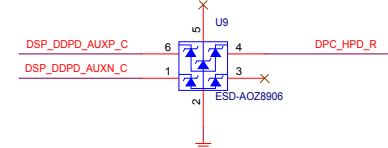
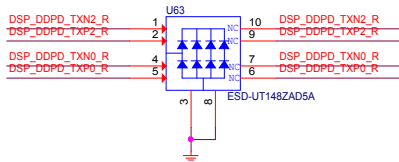
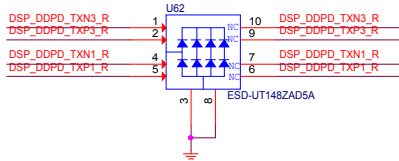
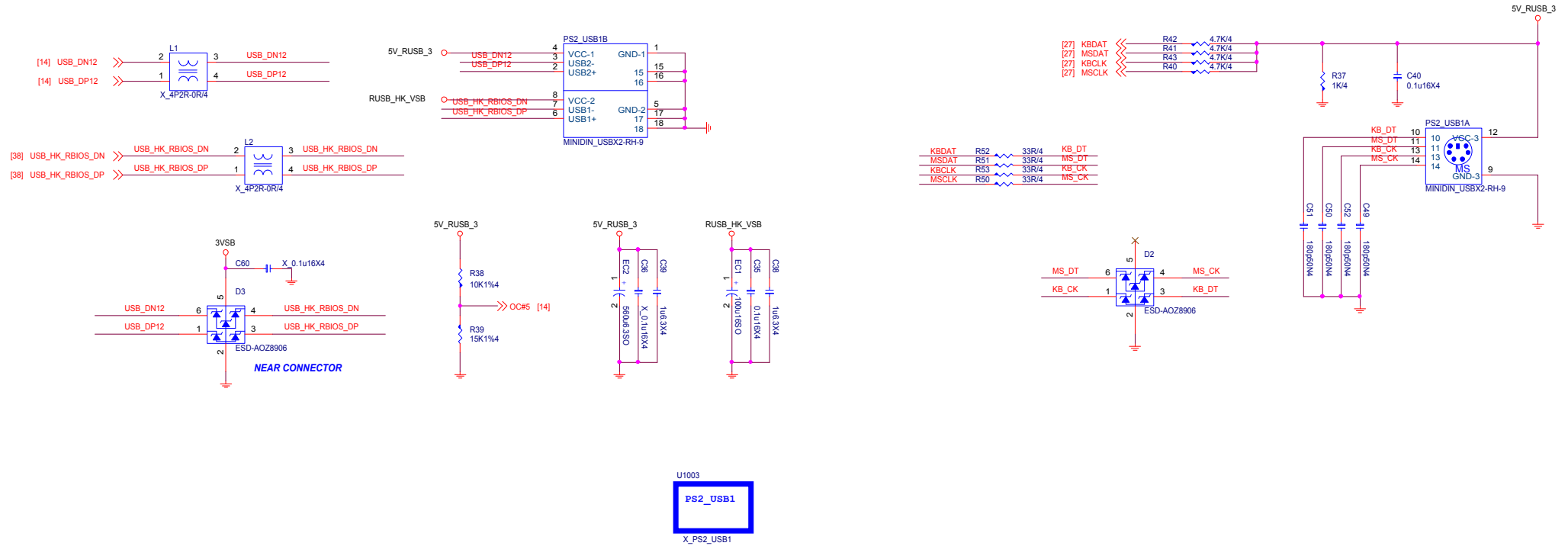


Figure 10: Recommended component values for the P-LM233-RH. The figure contains three circuit diagrams. The first diagram shows the HM_VREF pin connected to a 10k resistor (R485) and a 1k resistor (R492) to ground, with a 10k resistor (RT4) and a 0.1uF capacitor (C474) connected to the CPUTIN pin. The second diagram shows the HM_VREF pin connected to a 10k resistor (R468) and a 1k resistor (RT6) to the CPU socket, with a 0.1uF capacitor (C469) connected to the SKT_TIN pin. The third diagram shows the SKT_TIN pin connected to a 2200pF capacitor (C475) to ground, with a 0.1uF capacitor (C475) connected to the SKT_TIN pin. The third diagram also shows the SKT_TIN pin connected to a 2200pF capacitor (C475) to ground, with a 0.1uF capacitor (C475) connected to the SKT_TIN pin.



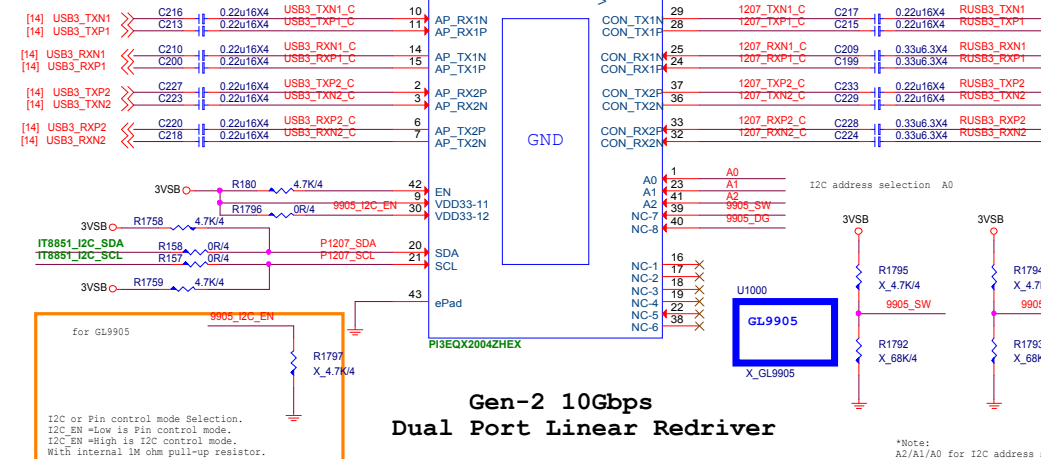
$$0.5A + 0.5A$$

PS2



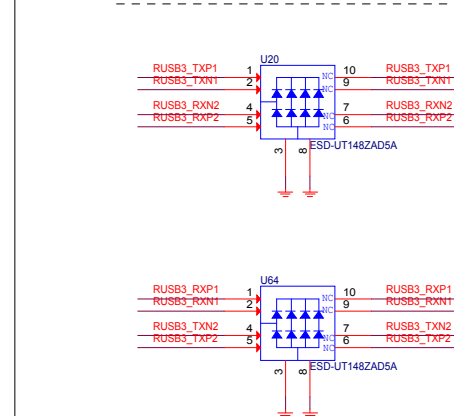
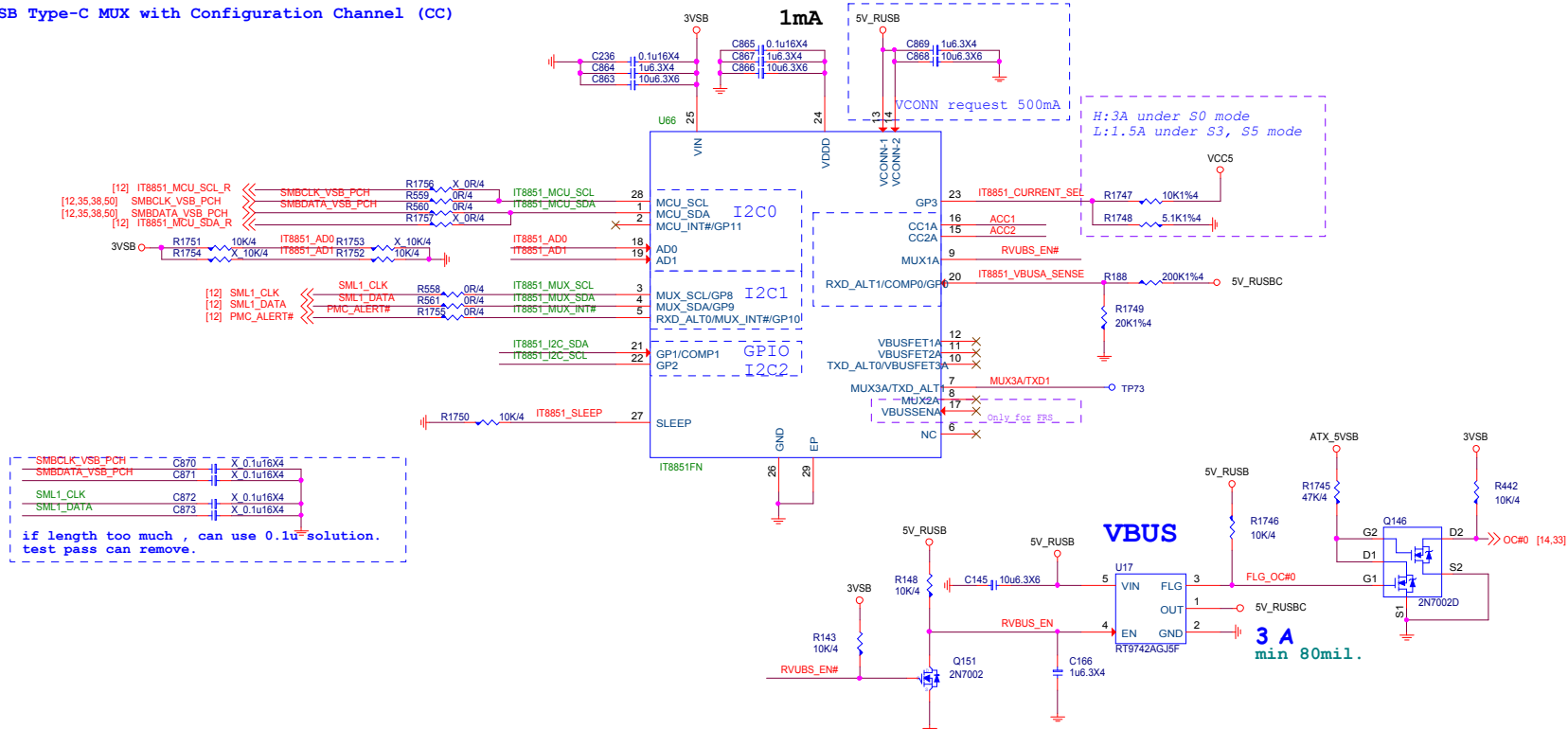
Rear USB3.1 Redriver

5 inch < Host to IC < 10 inch



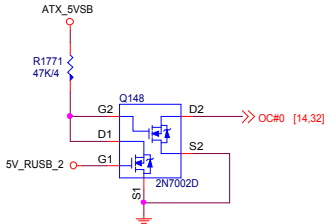
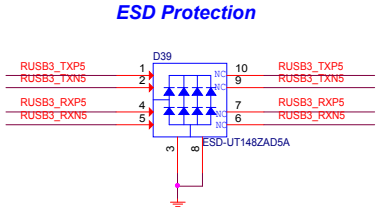
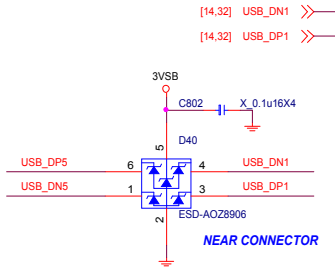
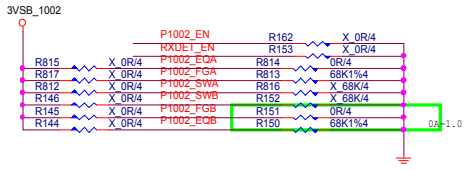
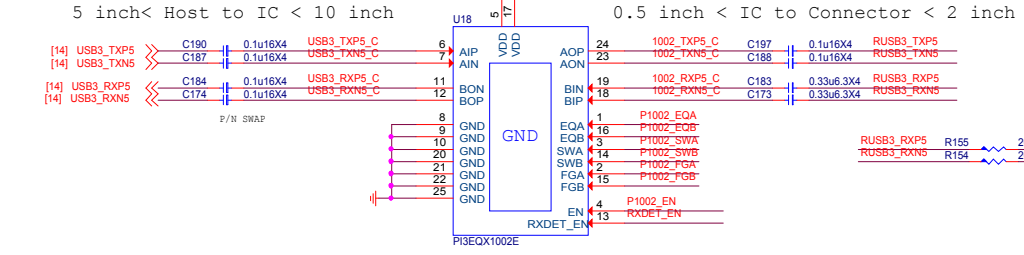
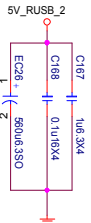
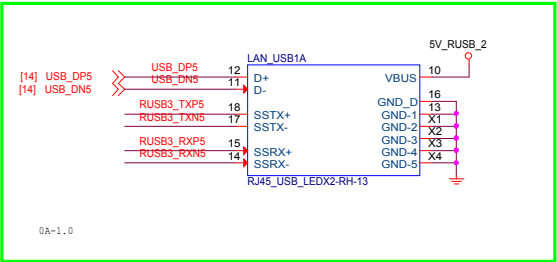
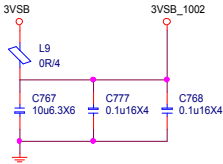
*Note:
A2/A1/A0 for I2C address selection

USB Type-C MUX with Configuration Channel (CC)



REAR USB3.0 GEN2 Type-A Re-Driver

167mA



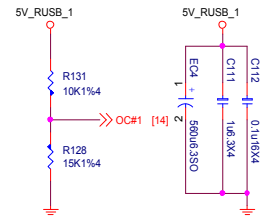
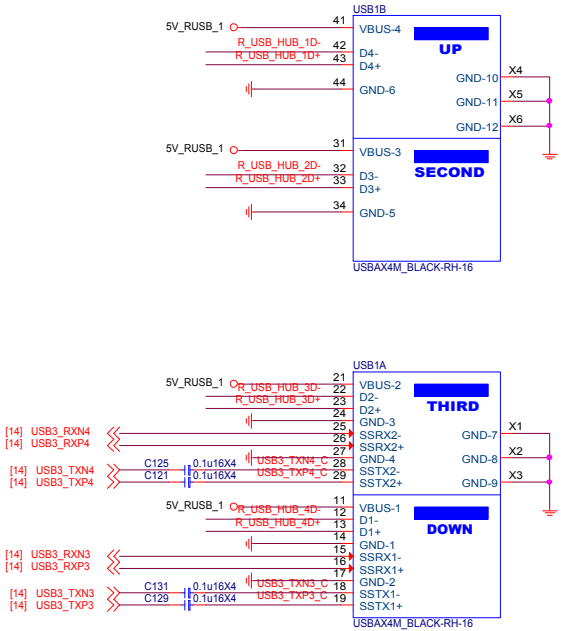
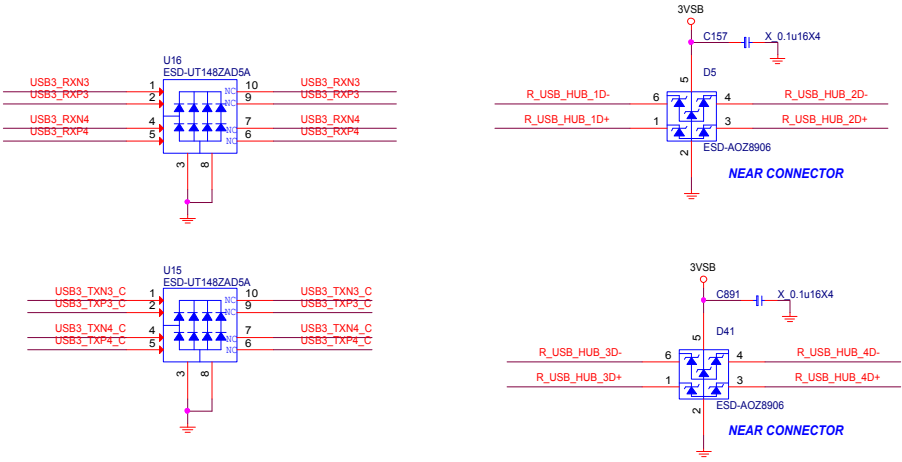
Receiver Detection Setting

RXDET_EN	Setting
0	Disabled
1	Enabled (Default)

Channel Enable Setting

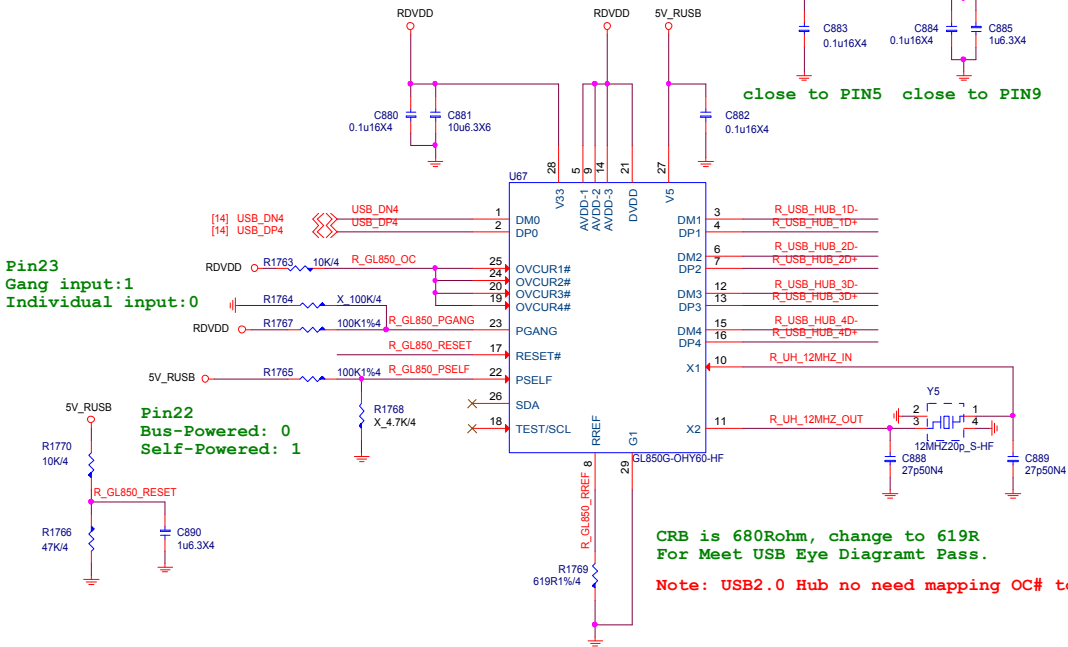
EN	Setting
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1	Enabled (Default)

Rear LAN1 USB3.1 GEN1 1.8A



Rear USB2.0 HUB

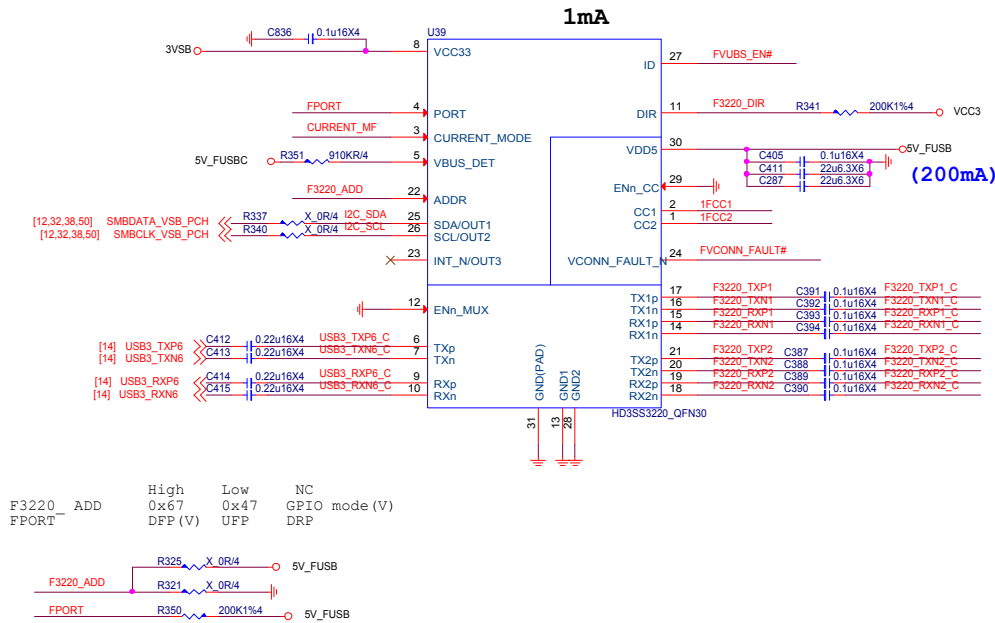
Note: Please connect to USB Power Source.



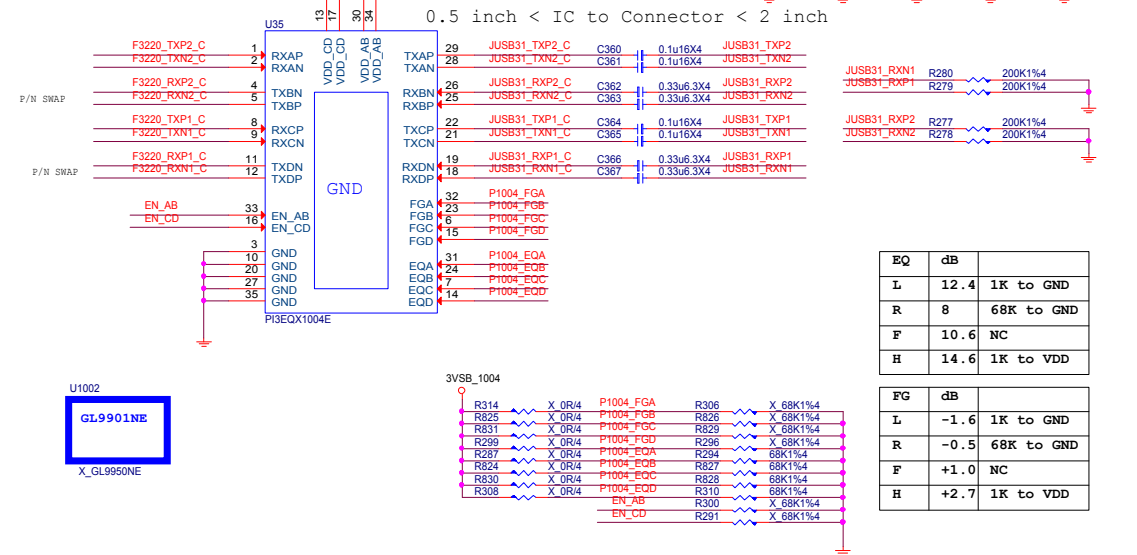
USB 3.1-Type-C

5 inch< Host to IC < 10 inch

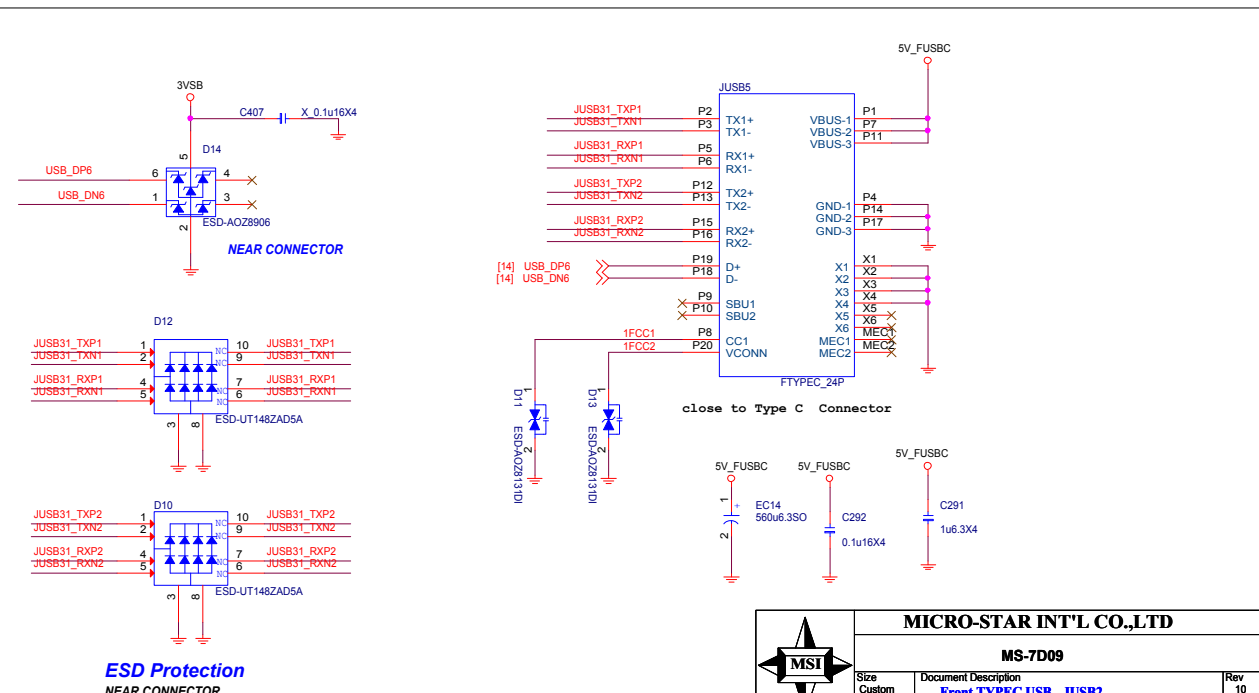
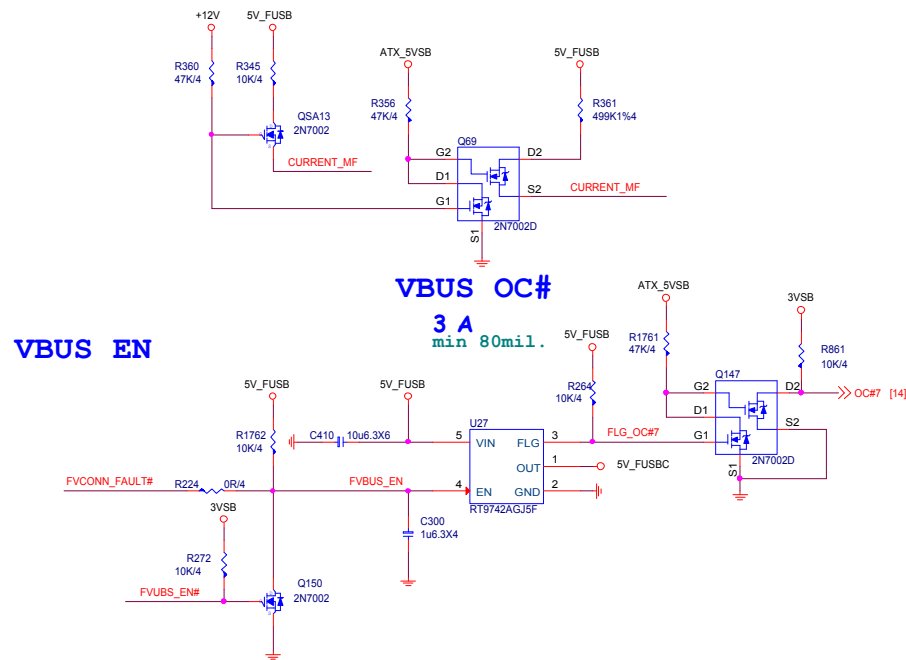
Host to IC < 6inch

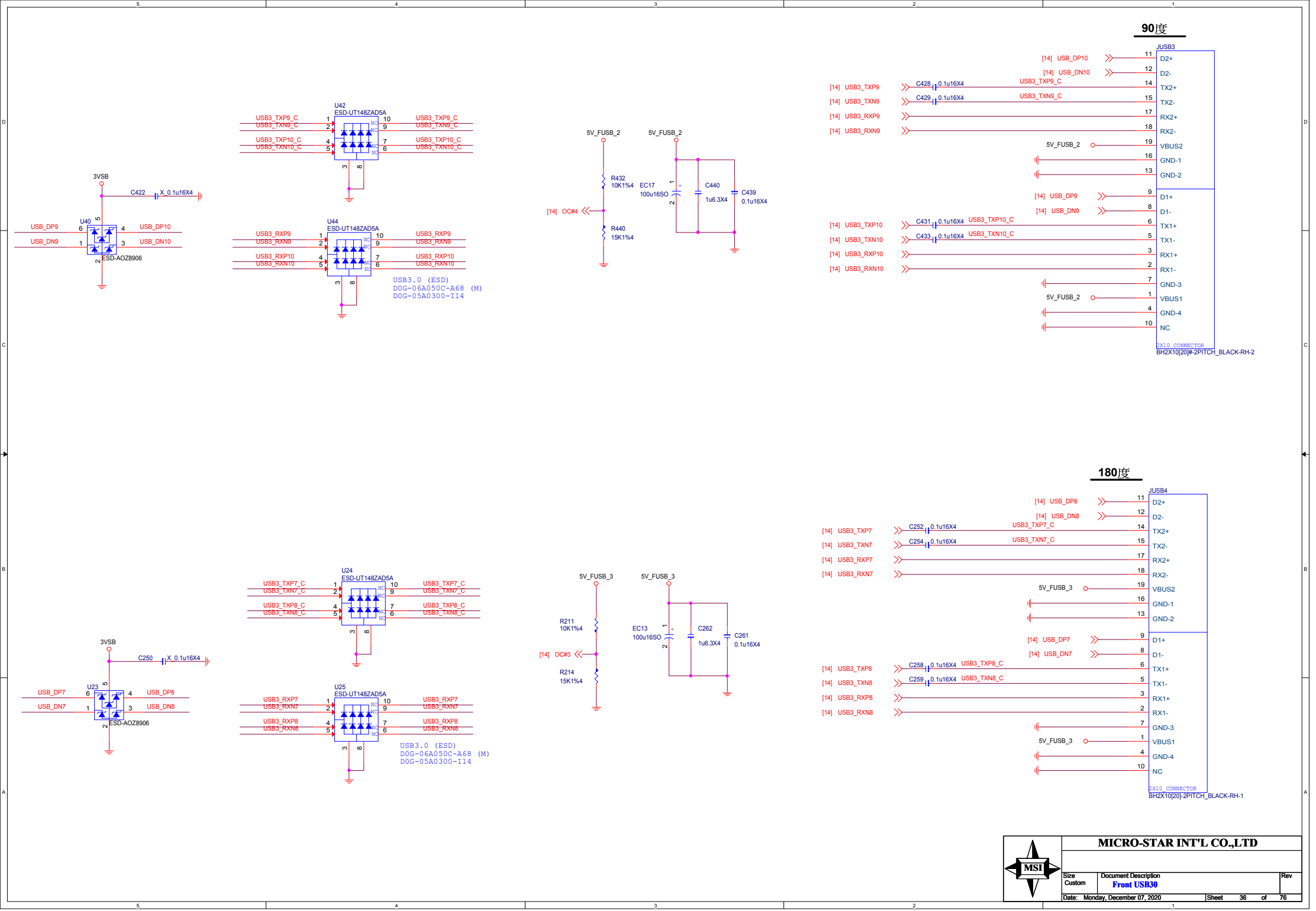


(334mA)



Current Mode





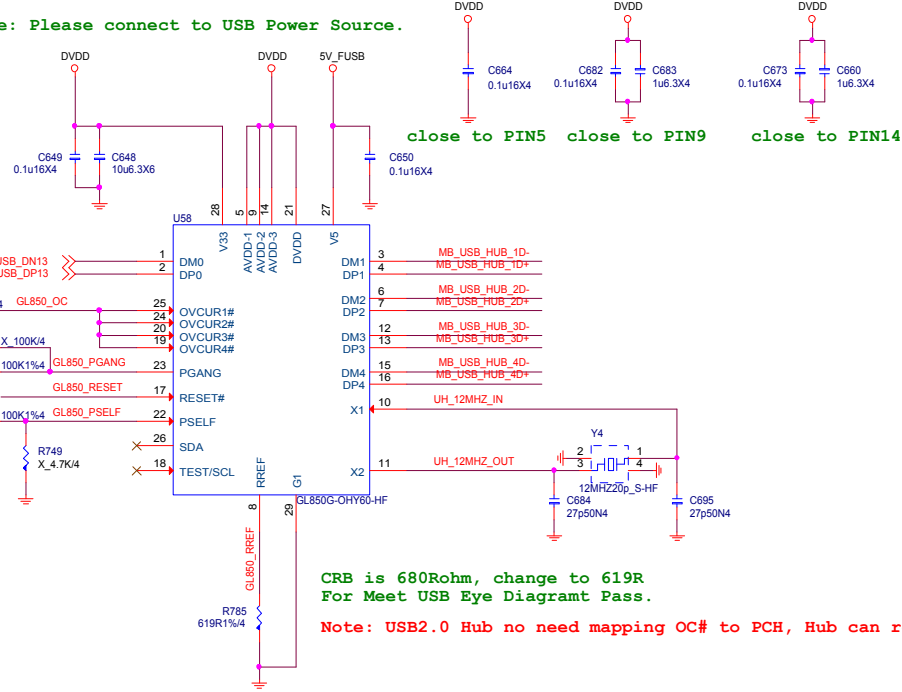
Front USB2.0 HUB

Note: Please connect to USB Power Source.

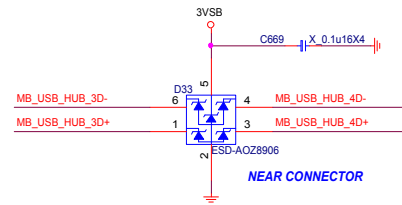
Pin23

Gang input:1

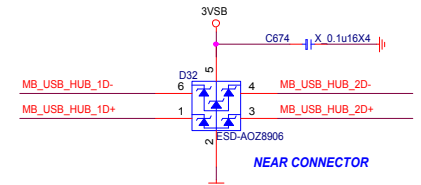
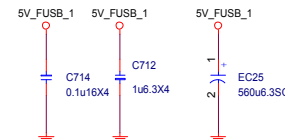
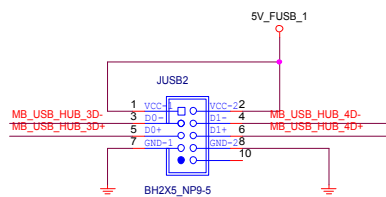
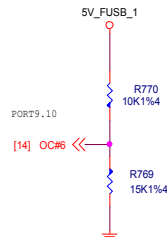
Individual input:0



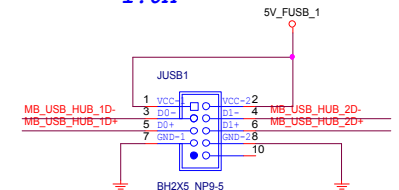
Rear USB2.0 PORT 9,10



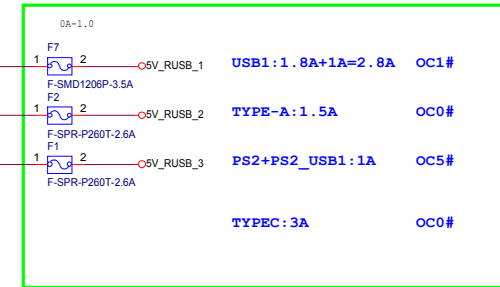
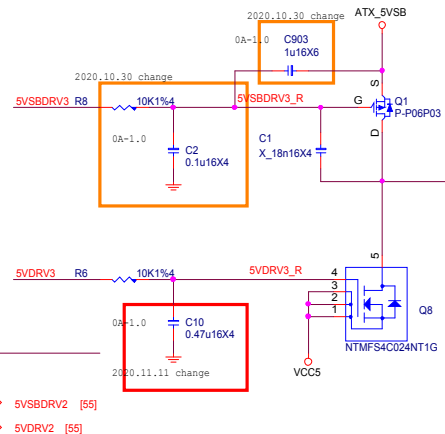
1.0A



1.0A



REAR USB PORT POWER



2020.10.30 change

0A-11.0

ATX_V5VSB

5VSBDRV3 R484 20K1% 4

5VSBDRV3_F

C904 1u16X6

C464 X_18n16X4

C465 0.1u16X4

Q78 P-P06P03

5V_FUSB

5VDRV3 R204 20K1% 4

5VDRV3_F

C278 0.1u16X4

Q30 NTMFS4C024NT1G

VCC5

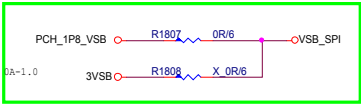
F6 2 1 F-SPR-P260T-2.6A 4XUSB20:2A OC6#

F5 2 1 F-SPR-P260T-2.6A 2XUSB30:1.8A OC3#

F3 2 1 F-SPR-P260T-2.6A 2XUSB30:1.8A OC4#

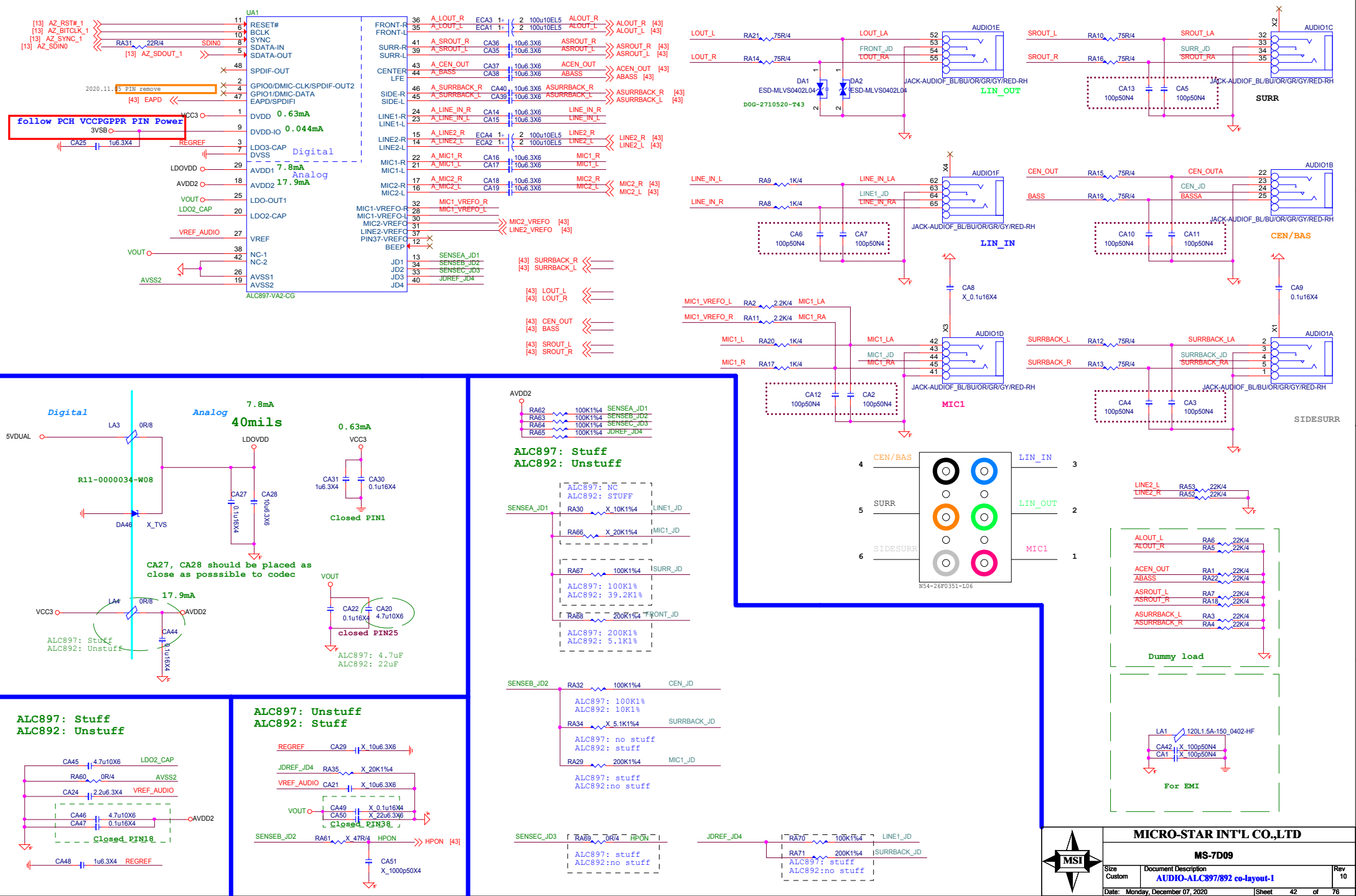
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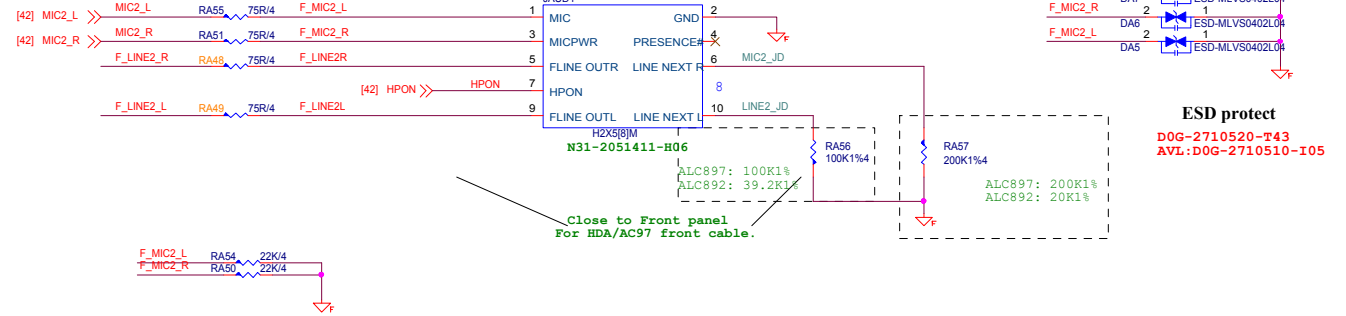
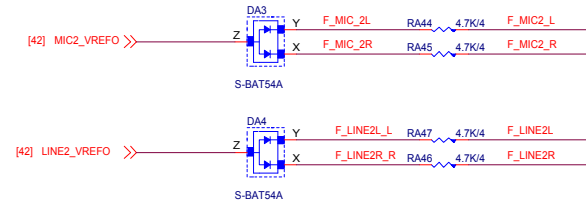




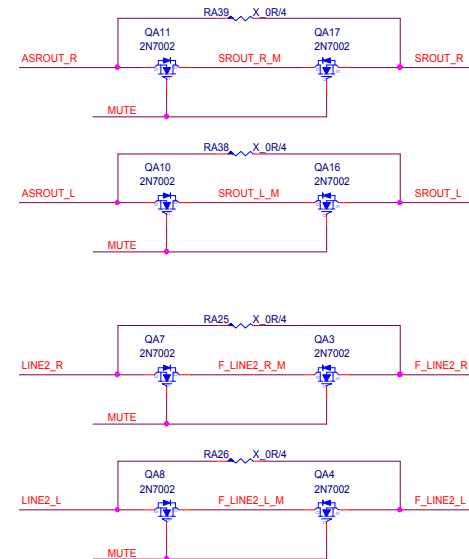
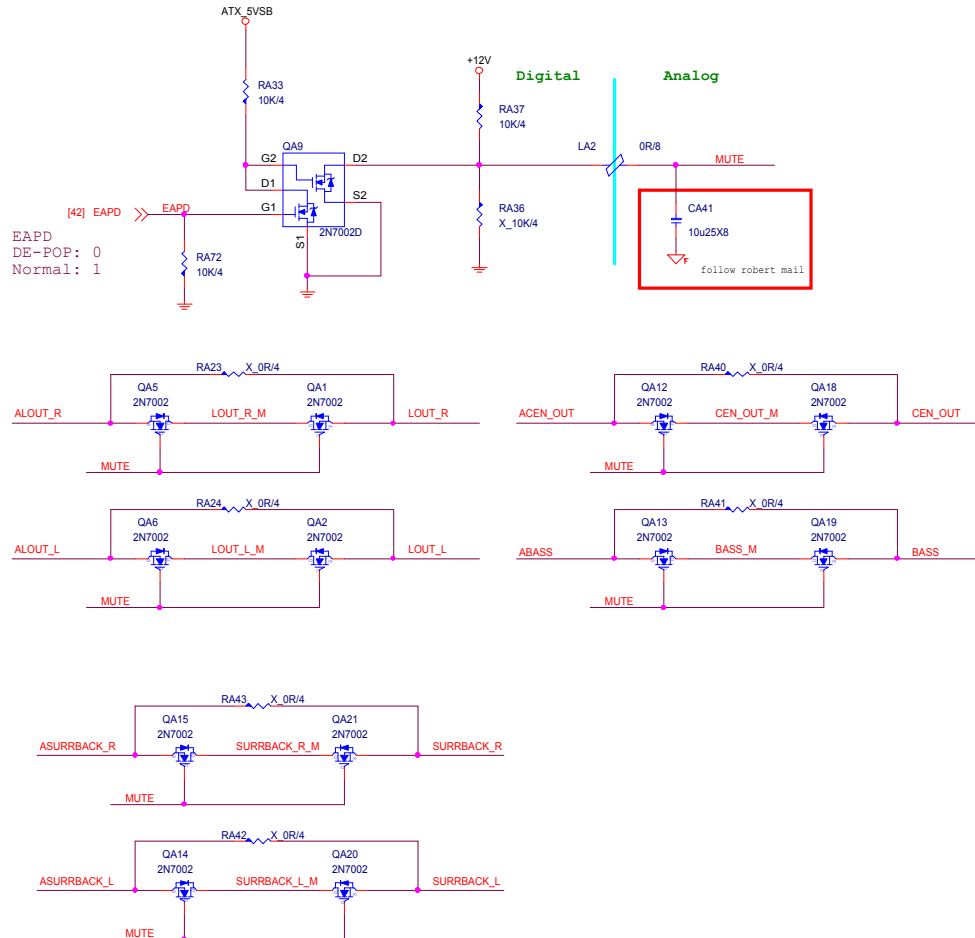
Intel I225V

ALC897





De-POP circuit



LOUT_L [42]

LOUT_R [42]

SROUT_R [42]

SROUT_R [42]

CEN_OUT [42]

BASS [42]

ALOUT_R [42]

ALOUT_L [42]

ASROUT_R [42]

ASROUT_L [42]

ACEN_OUT [42]

ABASS [42]

LINE2_R [42]

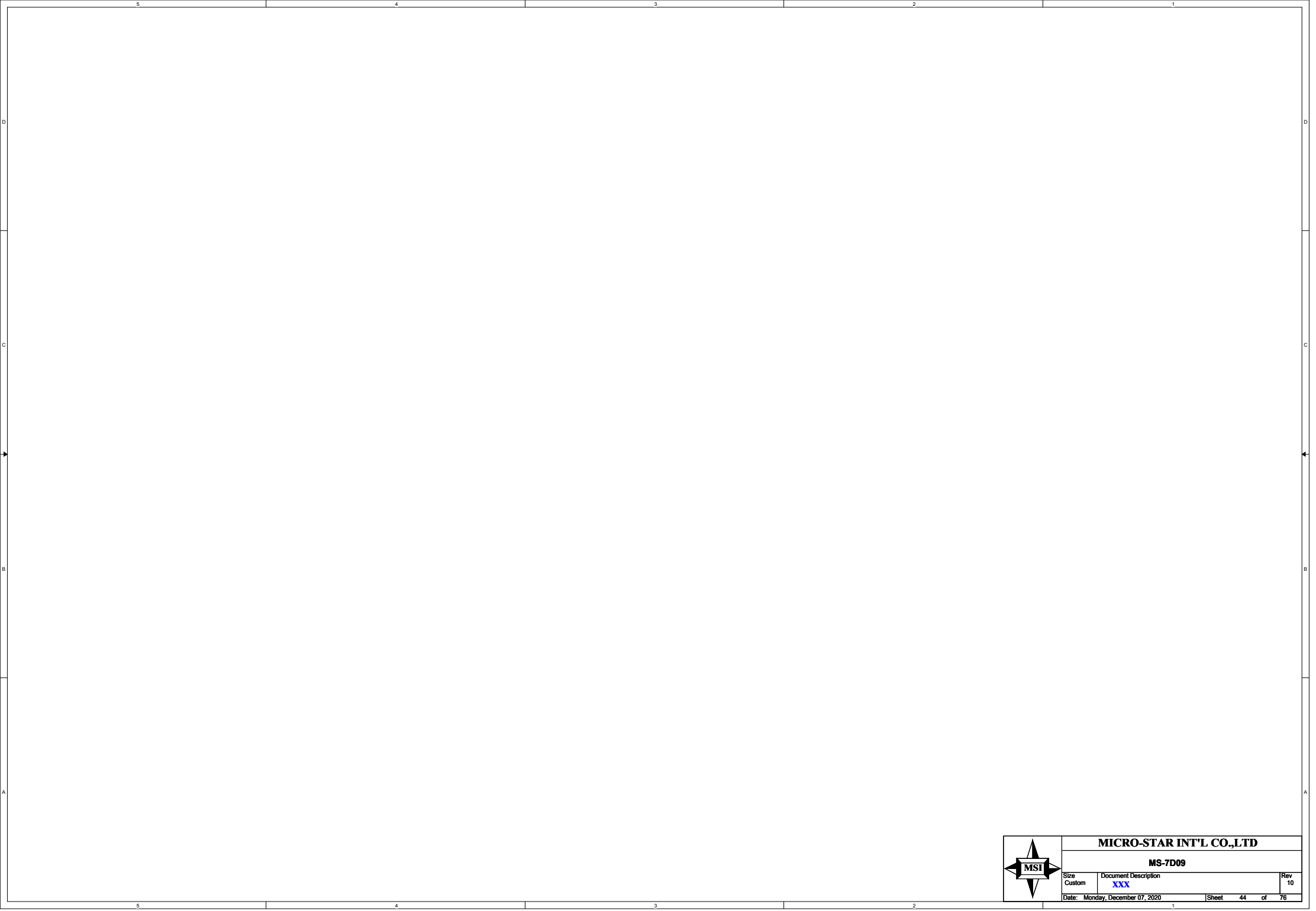
LINE2_L [42]

ASURRBACK_R [42]

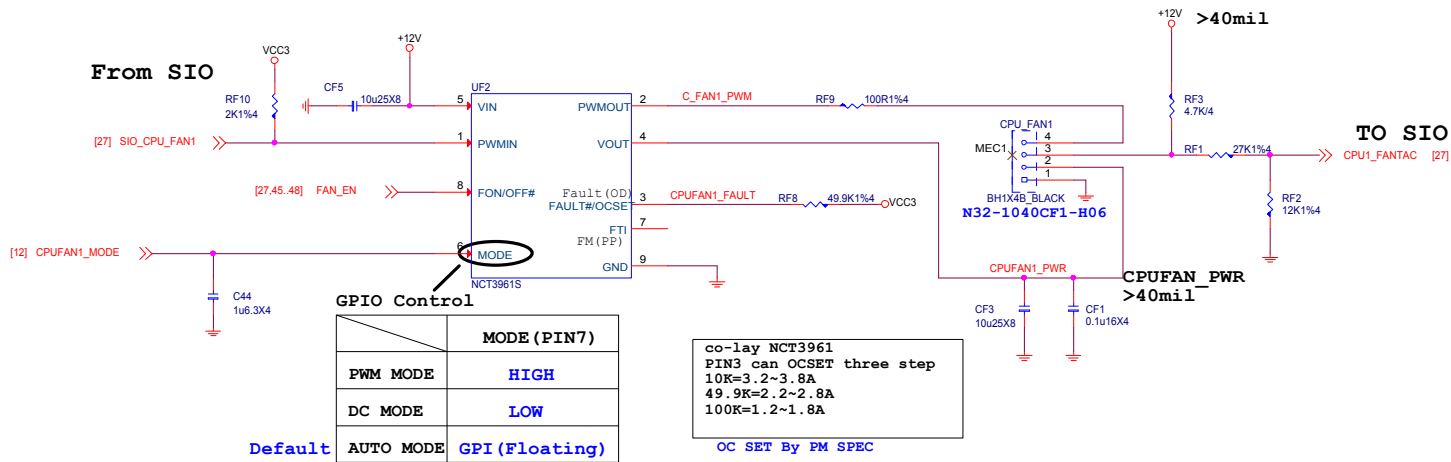
ASURRBACK_L [42]

SURRBACK_R [42]

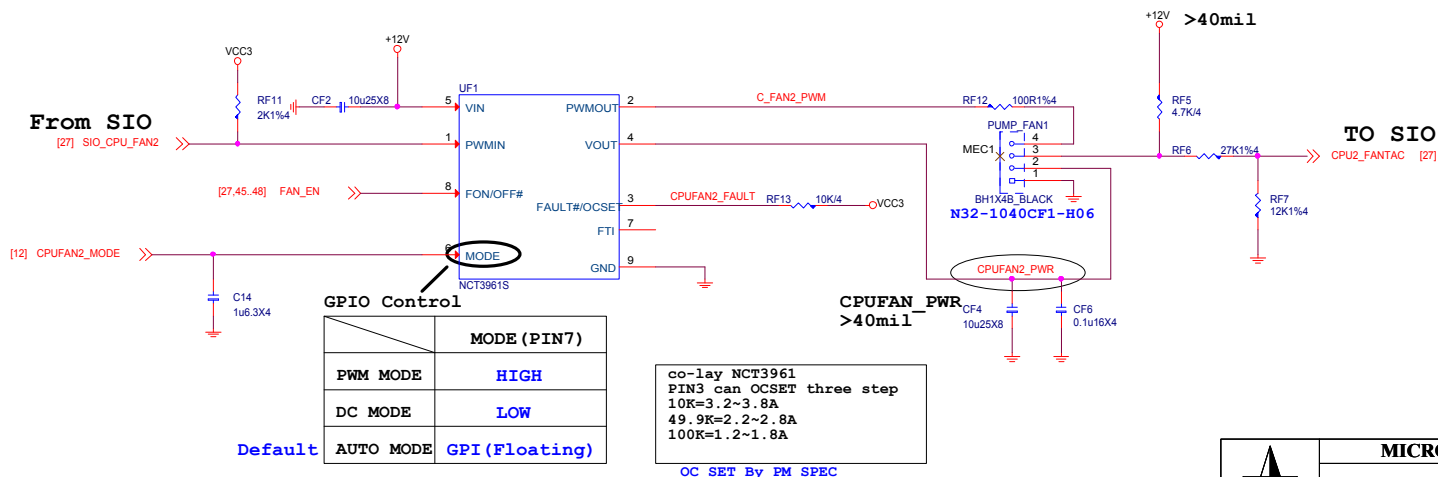
SURRBACK_L [42]



TYPE M:CPU FAN



TYPE M : PUMP FAN



TYPE M : SYSTEM FAN1

From SIO

[27] SIO_SYS1_FAN

[27,45..48] FAN_EN

[14] SYSFAN1_MODE

C417 1u6.3X4

GPIO Control

	MODE (PIN7)
PWM MODE	HIGH
DC MODE	LOW
AUTO MODE	GPI(Floating)

Default

Internall pull up 1.65V

UF4 VIN

PWMOUT

VOUT

FON/OFF#

FAULT#/OCSE

FTI

GND

S_FAN1_PWM

SYS_FAN1

MEC1

BH1X4B_BLACK

N32-1040CF1-H06

SYSFAN1_PWR

CF11 10u25X8

CF12 0.1u16X4

C22,C23,C263 close to FAN Connector

CPUFAN_PWR >40mil

TO SIO

SYS1_FANTAC [27]

TYPE M :SYSTEM FAN2

From SIO

[27] SIO_SYS2_FAN

[27,45..48] FAN_EN

[14] SYSFAN2_MODE

C678 1u6.3X4

GPIO Control

	MODE (PIN7)
PWM MODE	HIGH
DC MODE	LOW
AUTO MODE	GPI(Floating)

Default

Internall pull up 1.65V

UF7 VIN

PWMOUT

VOUT

FON/OFF#

FAULT#/OCSE

FTI

GND

S_FAN2_PWM

SYS_FAN2

MEC1

BH1X4B_BLACK

N32-1040CF1-H06

SYSFAN2_PWR

CF16 10u25X8

CF22 0.1u16X4

CPUFAN_PWR >40mil

TO SIO

SYS2_FANTAC [27]

MSI

MICRO-STAR INT'L CO.,LTD

MS-7D09

Size Custom

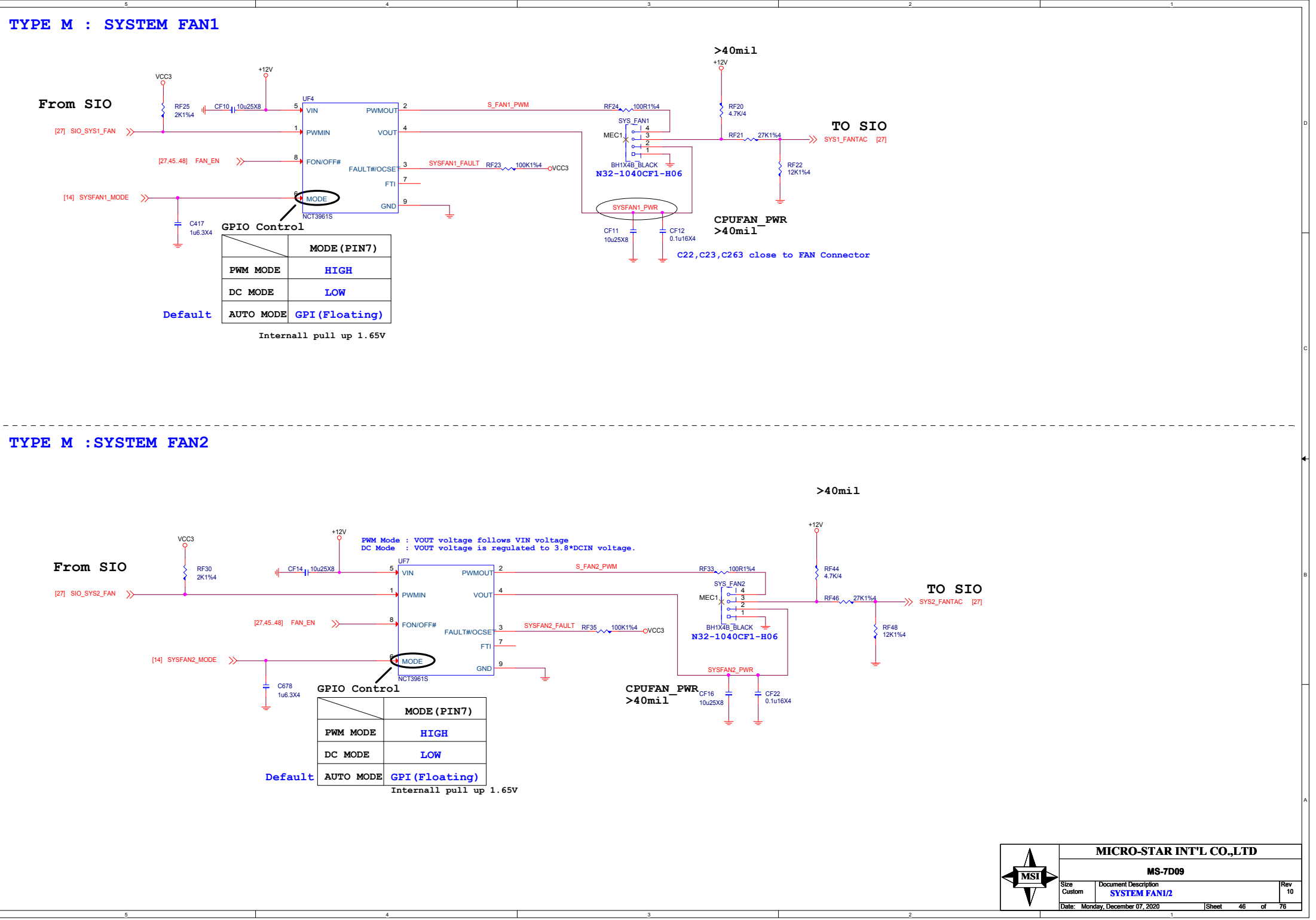
Document Description

SYSTEM FAN1/2

Date: Monday, December 07, 2020

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Rev 10



TYPE M : SYSTEM FAN1

From SIO

[27] SIO_SYS1_FAN

[27,45..48] FAN_EN

[14] SYSFAN1_MODE

C417 1u6.3X4

GPIO Control

	MODE (PIN7)
PWM MODE	HIGH
DC MODE	LOW
AUTO MODE	GPI(Floating)

Default

Internall pull up 1.65V

UF4 VIN

PWMOUT

VOUT

FON/OFF#

FAULT#/OCSE

FTI

GND

S_FAN1_PWM

SYS_FAN1

MEC1

BH1X4B_BLACK

N32-1040CF1-H06

SYSFAN1_PWR

CF11 10u25X8

CF12 0.1u16X4

C22,C23,C263 close to FAN Connector

CPUFAN_PWR >40mil

TO SIO

SYS1_FANTAC [27]

TYPE M :SYSTEM FAN2

From SIO

[27] SIO_SYS2_FAN

[27,45..48] FAN_EN

[14] SYSFAN2_MODE

C678 1u6.3X4

GPIO Control

	MODE (PIN7)
PWM MODE	HIGH
DC MODE	LOW
AUTO MODE	GPI(Floating)

Default

Internall pull up 1.65V

UF7 VIN

PWMOUT

VOUT

FON/OFF#

FAULT#/OCSE

FTI

GND

S_FAN2_PWM

SYS_FAN2

MEC1

BH1X4B_BLACK

N32-1040CF1-H06

SYSFAN2_PWR

CF16 10u25X8

CF22 0.1u16X4

CPUFAN_PWR >40mil

TO SIO

SYS2_FANTAC [27]

MSI

MICRO-STAR INT'L CO.,LTD

MS-7D09

Size Custom

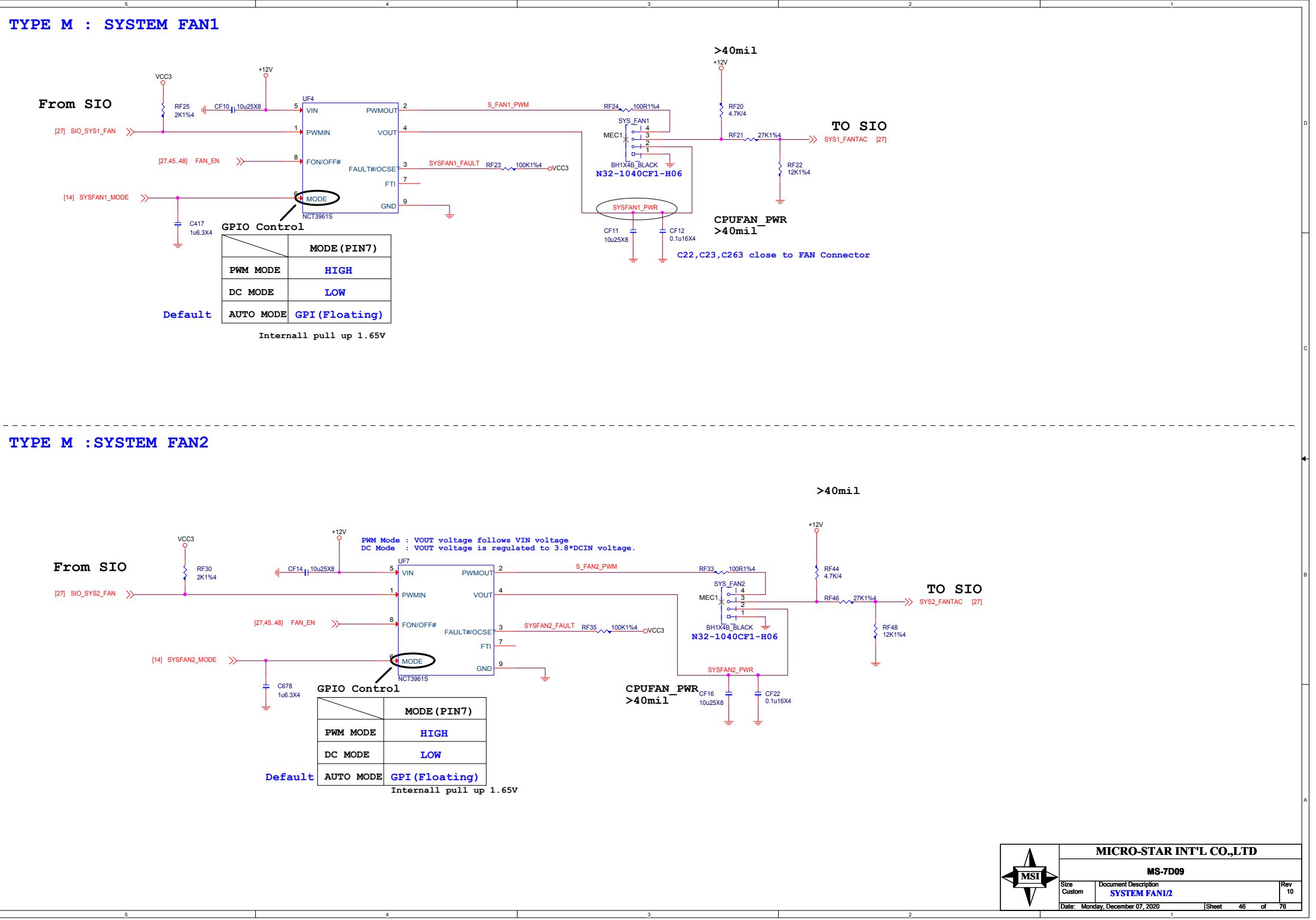
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SYSTEM FAN1/2

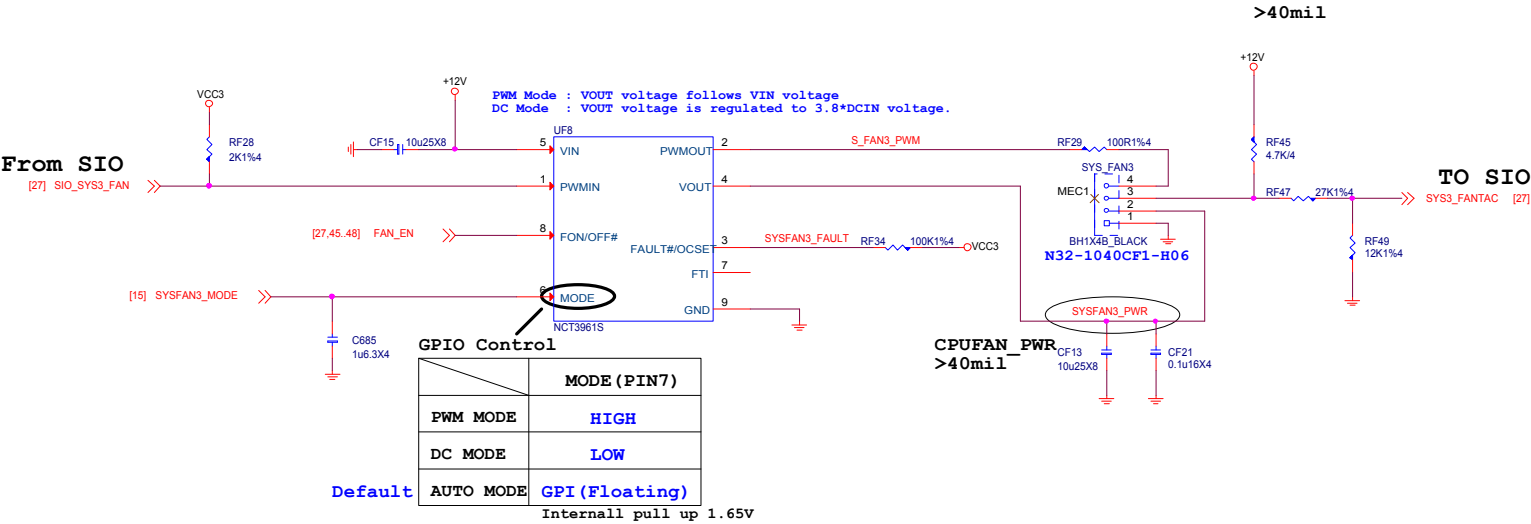
Date: Monday, December 07, 2020

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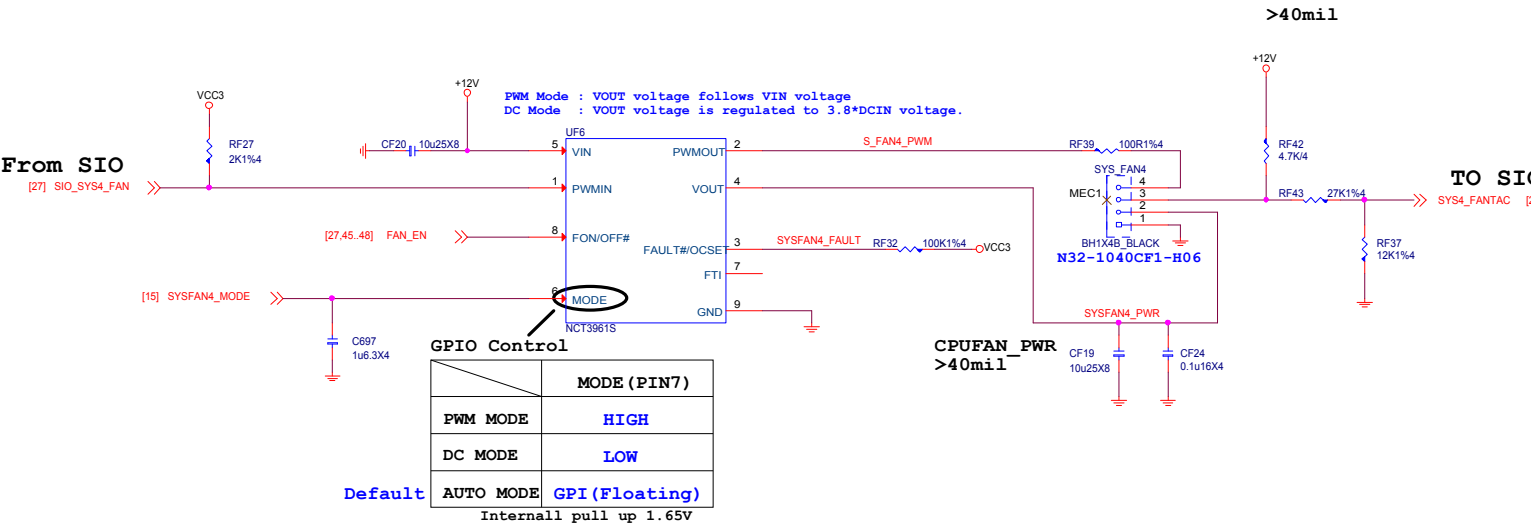
Rev 10



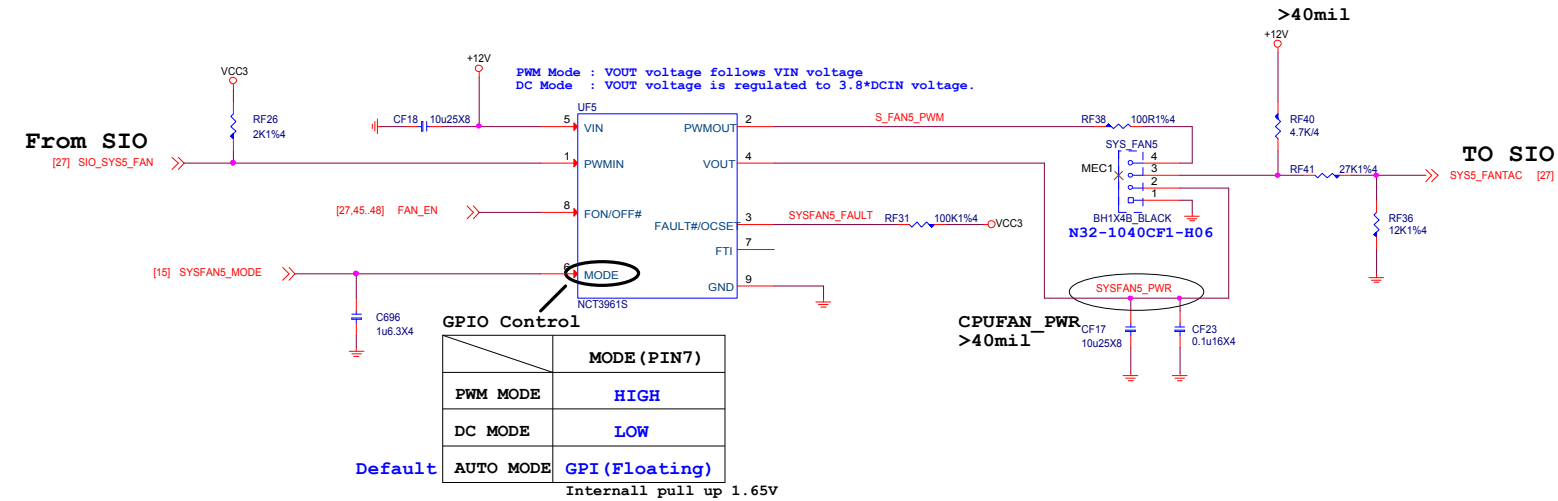
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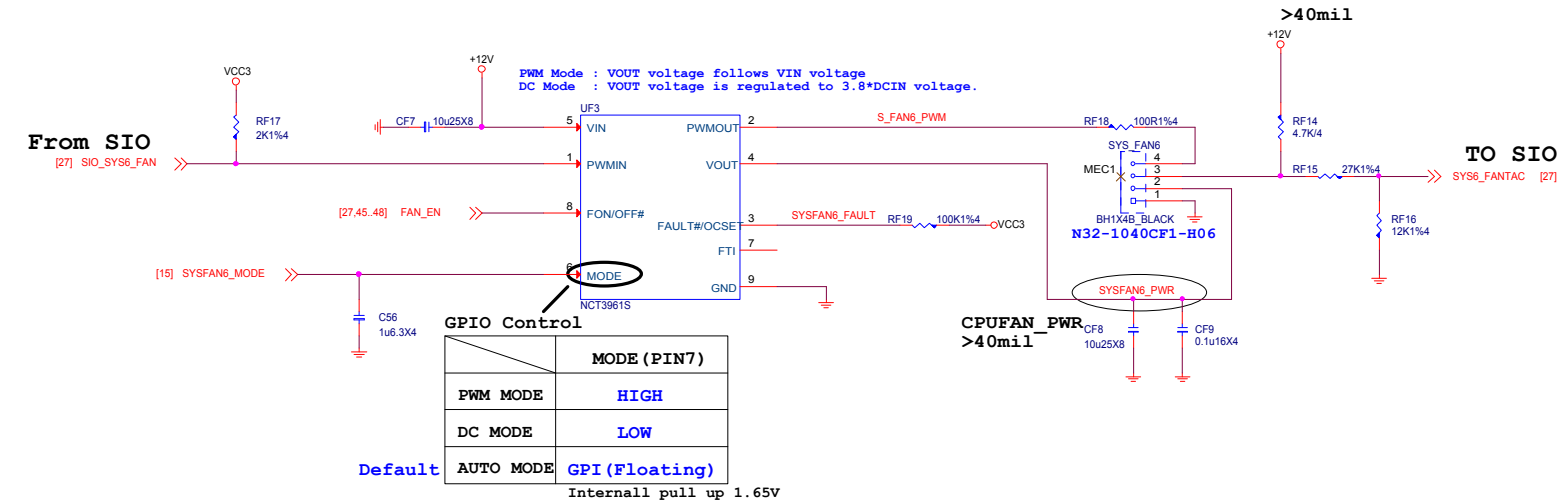
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TYPE M : SYSTEM FAN5



TYPE M : SYSTEM FAN6



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MS-7D09

Size
Custom

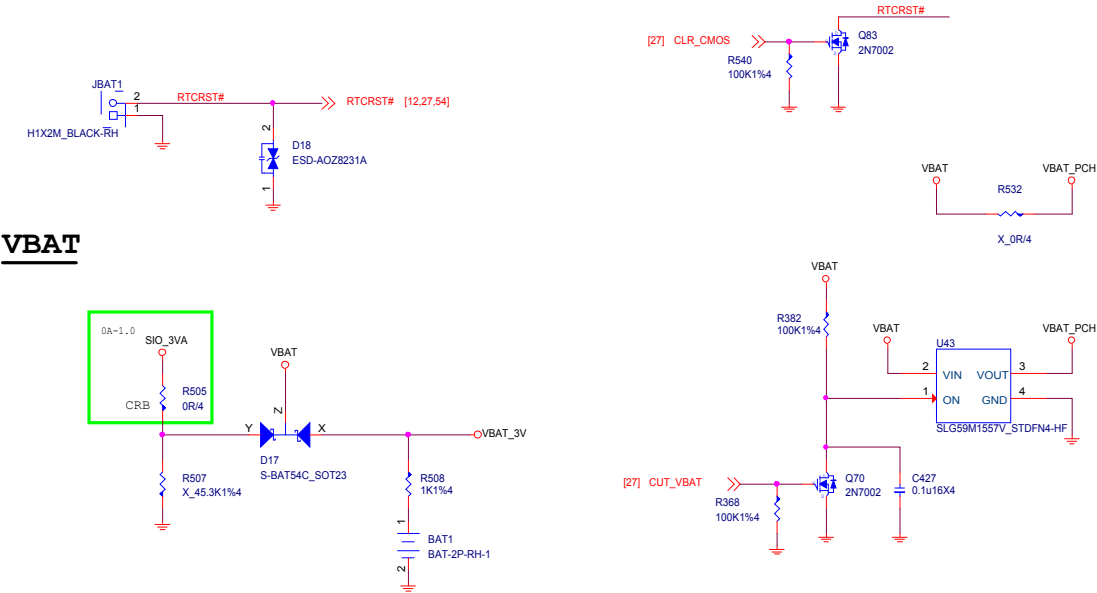
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SYSTEM FANS/6

Rev
10

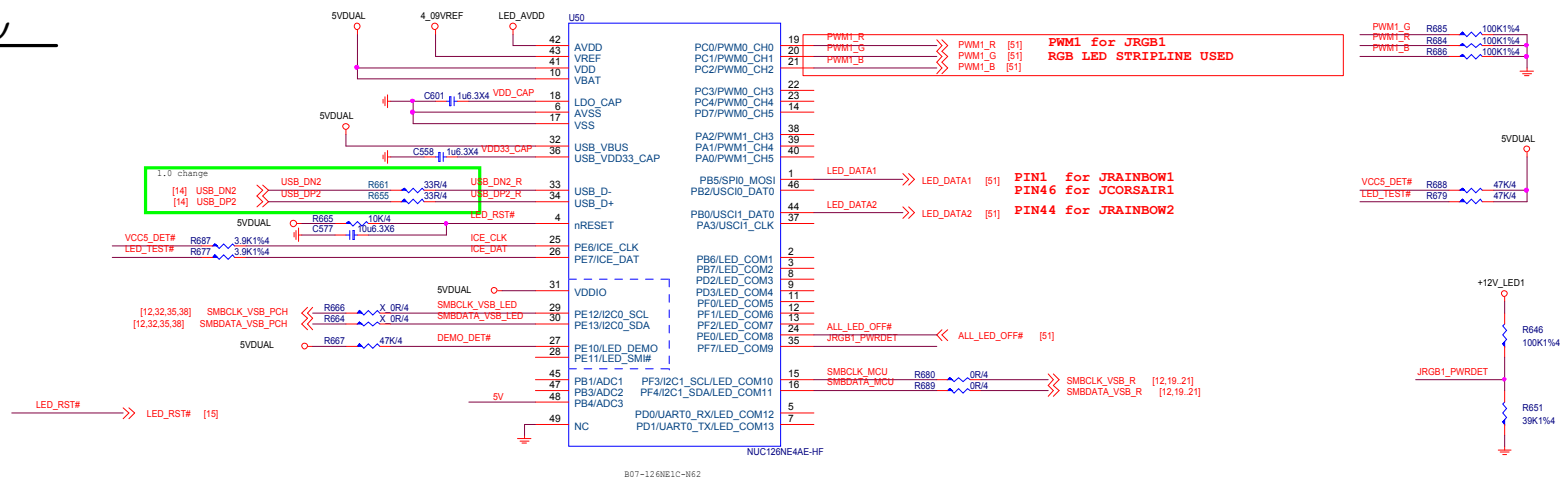
Date: Monday, December 07, 2020

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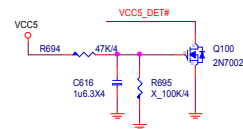
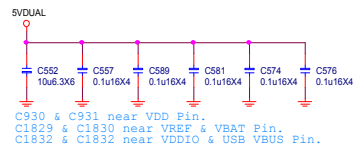
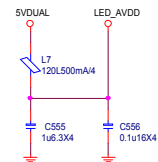
VBAT



**LED
MCU**



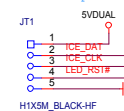
Control	Net Name	PWM USE
PCH	LED_DATA1	No Use
AUDIO Cover	LED_GPIO_01	No Use
MOS/IO cover	LED_GPIO_02	No Use
JRAINBOW1	LED_GPIO_03	No Use
JCORSAIR1	LED_DATA2	No Use
JRGB1/JRGB2	PWM1/ PWM2	PWM1/ PWM2
Board Side LED	COM 1~8	PWM3
Board Side LED	COM 9~16	PWM2



IF no JPWRLED1 & JPIPE_LED spec

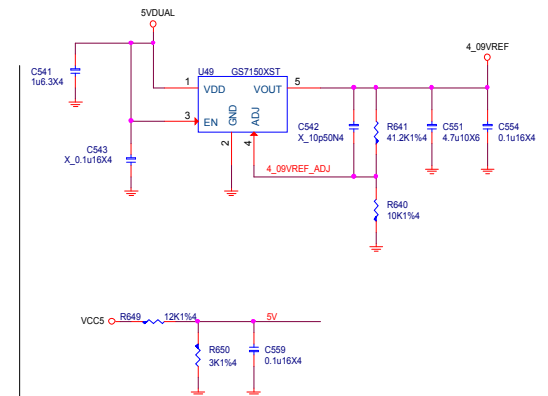
MCU can powered by 5VDUAL directly.
LED_VCC5 replace with 5VDUAL.

For FW update.

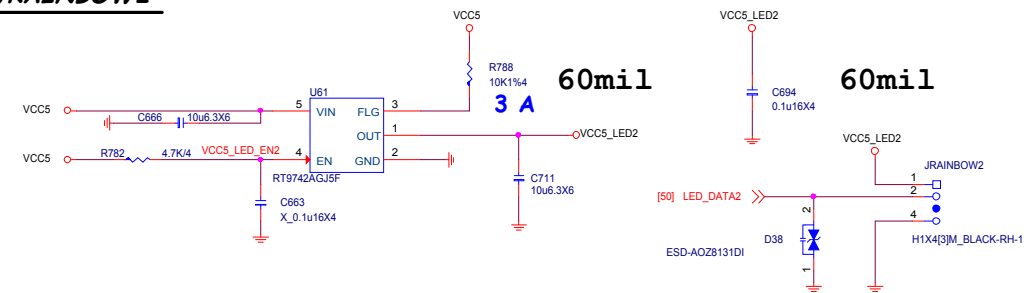


0A-1.0 Remove JF1

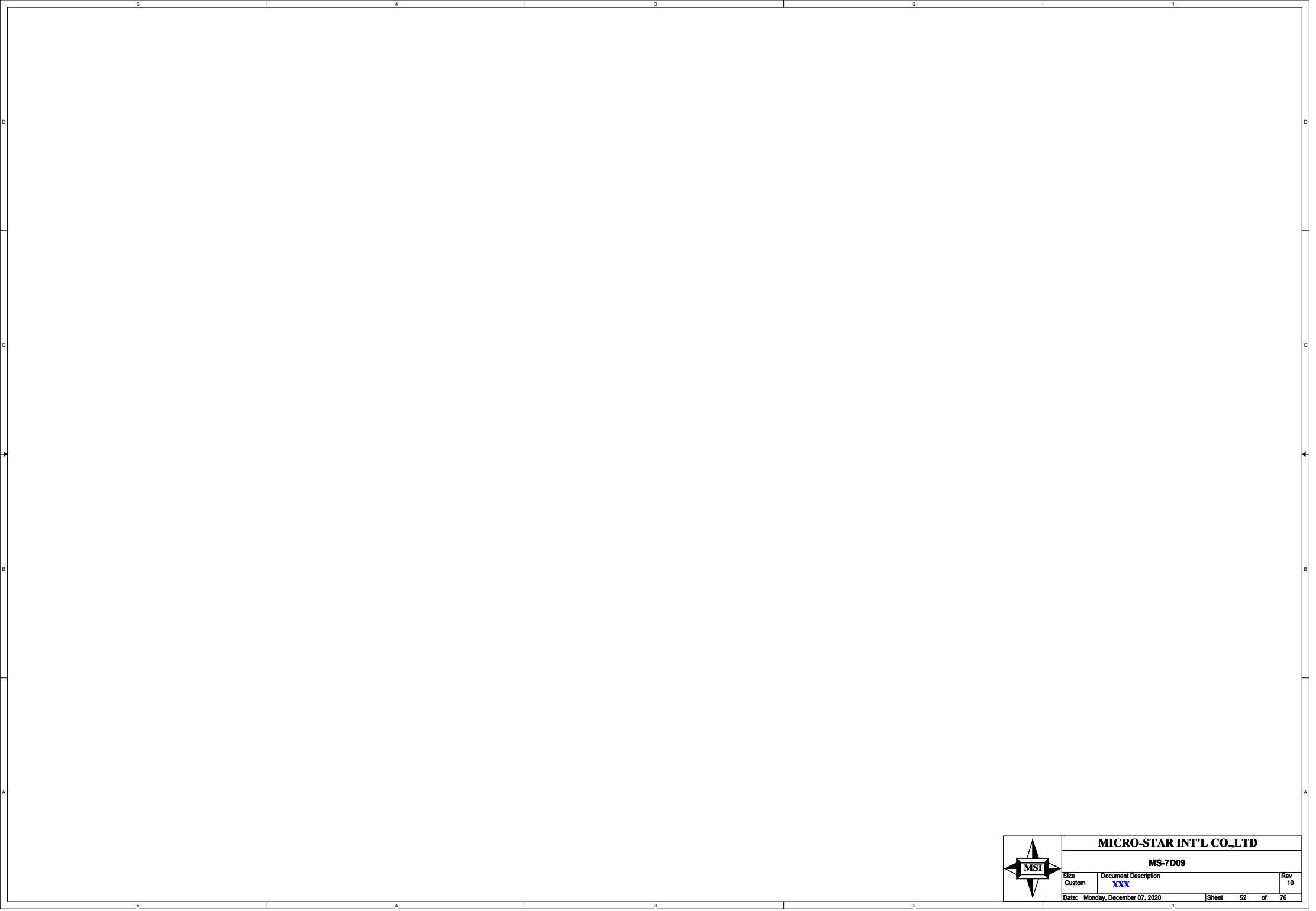
MCU Voltage Monitor



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JRAINBOW1

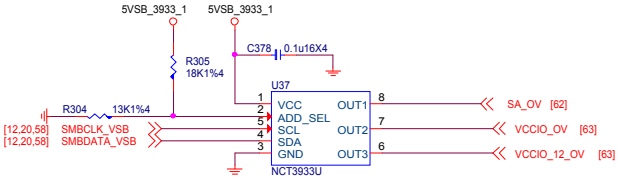
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UPI VOLTAGE CONSOLE

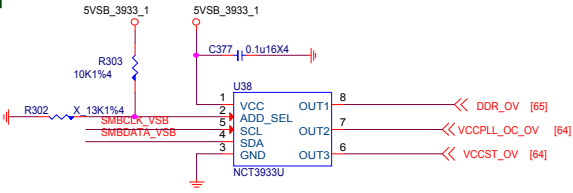
0x26: RH=18K, RL=13K



ADDRESS	0x2A	0x28	0x26	0x24	0x22	0x20
RH (KOhm)	OPEN	3.9	3	2.2	1.3	10
RL (KOhm)	10	1.3	2.3	3	3.9	OPEN
BUS_SEL	0%	25%	40%	60%	75%	100%

UPI VOLTAGE CONSOLE

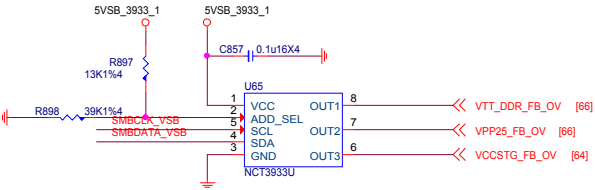
0x20: RH=10K, RL=OPEN



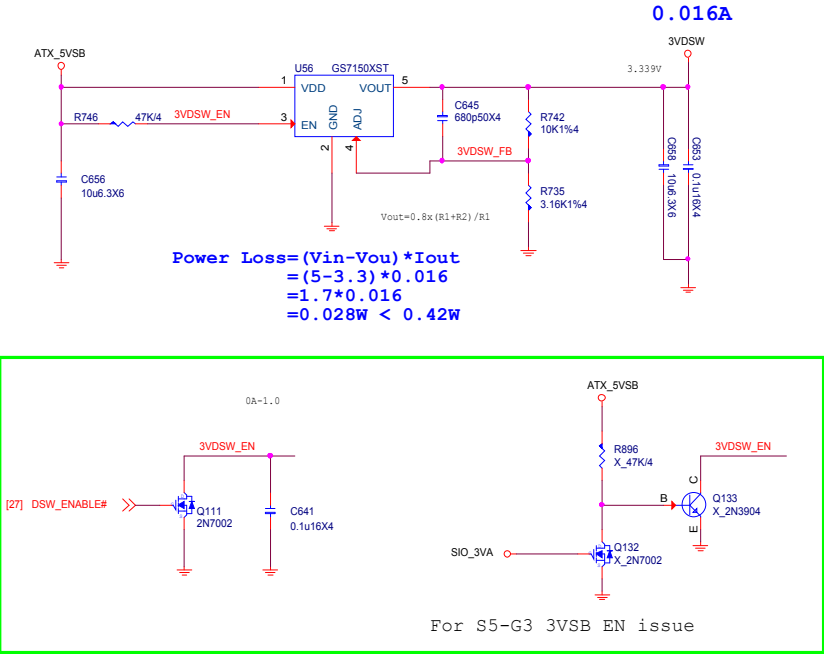
UPI VOLTAGE CONSOLE

0A=1.0

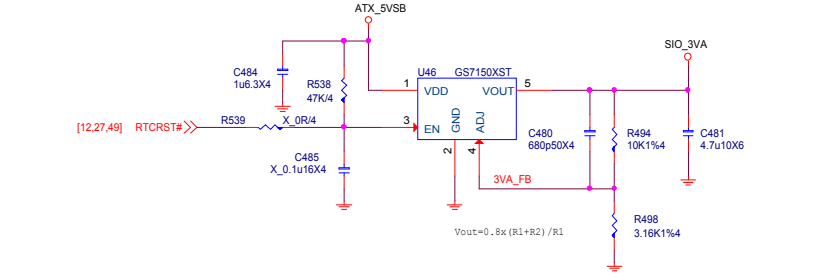
0x22: RH=13, RL=39K



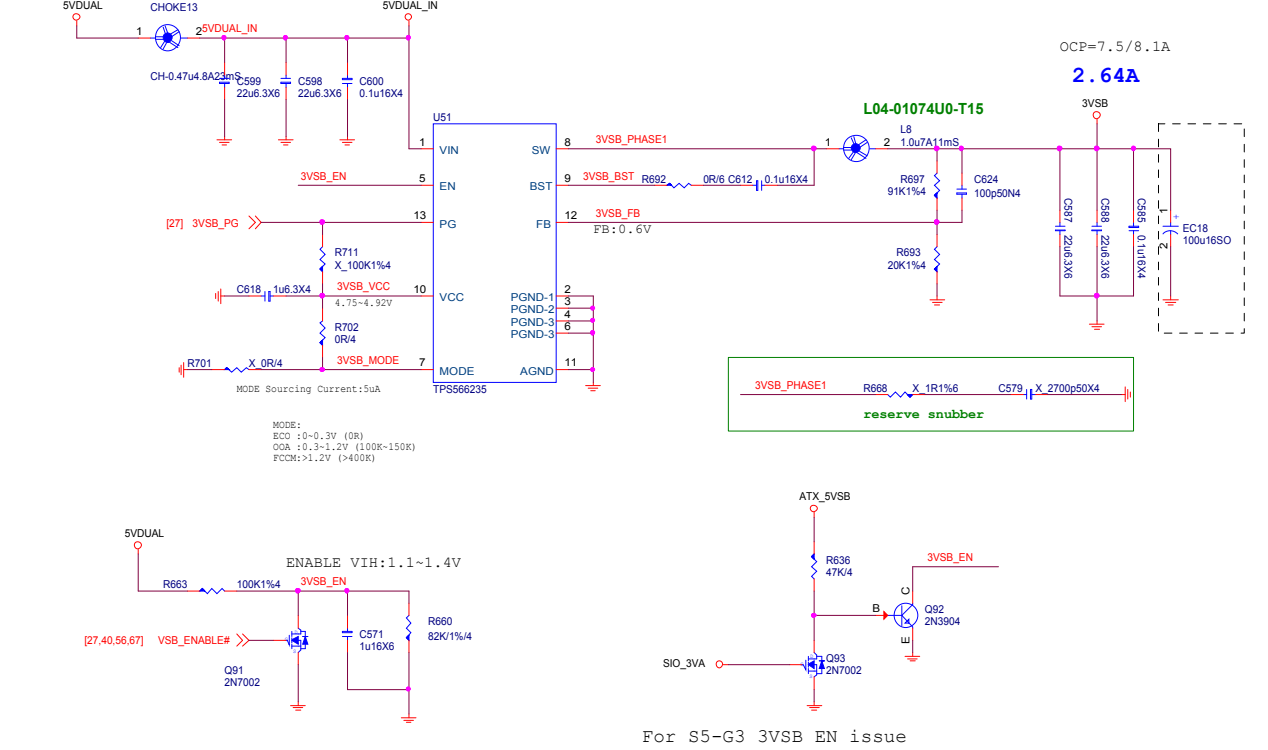
3VDSW



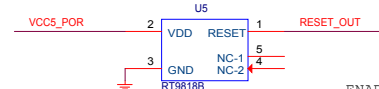
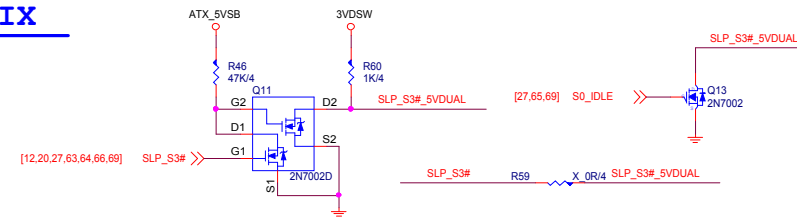
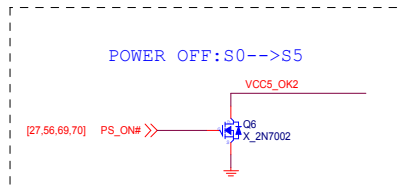
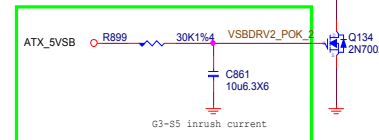
SIO 3VA



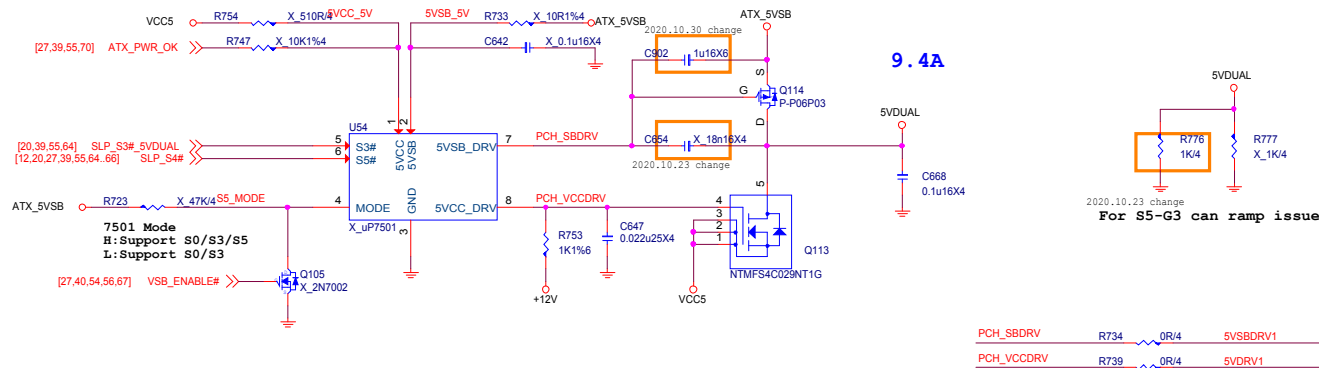
SYS 3VSB



S0IX

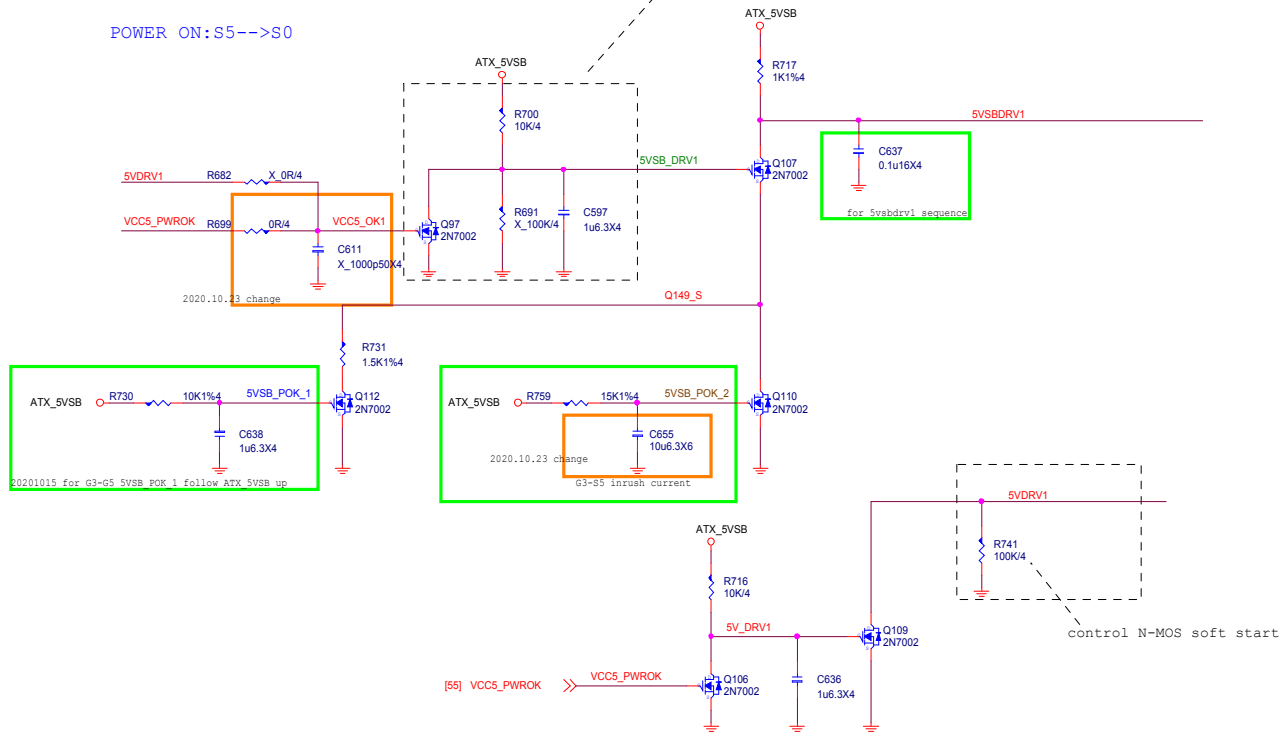
[illegible]

5VDUAL

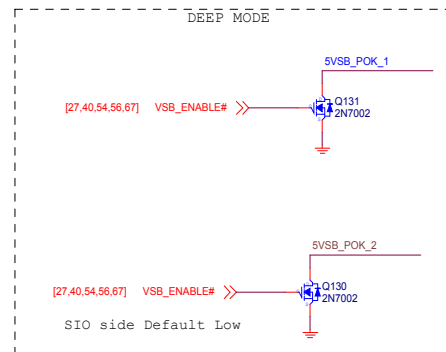
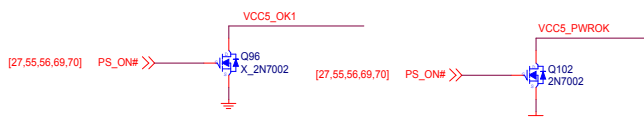


control P-MOS soft start and waiting ATX 5VSB ramp to 100%

POWER ON:S5-->S0



POWER OFF: S0-->S5



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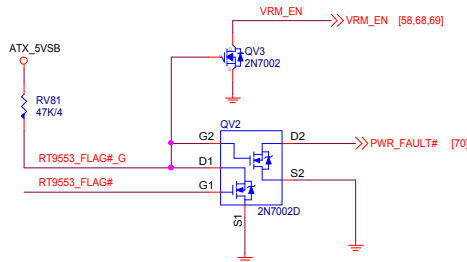
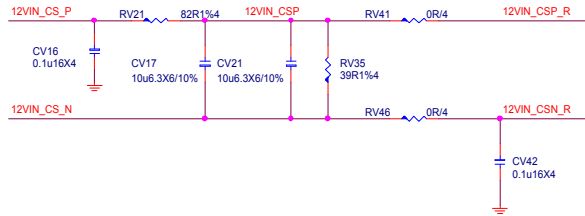
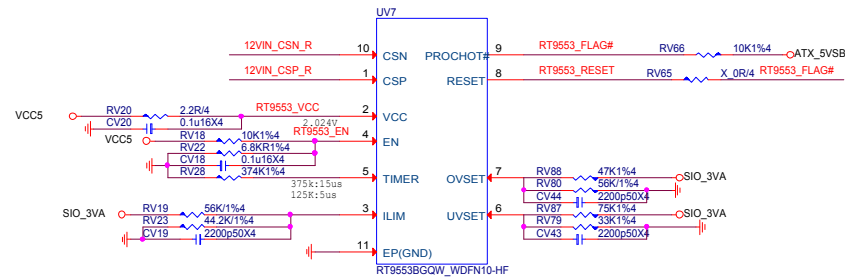
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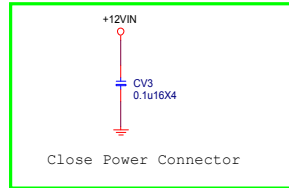
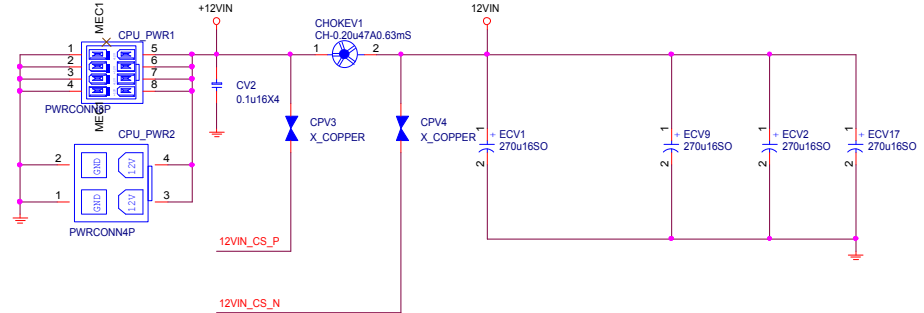
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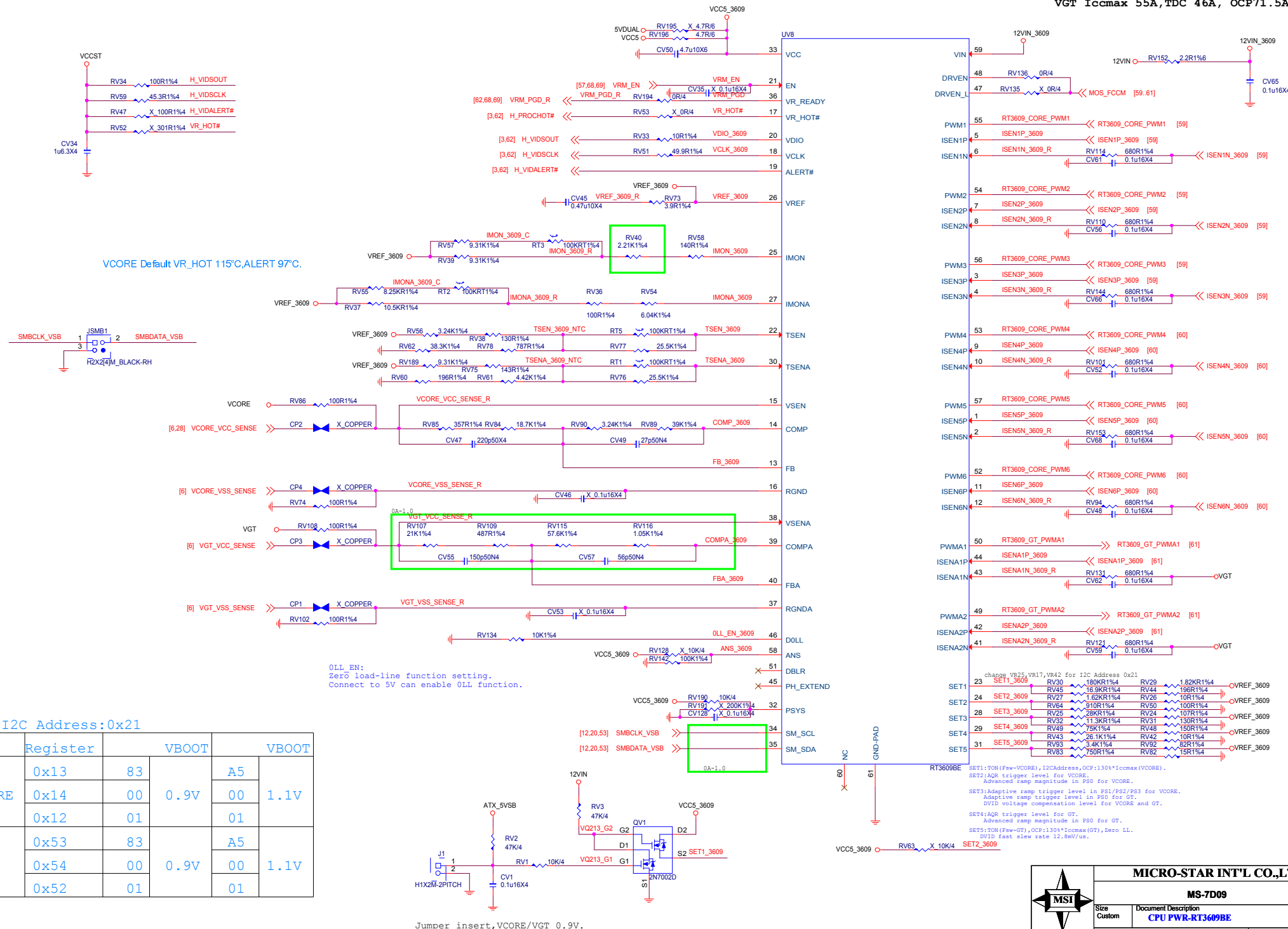
POWER METER

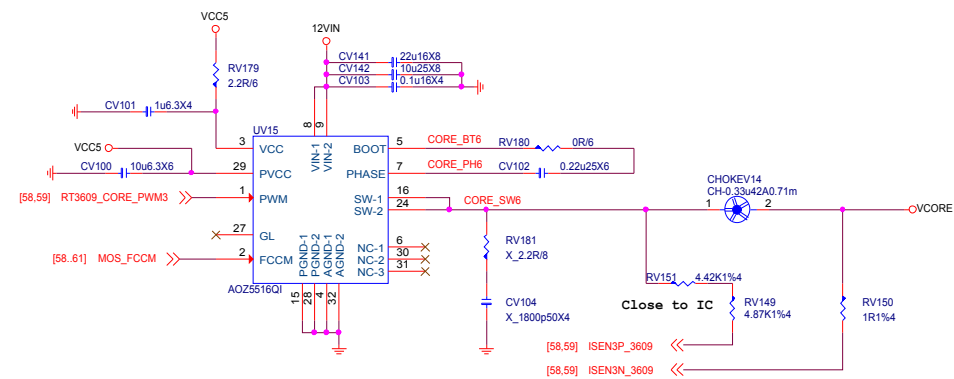
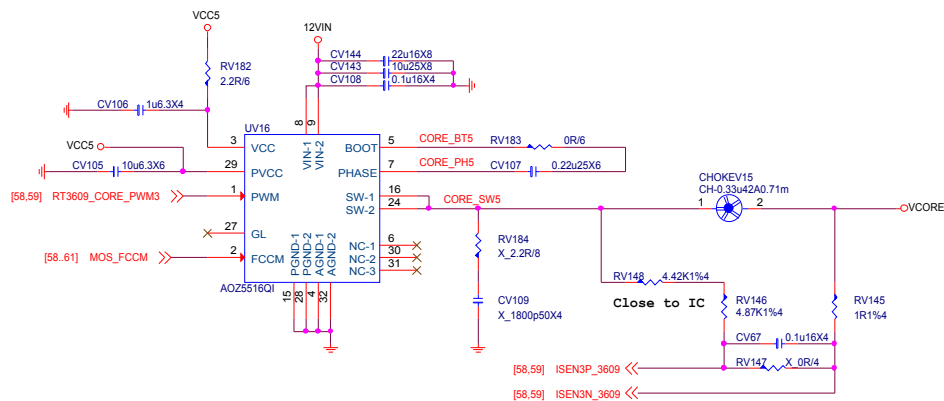
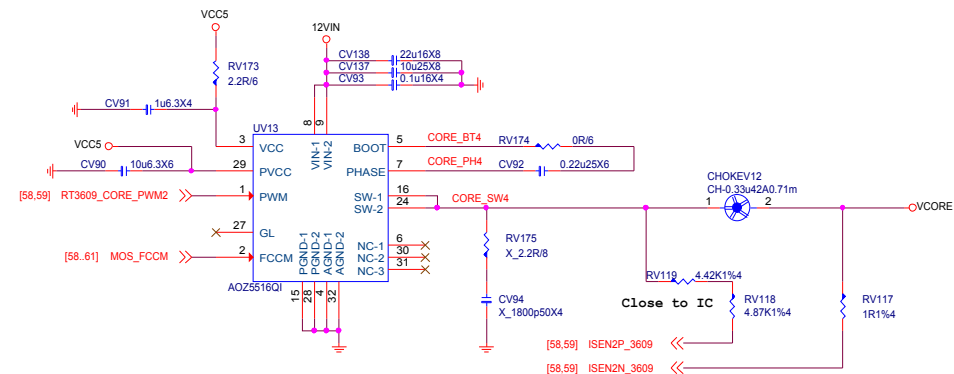
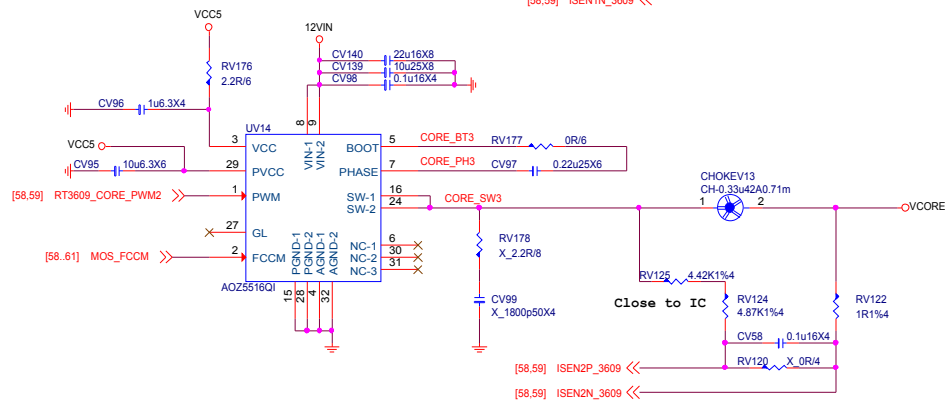
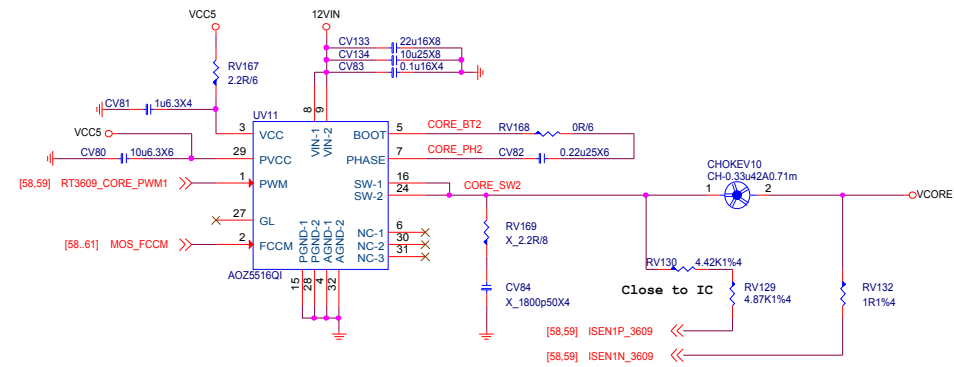
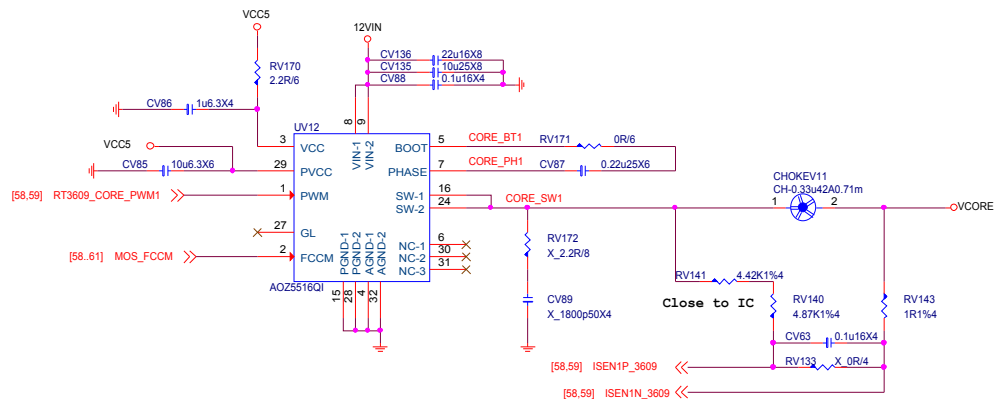
OCP: 45A for
8 core 193A
support

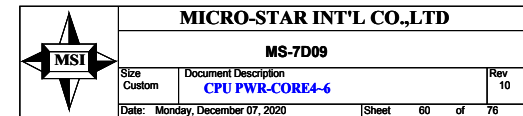
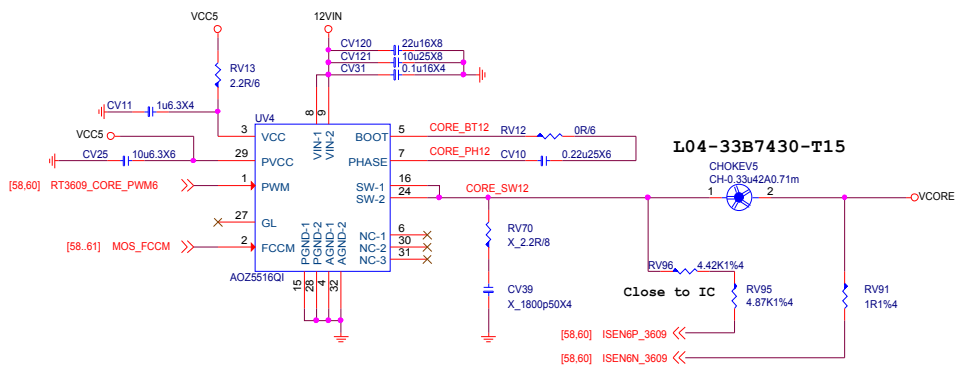
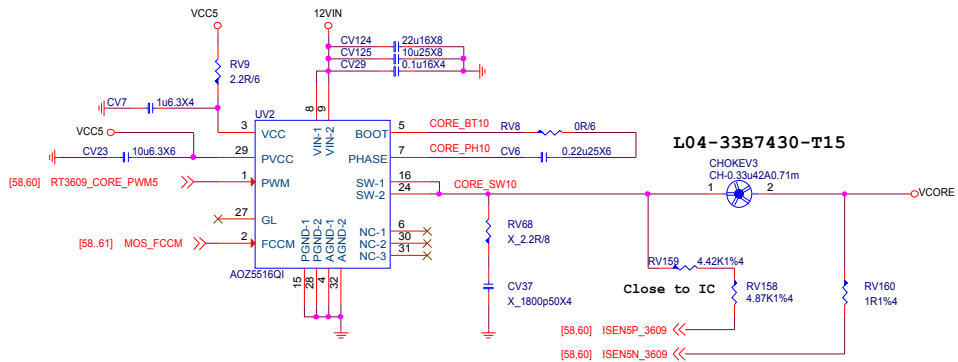
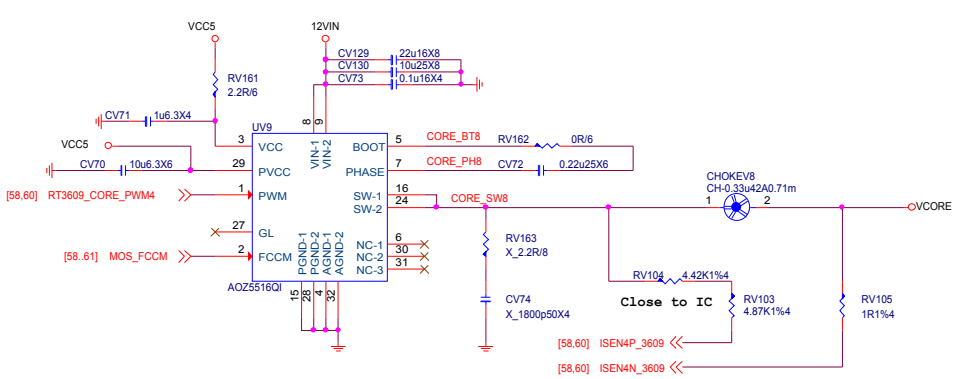


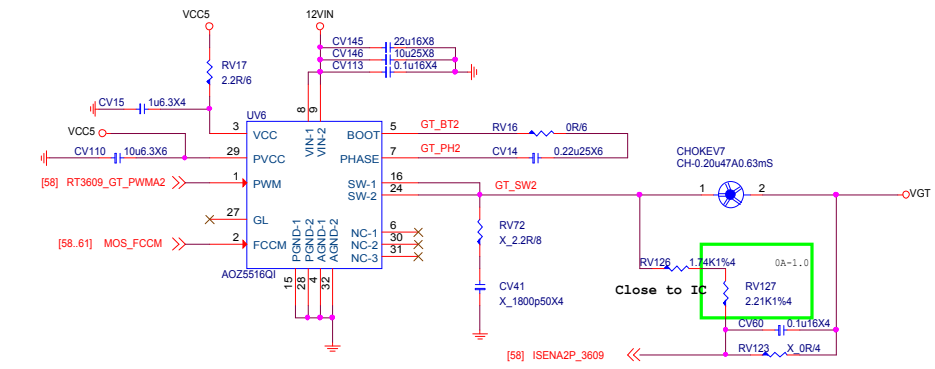
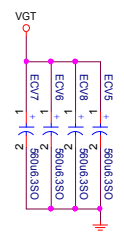
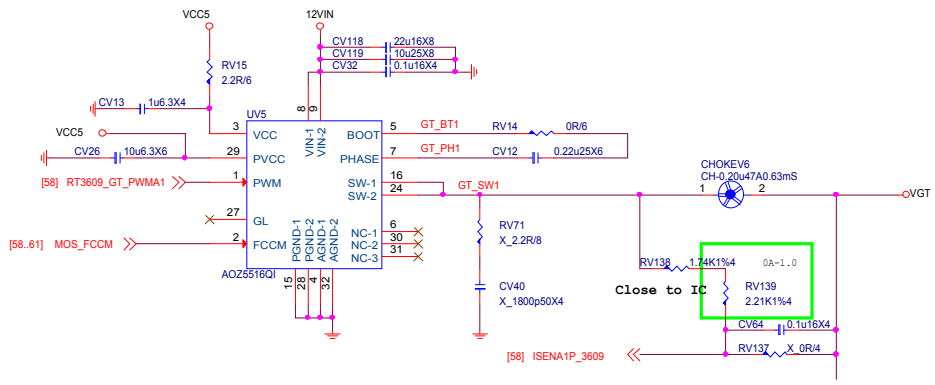
Iripple:12.991A
VCORE:9.855A
VCCSA:3.136A



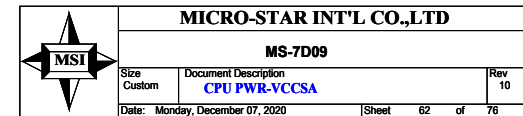




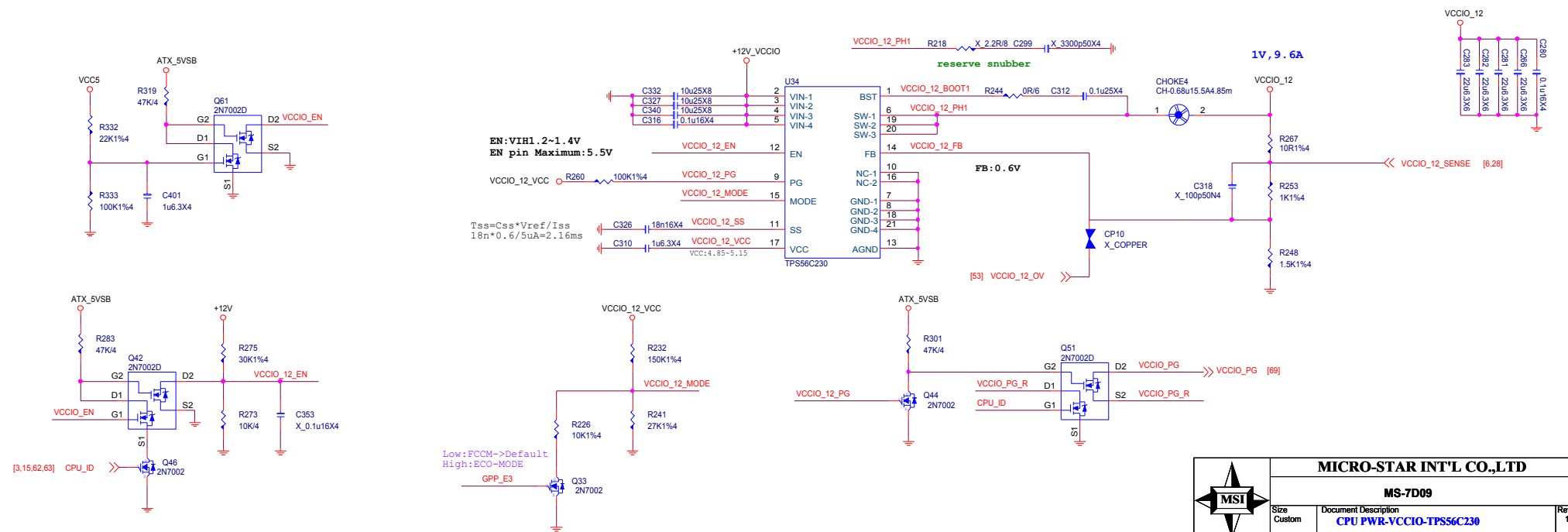
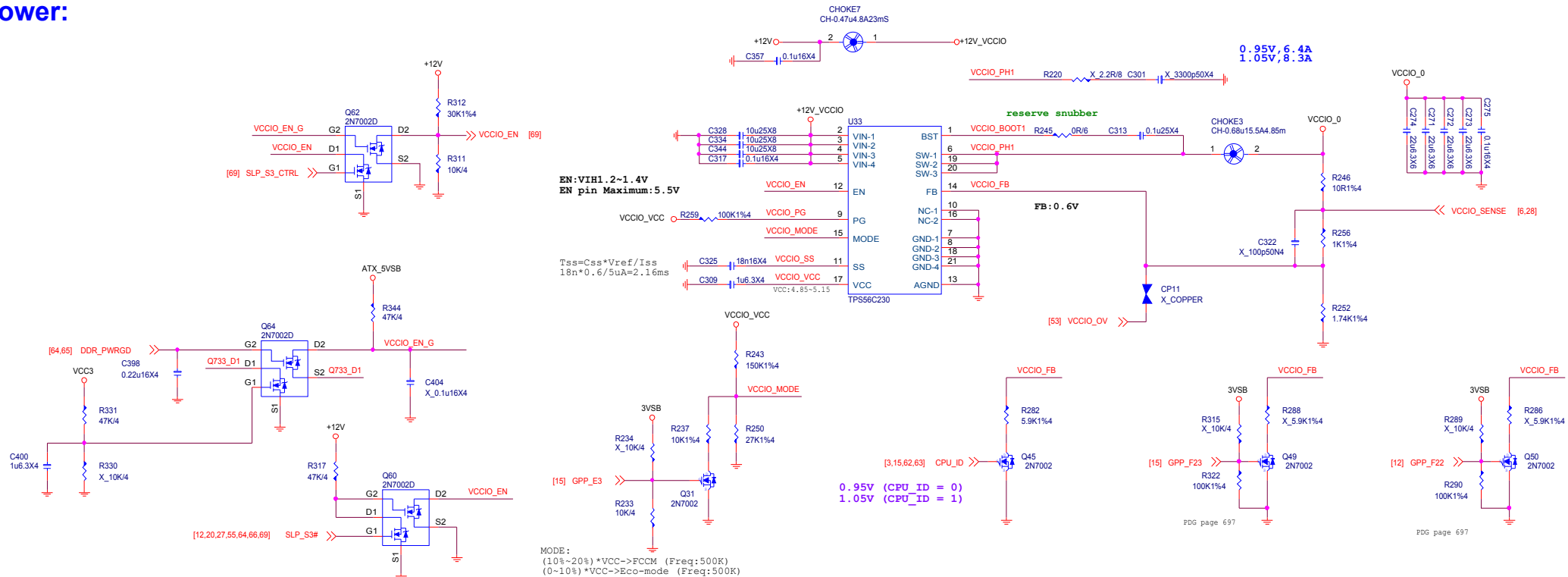




1.05V, ICC_max=22.1A



IO Power:



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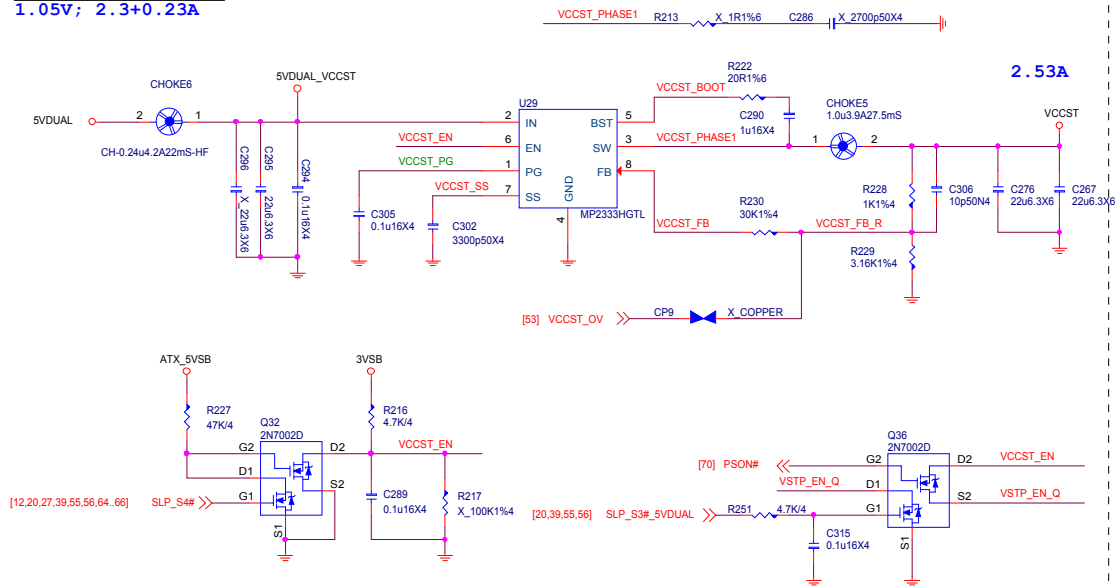
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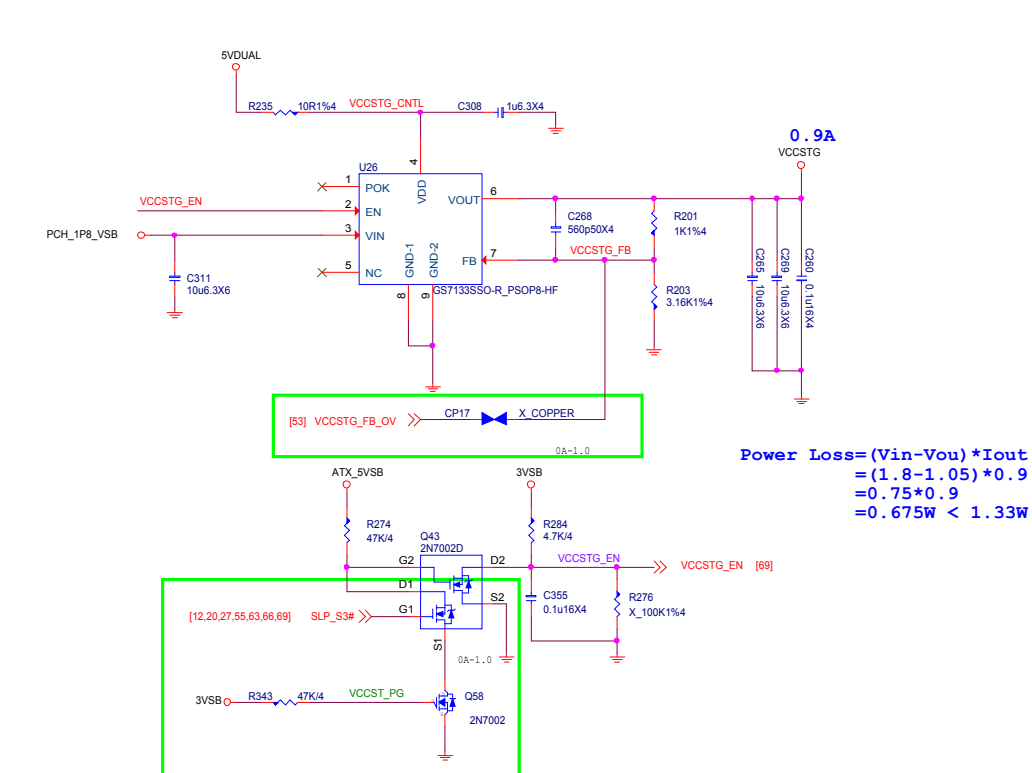
VCCST

1.05V; 2.3+0.23A



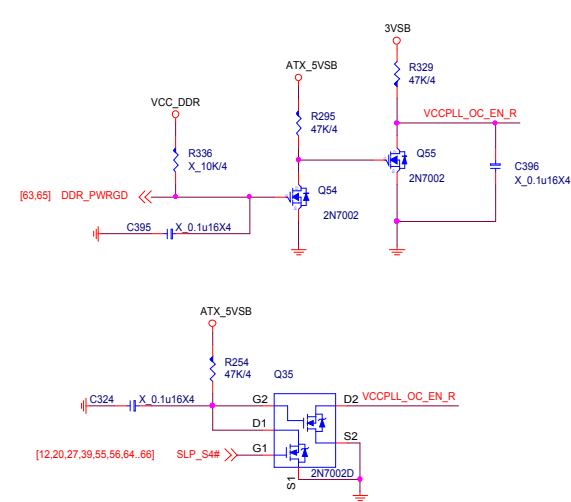
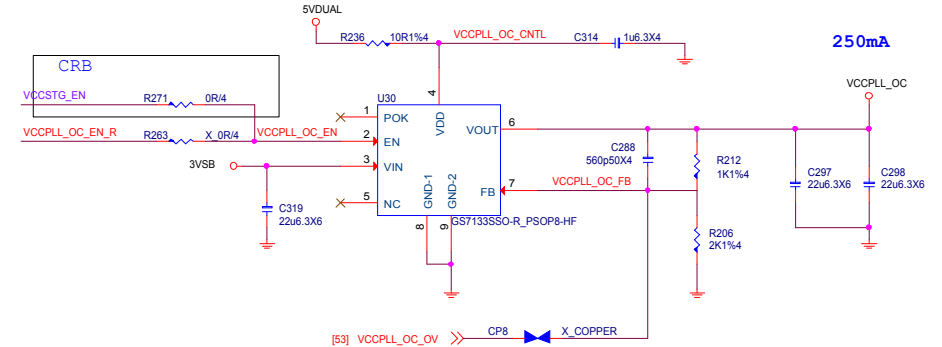
VCCSTG

1.05V; 0.9A



VCCPLL OC

1.2V; 250mA



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DDR4 Power:1.2V,14.4A OCP:18A

$R_{ocset} = 1.2 * I_{max} * R_{dson(1ow)} / I_{ocset}$
 $= 21.6 * 4mohm / 10uA$
 $= 8.64K$

4.3A FOR CPU

9.1A FOR 4DIMM

0.7A FOR DDR VTT

$I_{ocp} = R_{ocset} * I_{ocset} / R_{dson(1ow)}$
 $= 7.15K * 10uA / 3.3m$
 $= 21.66A$
 $I_{ocp} = R_{ocset} * I_{ocset} / R_{dson(max)}$
 $= 7.15K * 10uA / 4m$
 $= 17.88A$

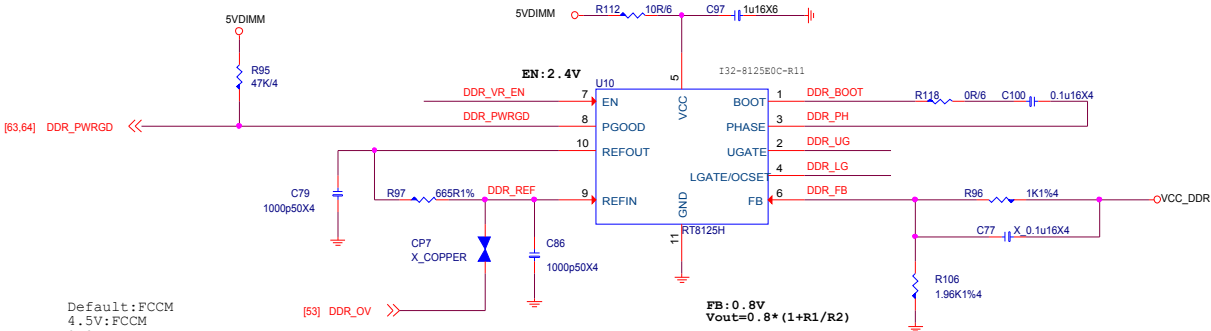
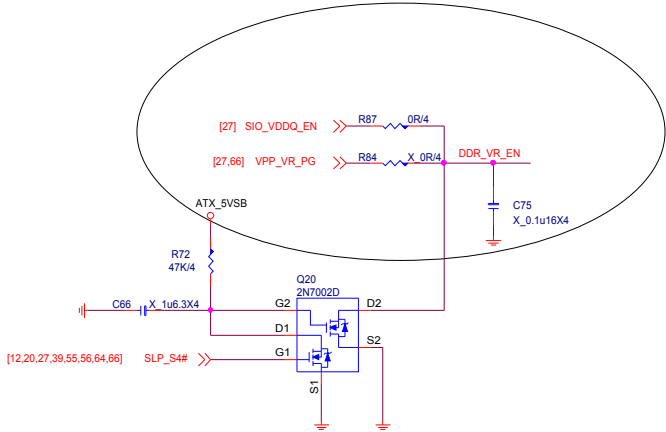
$R_{ocset} = 1.2 * I_{max} * R_{dson(1ow)} / I_{ocset}$
 $= 21.6 * 3.3mohm / 10uA$
 $= 7.12K$

Rdson(Low Side) 5V
D03-4C02403-005:3.3 ~4 mohm

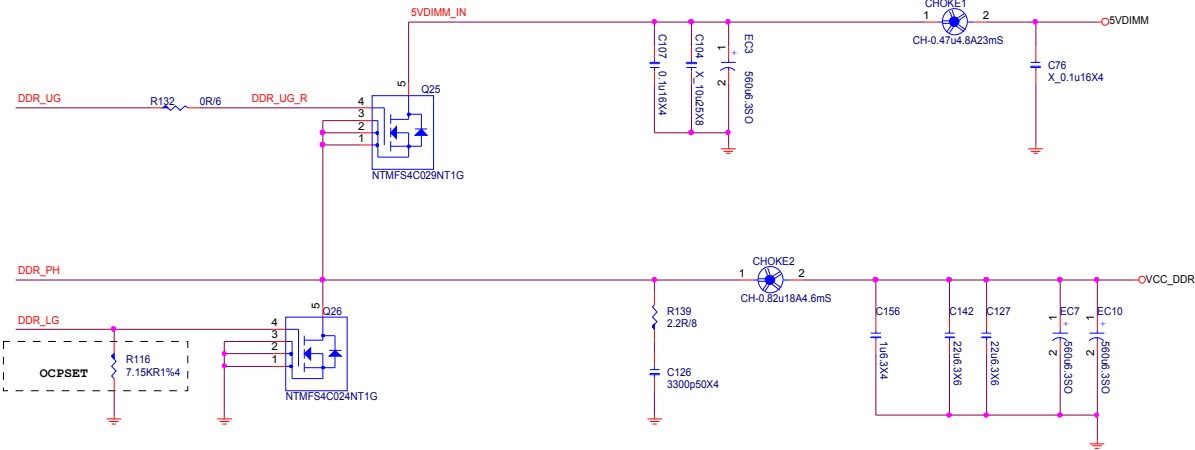
Note: Rocset(min)=5K

Default:FCCM
4.5V:FCCM
2.37V:DEM

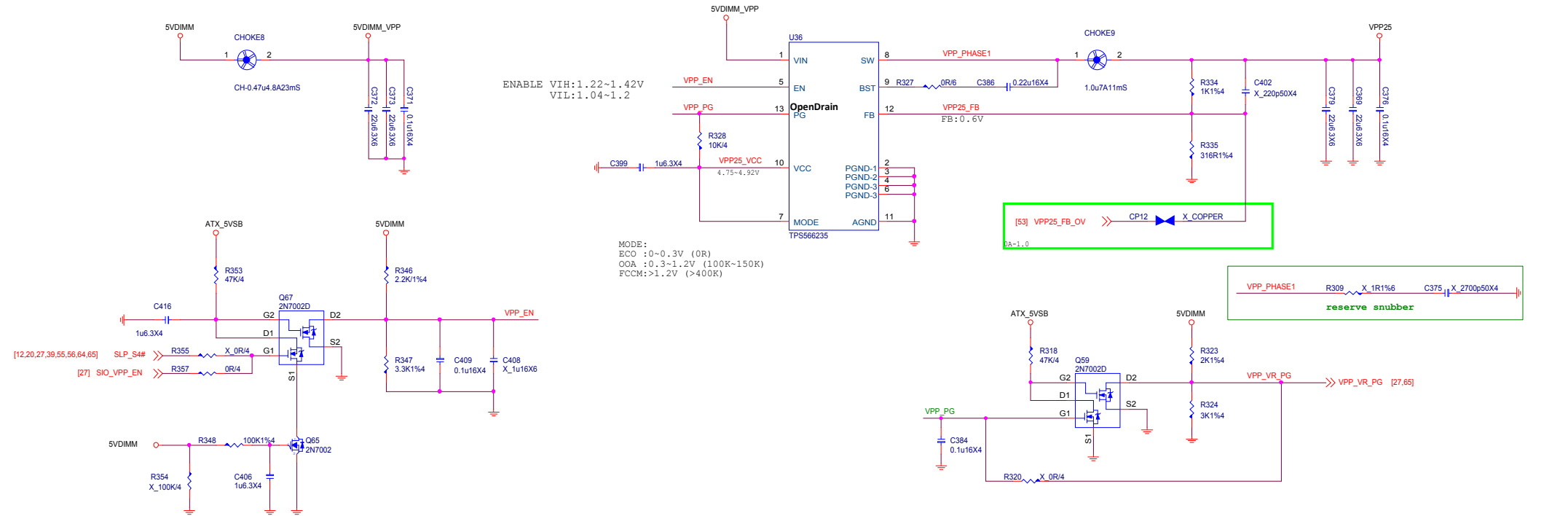
Default:FCCM
L:FCCM
H:DEM



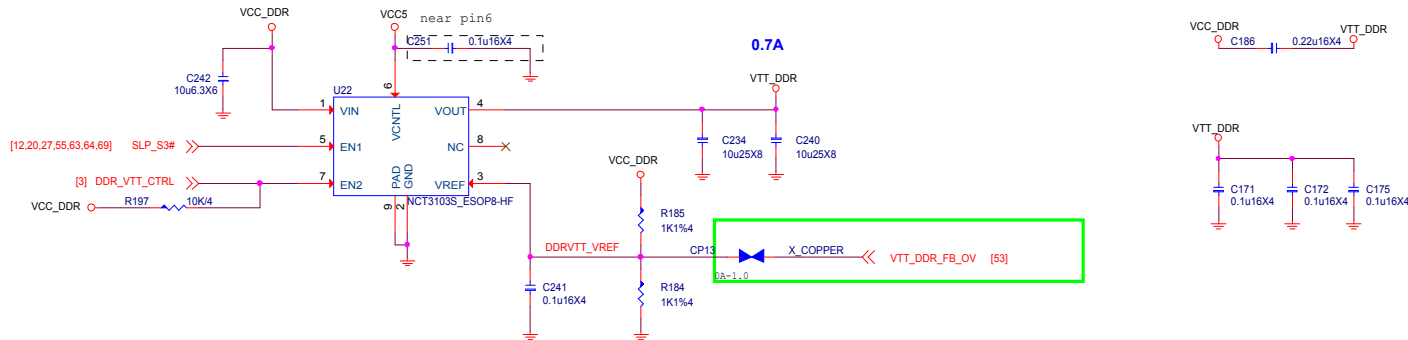
$I_{rms} = I_{out} * \sqrt{(V_{out}/V_{in}) * (1 - (V_{out}/V_{in}))}$
 $= 14.4 * 0.43$
 $= 6.192A$



VPP25 Power:2.5V,6A(For DIMM LED)



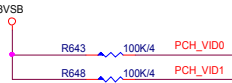
DDR VTT Power



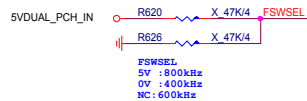
PCH Power:VCCIN AUX

1.8V 11.2A

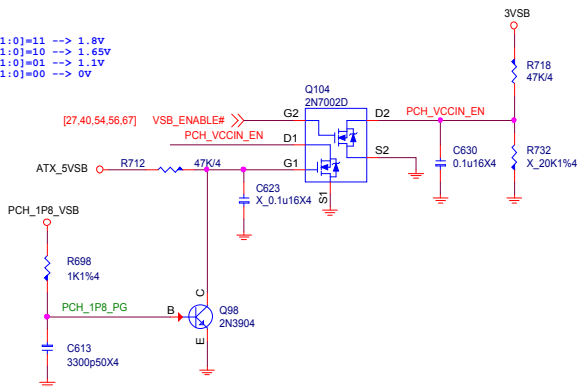
OCp=18.6A



VID[1:0]=11 --> 1.8V
VID[1:0]=10 --> 1.65V
VID[1:0]=01 --> 1.1V
VID[1:0]=00 --> 0V



VID[1:0]=11 --> 1.8V
VID[1:0]=10 --> 1.65V
VID[1:0]=01 --> 1.1V
VID[1:0]=00 --> 0V



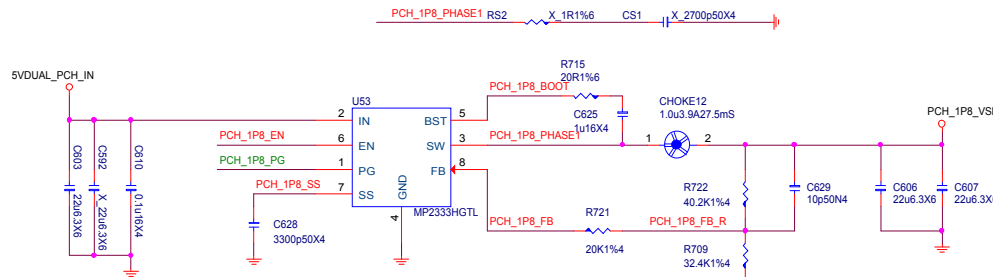
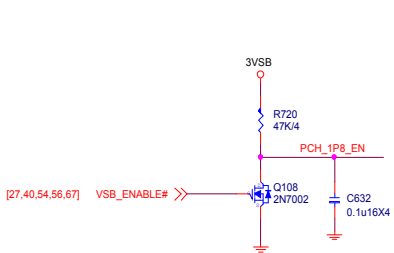
$$I_{rms} = I_{out} * \sqrt{((V_{out}/V_{in}) * (1 - (V_{out}/V_{in})))}$$

$$= 11.2 * 0.43$$

$$= 4.22A$$

PCH 1P8 VSB

1.8V; 2.048A



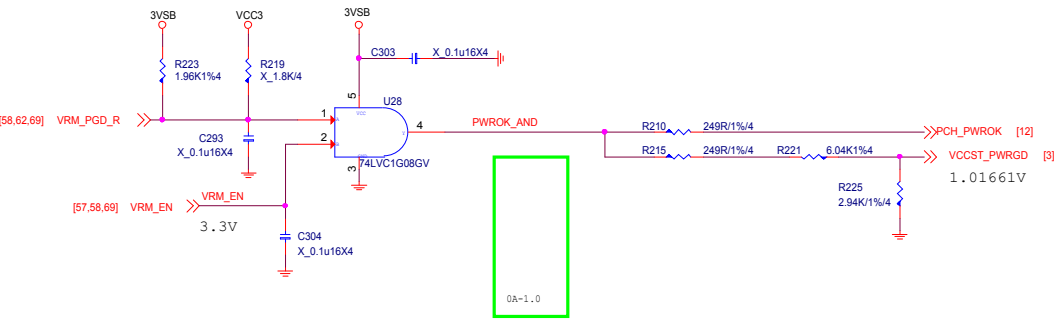
$$V_{out} = V_{ref} * (1 + R1606/R1607)$$

$$= 0.8 * (1 + 40.2K/32.4K)$$

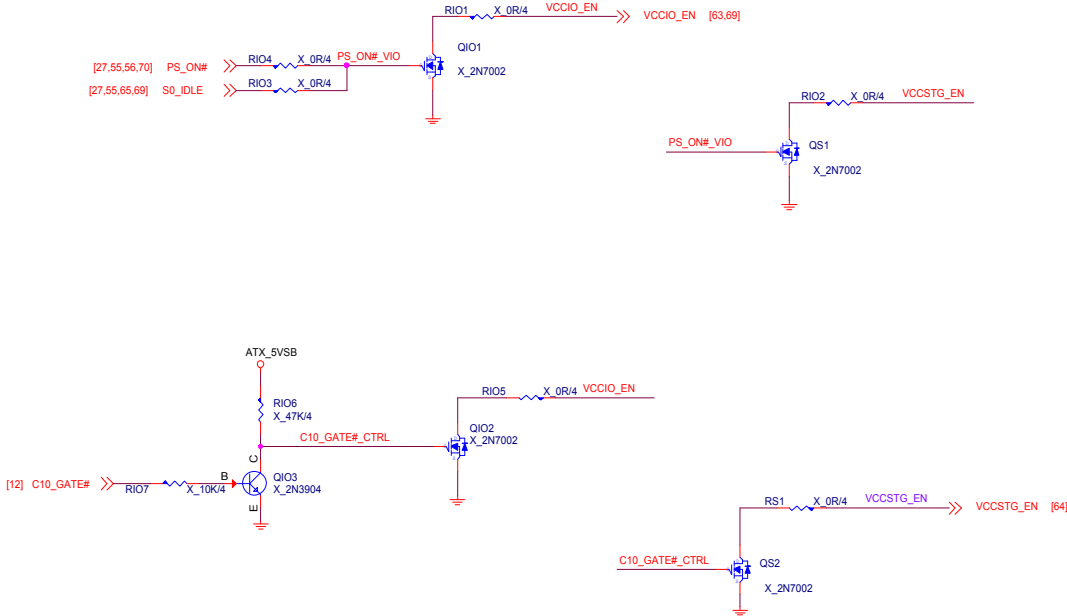
$$= 0.8 * 2.241$$

$$= 1.793V$$

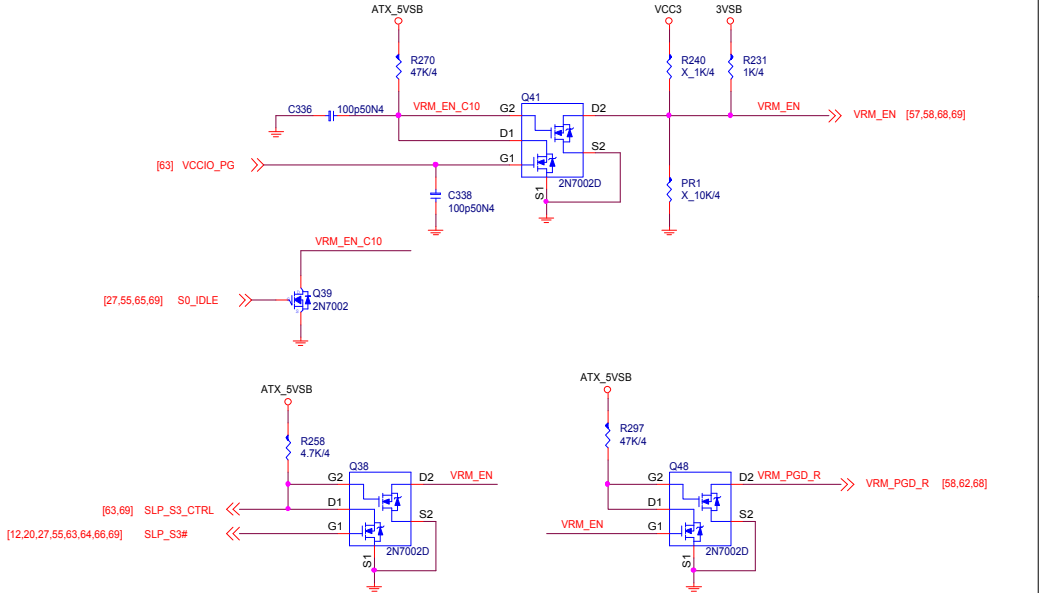
PCH PWROK/VCCST PWRGD



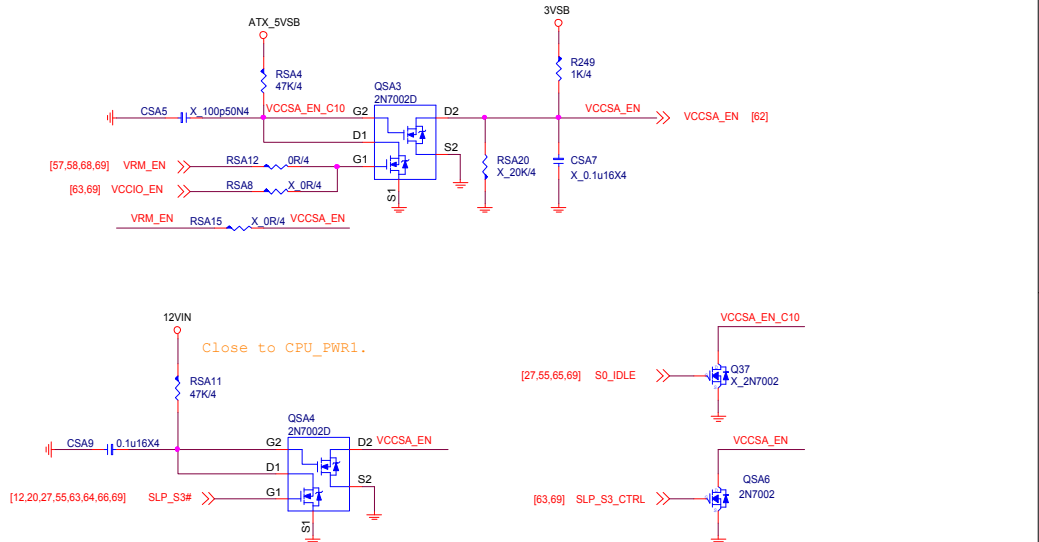
S0IX: VCCIO/VCCSTG OFF



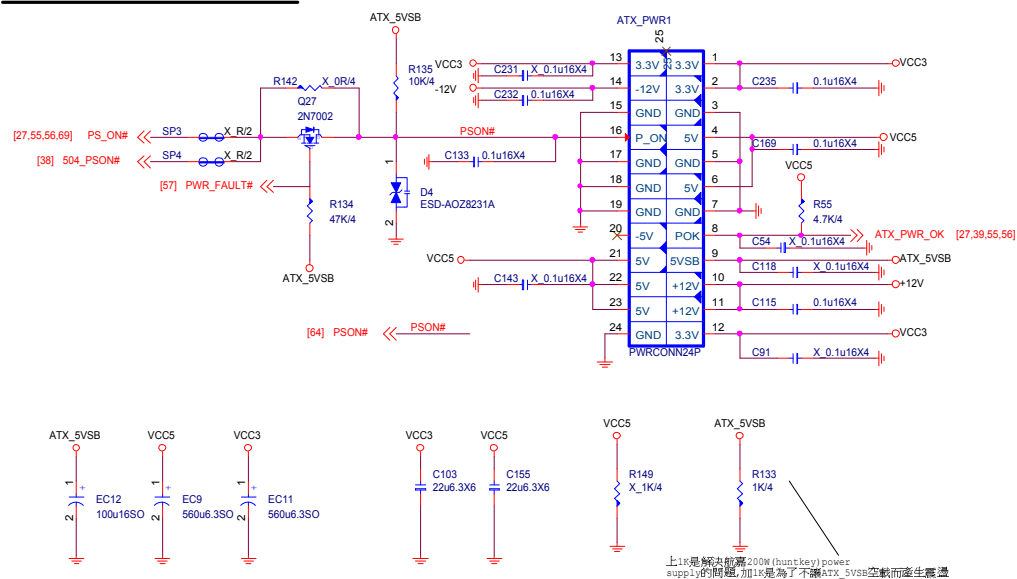
PWR-VRM-Sequence



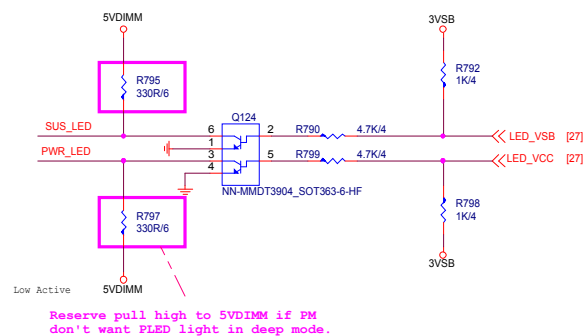
PWR-VCCSA-Sequence



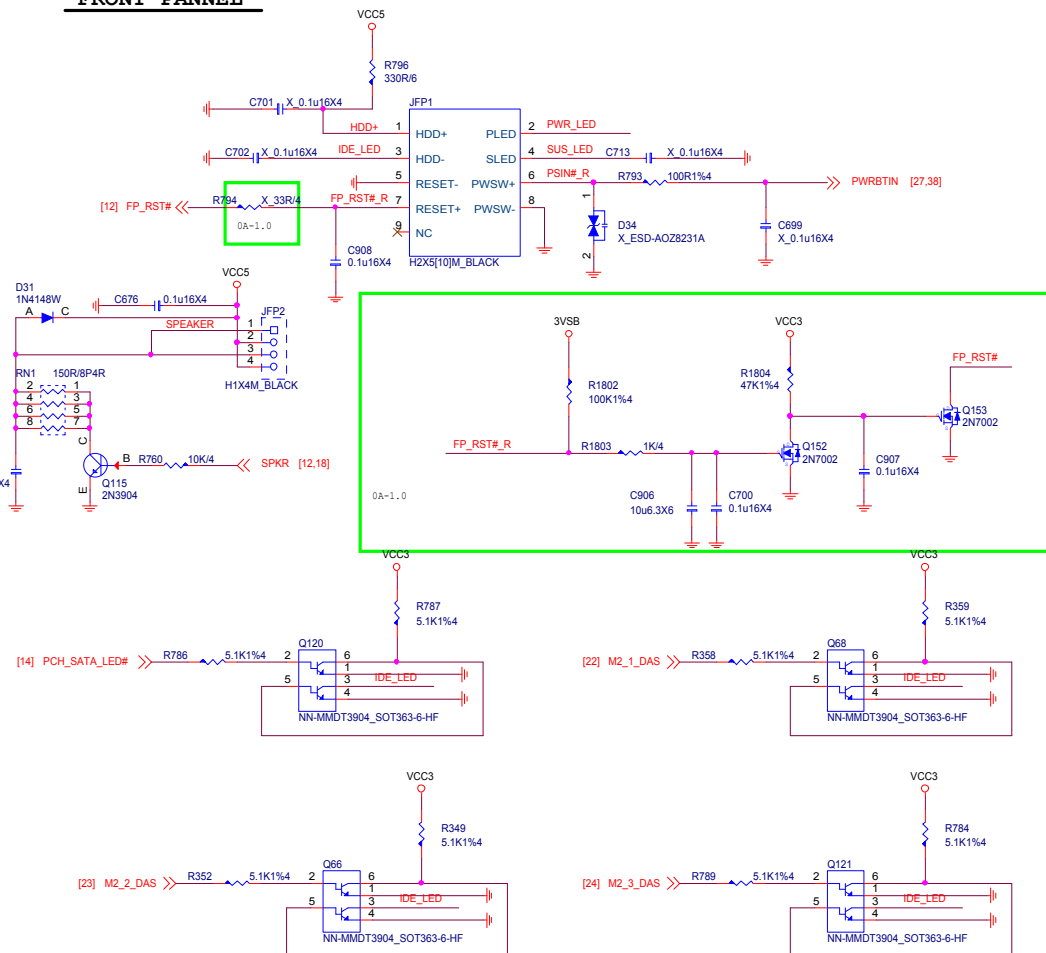
ATX POWER CONNECTOR



**Front Panel
LED**



FRONT PANNEL



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EZ Debug LED

